Class Notes

PH4113: General Theory of Relativity and Cosmology

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Course Information

§1 Marks Distribution

Mid-Semester Exams: 20 Marks.

End-Semester Exams: 50 Marks.

Internal Assignment: 30 Marks.

- ▷ Class Test: Best 2 of 3 (20 Marks),
- ▷ AACP: Assignment, Attendance, Class Performance (10 Marks).

§2 Syllabus

Basic overview of the course syllabus:

1. Basics of Riemannian Geometry,
2. Einstein's Field Equations,
3. Schwarzschild Solution,
4. Maxwell's Equations in Curved Spacetime,
5. Interior Solution of a Star,
6. Gravitational Waves,
7. Kiling Vectors and Space-Time Symmetries,
8. Cosmology.

§3 References

1. *A First Course in General Relativity* by Bernard Schutz[2]. This is the primary textbook for this course.
2. *Gravitation* by Misner, Thorne, and Wheeler[1]. The "Big Book", a classic reference for general relativity.

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Lec 01: What is "Relativity"?

Before we start with the course, let's first clarify what are we going to talk about for the next few months? To understand the motivation behind the theory of relativity, we need to ask ourselves a few questions:

- What is "relativity"?

  This is a broad term in physics, which came into use due to the work of Galileo and Newton. It refers to the idea that the *laws of physics* are the same in all inertial frames of reference.

  Operationally, this means that if you perform an experiment in one inertial frame and then repeat it in another inertial frame, the results will be the same. So in general, the `Principles of Relativity` is a mathematical framework that allows two observers in relative motion to compare their observations and measurements of the *same* physical event.

- What is "special" in special theory of relativity?

  The term *special* refers to the fact that the theory applies only to inertial frames of reference, which are frames that are not accelerating.

- What is "general" in general theory of relativity?

  The term *general* refers to the fact that the theory applies to all frames of reference, including those that are accelerating.

One thing to keep in mind is that general relativity is not a theory of gravity, rather gravity is a consequence of the general theory of relativity. In this course, we will see that how gravity is a *pseudo-force*.

§01.1 Laws of Physics

The modern picture of the laws of physics is a mathematical formalism that describes the behavior of physical systems. This way of thinking about physics was started with the work Newton in the 17th century. He formulated the laws of motion and the law of universal gravitation, which are now known as Newton's laws of motion and Newton's law of gravitation, respectively.

In standard physics textbooks, Newton's laws of motion are usually stated as follows:

- **First Law:** An object's velocity will remain constant unless acted upon by an external force.
- **Second Law:**
$$\vec{\mathbf{F}} = m\vec{\mathbf{a}}, \tag{01.1}$$
  where $\vec{\mathbf{F}}$ is the external force acting on the object, m is the mass of the object, and $\vec{\mathbf{a}}$ is the acceleration of the object.
- **Third Law:** For every action, there is an equal and opposite reaction.

  This applies in the case of two objects interacting with each other, where the force exerted by one object on the other is equal in magnitude and opposite in direction to the force exerted by the second object on the first *i.e.*
$$\vec{\mathbf{F}}_{12} = -\vec{\mathbf{F}}_{21}. \tag{01.2}$$

The Newton's laws of gravitation is stated as follows:

$$\vec{\mathbf{F}}_{12} = \frac{Gm_1m_2}{r^2}\hat{\boldsymbol{r}}, \tag{01.3}$$

where $\vec{\mathbf{F}}_{12}$ is the gravitational force exerted on object 1 by object 2, $\boldsymbol{r} = \vec{\mathbf{r}}_2 - \vec{\mathbf{r}}_1$ is the relative position vector between the two objects, G is the universal gravitational constant, and m_1 and m_2 are the masses of the two objects.

§01.1.1 Issues with Naive Formulation of Newton's Laws

The way we have stated the laws of motion, many tend to believe that second law implies the first law like this:

$$\vec{\mathbf{F}} = 0 \quad \Rightarrow \quad \vec{\mathbf{a}} = 0 \quad \Rightarrow \quad \text{constant velocity.}$$

However, this is not true. The correct way to state the first law (which is also known as the *law of inertia*) is as follows:

> An observer is said to be in an *inertial frame of reference* if and only if "body remains at rest or moves with constant velocity in the absence of any external force".

This means that the first law is a definition of an inertial frame of reference, and it does not follow from the second law. Now, the reformulated second and third laws are as follows:

- **Second Law:** For an inertial frame of reference, the acceleration of an object is directly proportional to the net external force acting on it and inversely proportional to its mass:

$$\vec{\mathbf{F}} = m\vec{\mathbf{a}}. \tag{01.4}$$

- **Third Law:** For an inertial frame of reference, the forces exerted by two objects on each other are equal in magnitude and opposite in direction.

Lec 02: Galilean Transformation

Consider a ball hanging from ceiling of a train by a string. We are sitting in the train and observing the ball. Then train starts accelerating, and we see the ball moving, but there is no external force acting on the ball. This may seem like a violation of Newton's second law, but it is not. Because we are not in an inertial frame of reference. To make sense of physical laws observed in different frames of reference, we need to find a common ground to compare them. For this purpose, Galileo introduced the Principle of Relativity, which states that

> "*The laws of physics are the same in all inertial frames of reference.*"

That time 'laws of physics' meant Newton's laws of motion, and law of gravitation. Newton later formulated the transformation between two inertial frames of reference, which is now known as the Galilean Transformation.

§02.1 Galilean Transformation

Before we proceed, we need to talk about "*time*," according to Newton and Galileo, there is a universal time that is the same for all observers, regardless of their state of motion.

Let us consider two inertial frames of reference, S (lab frame) and S' (moving frame), which are moving with respect to each other with a constant velocity $\vec{\mathbf{v}}$.

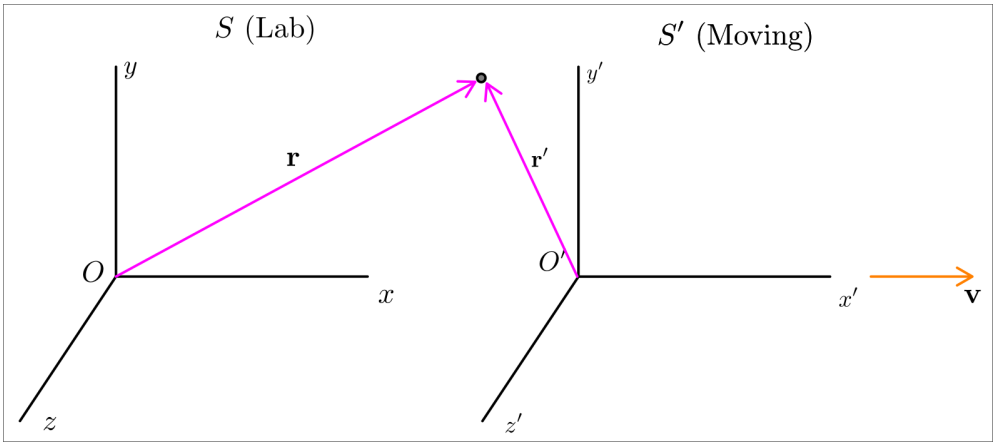


Figure 02.1: Two inertial frames of reference S and S' moving with respect to each other with constant velocity $\vec{\mathbf{v}}$ (assumed to be along the x-axis).

Let $\vec{\mathbf{R}}$ be the position vector of origin of S' (denoted by O') with respect to S (denoted by O), and assume at $t = 0$, $\vec{\mathbf{R}} = 0$. Then we have

$$\vec{\mathbf{R}} = \vec{\mathbf{v}}t.$$

Now, let we have a point particle, which is at position $\vec{\mathbf{r}}$ in S and at position $\vec{\mathbf{r}}'$ in S'. Then we have

$$\vec{\mathbf{r}}' = \vec{\mathbf{r}} - \vec{\mathbf{R}} = \vec{\mathbf{r}} - \vec{\mathbf{v}}t \quad \text{and} \quad t' = t. \tag{02.1}$$

This is the Galilean Transformation between two inertial frames of reference S and S'. In coordinate form, we can write the transformation as

$$\begin{aligned} t' &= t, \\ x' &= x - v_x t, \\ y' &= y - v_y t, \\ z' &= z - v_z t. \end{aligned} \quad \equiv \quad \begin{pmatrix} t' \\ x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -v_x & 1 & 0 & 0 \\ -v_y & 0 & 1 & 0 \\ -v_z & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} t \\ x \\ y \\ z \end{pmatrix}. \tag{02.2}$$

Let us now check whether the laws of physics are invariant under this transformation.

§02.1.1 Invariance of "Laws of Physics"

Law of Gravitation

Consider two point particles with masses m_1 and m_2 at positions $\vec{\mathbf{r}}_1$ and $\vec{\mathbf{r}}_2$ in S and $\vec{\mathbf{r}}'_1$ and $\vec{\mathbf{r}}'_2$ in S'.

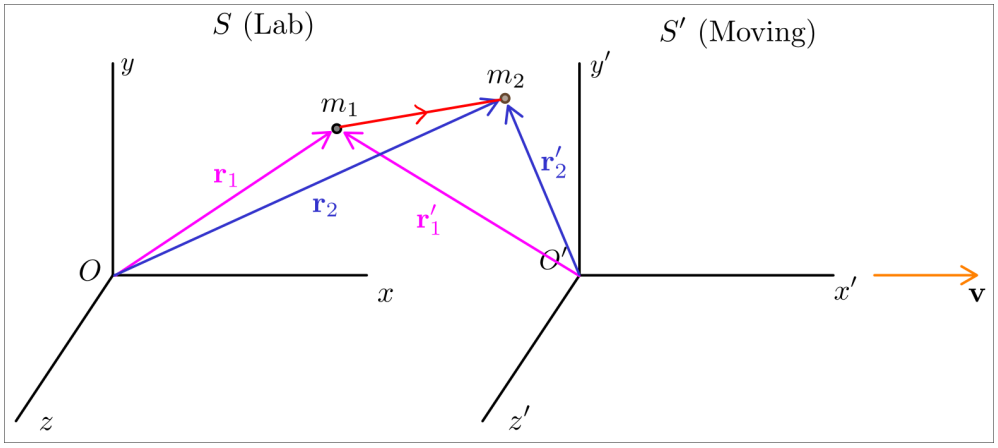


Figure 02.2: Two point particles with masses m_1 and m_2 at positions $\vec{\mathbf{r}}_1$ and $\vec{\mathbf{r}}_2$ in S and $\vec{\mathbf{r}}'_1$ and $\vec{\mathbf{r}}'_2$ in S'.

Let $\vec{\boldsymbol{r}}$ be the relative position vector between them, $\vec{\boldsymbol{r}} = \vec{\mathbf{r}}_2 - \vec{\mathbf{r}}_1$. Then

$$\begin{aligned}\vec{\boldsymbol{r}} &= \vec{\mathbf{r}}_2 - \vec{\mathbf{r}}_1 \\ &= (\vec{\mathbf{r}}'_2 + \vec{\mathbf{v}}t') - (\vec{\mathbf{r}}'_1 + \vec{\mathbf{v}}t') \\ &= \vec{\mathbf{r}}'_2 - \vec{\mathbf{r}}'_1 \\ &= \vec{\boldsymbol{r}}'.\end{aligned}$$

Now, the law of gravitation states that the gravitational force between the two particles is given by

$$\vec{\mathbf{F}}_{12} = \frac{Gm_1m_2}{r^2}\hat{\boldsymbol{r}},$$

since $\vec{\boldsymbol{r}} = \vec{\boldsymbol{r}}'$, we have

$$\vec{\mathbf{F}}'_{12} = \frac{Gm_1m_2}{r'^2}\hat{\boldsymbol{r}}' = \vec{\mathbf{F}}_{12}. \tag{02.3}$$

This means that the law of gravitation is invariant under the Galilean transformation.

Newton's Laws of Motion

Let's start with the third law of motion, that time only force known to us was the gravitational force, which is invariant under the Galilean transformation, so we have

$$\vec{\mathbf{F}}_{12} = -\vec{\mathbf{F}}_{21} \quad \Rightarrow \quad \vec{\mathbf{F}}'_{12} = -\vec{\mathbf{F}}'_{21}. \tag{02.4}$$

This means that the third law of motion is invariant under the Galilean transformation.

Now, let's consider the second law of motion, let mass m_1 is under the influence of an external force $\vec{\mathbf{F}}_{12}$ (gravitational force) in S, then we have

$$\vec{\mathbf{F}}_{12} = m_1\frac{\mathrm{d}^2\vec{\mathbf{r}}_1}{\mathrm{d}t^2}.$$

We have

$$\begin{aligned}&\vec{\mathbf{r}}_1 = \vec{\mathbf{r}}'_1 + \vec{\mathbf{v}}t' \\ \Rightarrow \quad &\frac{\mathrm{d}\vec{\mathbf{r}}_1}{\mathrm{d}t} = \frac{\mathrm{d}\vec{\mathbf{r}}'_1}{\mathrm{d}t'}\frac{\mathrm{d}t'}{\mathrm{d}t} + \vec{\mathbf{v}}\frac{\mathrm{d}t'}{\mathrm{d}t}\end{aligned}$$

$$\Rightarrow \quad \frac{\mathrm{d}\vec{\mathbf{r}}_1}{\mathrm{d}t} = \frac{\mathrm{d}\vec{\mathbf{r}}'_1}{\mathrm{d}t'} + \vec{\mathbf{v}}$$
$$\Rightarrow \quad \frac{\mathrm{d}^2\vec{\mathbf{r}}_1}{\mathrm{d}t^2} = \frac{\mathrm{d}^2\vec{\mathbf{r}}'_1}{\mathrm{d}t'^2} + 0.$$

Thus, we have

$$\vec{\mathbf{F}}'_{12} = m_1 \frac{\mathrm{d}^2\vec{\mathbf{r}}'_1}{\mathrm{d}t'^2} = m_1 \frac{\mathrm{d}^2\vec{\mathbf{r}}_1}{\mathrm{d}t^2} = \vec{\mathbf{F}}_{12}. \tag{02.5}$$

This means that the second law of motion is invariant under the Galilean transformation.

§02.1.2 Galilean Velocity Addition

Consider a particle moving with velocity $\vec{\mathbf{u}}$ in S and with velocity $\vec{\mathbf{u}}'$ in S'. Then we have

$$\begin{aligned}\vec{\mathbf{u}} = \frac{\mathrm{d}\vec{\mathbf{r}}}{\mathrm{d}t} &= \frac{\mathrm{d}\vec{\mathbf{r}}'}{\mathrm{d}t'}\frac{\mathrm{d}t'}{\mathrm{d}t} + \vec{\mathbf{v}}\frac{\mathrm{d}t'}{\mathrm{d}t} \\ &= \frac{\mathrm{d}\vec{\mathbf{r}}'}{\mathrm{d}t'} + \vec{\mathbf{v}} \\ \Rightarrow \quad \vec{\mathbf{u}} &= \vec{\mathbf{u}}' + \vec{\mathbf{v}}.\end{aligned}$$

However, there is an issue with this velocity addition. If we consider velocity of light, so we have

$$c = c' + v,$$

which means that $c \neq c'$ if $v \neq 0$, which means that the speed of light changes with respect to the observer's frame of reference. After experimental results due to Michelson-Morley, it was found that the speed of light is a universal constant, which means that there is something wrong with the Galilean transformation. This led to the development of the theory of special relativity (specifically development of Lorentz transformation).

We know that in special relativity, length contracts in the direction of motion, which means Newton's laws of gravitation is not invariant under Lorentz transformation. This led to the development of general relativity.

Lec 03: Lorentz Transformation

In 1892, Lorentz introduced a transformation that relates the coordinates of events as measured in two inertial frames moving relative to each other at a constant velocity such that the speed of light is invariant in both frames. This transformation is now known as the Lorentz Transformation (L.T.).

$$\begin{aligned} t' &= \gamma\left(t - \frac{v}{c^2}x\right), \\ x' &= \gamma(x - vt), \\ y' &= y, \\ z' &= z, \end{aligned} \tag{03.1}$$

where $\gamma = 1\Big/\sqrt{1 - \frac{v^2}{c^2}}$ is the Lorentz factor, c is the speed of light, and v is the relative velocity between the two frames along the x-axis.

With this transformation, we have the following velocity addition formula:

$$\begin{aligned} u' = \frac{\mathrm{d}x'}{\mathrm{d}t'} &= \frac{\mathrm{d}x'/\mathrm{d}t}{\mathrm{d}t'/\mathrm{d}t} \\ &= \frac{\gamma(\mathrm{d}x/\mathrm{d}t - v)}{\gamma\left(1 - \frac{v}{c^2}\,\mathrm{d}x/\mathrm{d}t\right)} \\ &= \frac{u - v}{1 - \frac{uv}{c^2}}. \end{aligned}$$

Assignment 01.1 (Velocity Addition Formula)**.**
Consider two inertial frames S and S' moving with respect to each other with a constant velocity v along the x-axis. If an object is moving with a velocity u' in frame S', show that its velocity u in frame S is given by the velocity addition formula:

$$u = \frac{u' + v}{1 + \frac{u'v}{c^2}}.$$

It is easy to see that with these new velocity addition formula, the speed of light is invariant in both frames, *i.e.*,

$$u = \frac{c + v}{1 + \frac{cv}{c^2}} = c.$$

§03.1 Implications of L.T. on "Laws of Physics"

We have already seen that the laws of physics are invariant under the Galilean transformation, so we need to check whether they are invariant under the Lorentz transformation.

§03.1.1 Newton's Law of Motion

Second Law of Motion

Clearly,

$$\frac{\mathrm{d}^2x'}{\mathrm{d}t'^2} \neq \frac{\mathrm{d}^2x}{\mathrm{d}t^2},$$

so the external force acting on a particle observed in two inertial frames is not the same *i.e.* $F' \neq F$. This means that Newton's second law is not invariant under the Lorentz transformation.

The Special Theory of Relativity developed by Einstein in 1905, reconciles this by introducing the concept of 4-Force and 4-Momentum.

Third Law of Motion

Lorentz transformation implies that the maximum speed of any object (or signal) is c. If we consider two particles A and B interacting with each other, say particle A exerts a force $\vec{\mathbf{F}}_{BA}$ on particle B it will take a finite time for the force to propagate from A to B. After the force has been applied, particle B will exert a force $\vec{\mathbf{F}}_{AB}$ on particle A. Since there is a finite time delay in the propagation of the force, the forces will not be equal and opposite, (see fig. 03.1).

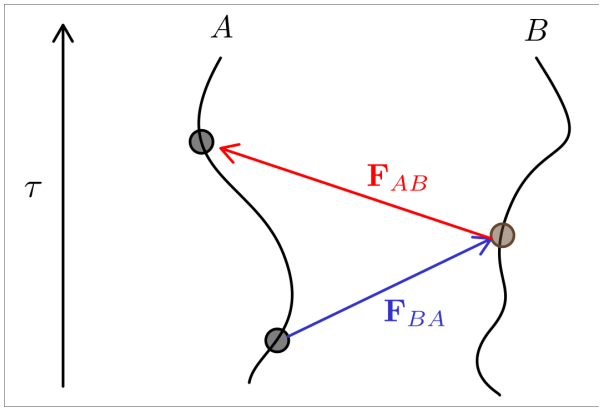


Figure 03.1: Illustration of the third law of motion in the context of Lorentz transformation.

Thus, the third law of motion is not invariant under the Lorentz transformation.

Similarly, we can reconcile the third law of motion with the Special Theory of Relativity by introducing the notion of fields. In this framework, there is no particle-to-particle interaction, but rather a particle-to-field and field-to-field interaction. The forces are mediated by fields, which propagate at the speed of light, ensuring that the third law of motion is satisfied in a relativistic context. An early example of this is the *electromagnetic theory*, where the interaction between charged particles is mediated by the electromagnetic field.

§03.1.2 Newton's Law of Gravitation

An immediate consequence of the Lorentz transformation is that the spatial separation between two points in space is not invariant, *i.e.* for a moving observer, the distance between two points in space will appear contracted along the direction of motion. Since the gravitational force is proportional to the inverse square of the distance between two masses, the gravitational force will not be invariant under the Lorentz transformation.

§03.2 Spacetime Interval

It is easy to see that the spatial distance is not invariant under the Lorentz transformation, but we can define a new quantity called the spacetime interval as

$$\mathrm{d}s^2 = -c^2\,\mathrm{d}t^2 + \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2\,. \tag{03.2}$$

Assignment 01.2 (Invariance of Spacetime Interval)**.**
Show that the spacetime interval is invariant under the Lorentz transformation, *i.e.* show that

$$\mathrm{d}s'^2 = \mathrm{d}s^2\,.$$

In contrast to the spatial distance, the spacetime interval can be any real number, depending on the sign of $\mathrm{d}s^2$:

- If $\mathrm{d}s^2 > 0$, the interval is called `space-like`, and the two events can be causally disconnected.
- If $\mathrm{d}s^2 < 0$, the interval is called `time-line`, and the two events can be causally connected.
- If $\mathrm{d}s^2 = 0$, the interval is called `light-like` or `null`, and the two events are connected by a light signal.

For light, we have

$$\frac{\mathrm{d}r}{\mathrm{d}t} = c \quad \Rightarrow \quad c^2\,\mathrm{d}t^2 = \mathrm{d}r^2 \quad \Rightarrow \quad \mathrm{d}s^2 = 0.$$

Thus, the trajectory of light in spacetime is a null curve, *i.e.* the spacetime interval between any two points on this curve is zero.

We know that every massive particle moves slower than light, so we have

$$\frac{\mathrm{d}r}{\mathrm{d}t} < c \quad \Rightarrow \quad c^2\,\mathrm{d}t^2 > \mathrm{d}r^2 \quad \Rightarrow \quad \mathrm{d}s^2 < 0.$$

Thus, all massive bodies move along time-like trajectories in spacetime, *i.e.* the spacetime interval between any two points on this curve is time-like.

Lec 04: 4-Vectors or Contravariant Vectors

Let's start by writing the infinitesimal version of L.T.

$$\begin{aligned} \mathrm{d}t' &= \gamma\left(\mathrm{d}t - \frac{v}{c^2}\,\mathrm{d}x\right), \\ \mathrm{d}x' &= \gamma(\mathrm{d}x - v\,\mathrm{d}t). \end{aligned} \tag{04.1}$$

We can express this in a more compact form using the matrix notation:

$$\begin{pmatrix} \mathrm{d}t' \\ \mathrm{d}x' \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma v/c^2 \\ -\gamma v & \gamma \end{pmatrix} \begin{pmatrix} \mathrm{d}t \\ \mathrm{d}x \end{pmatrix}.$$

If we look at the matrix carefully, we can see that it is Jacobian of the transformation from the unprimed to the primed coordinates. We can write this as:

$$J = \frac{\partial(t', x')}{\partial(t, x)} \equiv \begin{pmatrix} \frac{\partial t'}{\partial t} & \frac{\partial t'}{\partial x} \\ \frac{\partial x'}{\partial t} & \frac{\partial x'}{\partial x} \end{pmatrix}$$

Define 4-dimensional "space-time" with following coordinates:

$$\begin{cases} x^0 = ct, \\ x^1 = x, \\ x^2 = y, \\ x^3 = z. \end{cases} \xrightarrow{\text{position vector}} \bar{\mathbf{x}} = \begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix} = \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix}. \tag{04.2}$$

With this, we can write the transformation as[1]:

$$\mathrm{d}\bar{\mathbf{x}}' = \begin{pmatrix} \mathrm{d}x'^0 \\ \mathrm{d}x'^1 \\ \mathrm{d}x'^2 \\ \mathrm{d}x'^3 \end{pmatrix} = \underbrace{\begin{pmatrix} \frac{\partial x'^0}{\partial x^0} & \cdots & \frac{\partial x'^0}{\partial x^3} \\ \vdots & \ddots & \vdots \\ \frac{\partial x'^3}{\partial x^0} & \cdots & \frac{\partial x'^3}{\partial x^3} \end{pmatrix}}_{\Lambda} \begin{pmatrix} \mathrm{d}x^0 \\ \mathrm{d}x^1 \\ \mathrm{d}x^2 \\ \mathrm{d}x^3 \end{pmatrix} \qquad \equiv \qquad \mathrm{d}\bar{\mathbf{x}}' = \Lambda\,\mathrm{d}\bar{\mathbf{x}}\,. \tag{04.3}$$

For the Lorentz transformation, this transformation matrix is given by:

$$\begin{aligned} x'^0 &= \gamma\left(x^0 - \beta x^1\right), \\ x'^1 &= \gamma\left(x^1 - \beta x^0\right), \\ x'^2 &= x^2, \\ x'^3 &= x^3, \end{aligned} \qquad \text{thus,} \qquad \Lambda = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Let's now consider each component's transformation (see eqn. (04.3)):

$$\mathrm{d}x'^\mu = \frac{\partial x'^\mu}{\partial x^0}\,\mathrm{d}x^0 + \frac{\partial x'^\mu}{\partial x^1}\,\mathrm{d}x^1 + \frac{\partial x'^\mu}{\partial x^2}\,\mathrm{d}x^2 + \frac{\partial x'^\mu}{\partial x^3}\,\mathrm{d}x^3 = \sum_{\nu=0}^{3} \frac{\partial x'^\mu}{\partial x^\nu}\,\mathrm{d}x^\nu\,.$$

From this point, we will use the Einstein summation convention, 'repeated indices (twice) are summed over (must be on the same side of the equation).' Note that the summed indices are one upper and one lower index (no exceptions). With this, we can write the transformation of infinitesimal displacement as:

$$\mathrm{d}x'^\mu = \frac{\partial x'^\mu}{\partial x^\nu}\,\mathrm{d}x^\nu = \Lambda^\mu{}_\nu\,\mathrm{d}x^\nu\,.$$

[1] For 3-dimensional vectors, we use the notation $\vec{\mathbf{x}}$. For 4-vectors, we use the notation $\bar{\mathbf{x}}$.

§04.1 Contravariant Vectors

Definition 04.1 (Contravariant Vectors):
If a mathematical object transforms under a coordinate transformation like a differential displacement $\mathrm{d}\bar{\mathbf{x}}$ (*i.e.* $\mathrm{d}\bar{\mathbf{x}} \to \mathrm{d}\bar{\mathbf{x}}' = \Lambda\,\mathrm{d}\bar{\mathbf{x}}$), then it is called a *contra-variant vector*, or *contra-vector*, or a *tensor of rank (1, 0)*.

It is important to note that the notion of contra-vectors changes with the coordinate system and the transformation. For example, in Newtonian mechanics, the velocity vector $\vec{\mathbf{v}} = \frac{\mathrm{d}\vec{\mathbf{x}}}{\mathrm{d}t}$ is a contra-vector under G.T.

Problem (Velocity vector under G.T.)**.**
Show that the velocity vector $\vec{\mathbf{v}} = \frac{\mathrm{d}\vec{\mathbf{x}}}{\mathrm{d}t}$ is a contra-vector under Galilean transformation.

Solution.
In Galilean transformation, the transformation matrix is a 4×4 matrix (see eqn. (02.2)), which can be written as:

$$\begin{aligned} x'^0 &= x^0, \\ x'^1 &= x^1 - (v^1/c)x^0, \\ x'^2 &= x^2 - (v^2/c)x^0, \\ x'^3 &= x^3 - (v^3/c)x^0. \end{aligned} \qquad \text{transformation matrix: } \Lambda = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -v^1/c & 1 & 0 & 0 \\ -v^2/c & 0 & 1 & 0 \\ -v^3/c & 0 & 0 & 1 \end{pmatrix}. \tag{04.4}$$

The '4-velocity' is defined as the derivative of the position vector with respect to the time coordinate (as in G.T. time coordinate is the same in both frames):

$$u^\mu = \frac{\mathrm{d}x^\mu}{\mathrm{d}t} = \begin{pmatrix} c \\ u^1 \\ u^2 \\ u^3 \end{pmatrix} \quad ; \quad u'^\mu = \frac{\mathrm{d}x'^\mu}{\mathrm{d}t'} = \begin{pmatrix} c \\ u'^1 \\ u'^2 \\ u'^3 \end{pmatrix}.$$

We know that the differential displacement transforms as:

$$\mathrm{d}x'^\mu = \Lambda^\mu{}_\nu\,\mathrm{d}x^\nu\,.$$

We have $\mathrm{d}x^\nu = u^\mu\,\mathrm{d}t$, and $\mathrm{d}x'^\mu = u'^\mu\,\mathrm{d}t'$. Thus, we can write:

$$\begin{aligned} u'^\mu\,\mathrm{d}t' &= \Lambda^\mu{}_\nu u^\nu\,\mathrm{d}t \\ \Rightarrow\ u'^\mu &= \Lambda^\mu{}_\nu u^\nu \frac{\mathrm{d}t}{\mathrm{d}t'}. \end{aligned}$$

Now, we know that $\mathrm{d}t' = \mathrm{d}t$, so we can write:

$$u'^\mu = \Lambda^\mu{}_\nu u^\nu.$$

This shows that the '4-velocity' transforms like a contra-vector under Galilean transformation. ◂

Assignment 01.3 (4-acceleration)**.**
Show that the '4-acceleration' $a^\mu := \frac{\mathrm{d}u^\mu}{\mathrm{d}t}$ is a contra-vector under Galilean transformation.

But these definitions of '4-velocity' and '4-acceleration' are problematic, as they are not invariant under Lorentz transformation.

Problem.
Show that the $\tilde{4}$-velocity[2] $u^\mu = \frac{\mathrm{d}x^\mu}{\mathrm{d}t}$ is not a contra-vector under Lorentz transformation.

[2] This is not a '4-velocity' in the sense of special relativity, as it does not transform like a contra-vector under Lorentz transformation. The actual '4-velocity' is defined as $u^\mu = \frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}$, where τ is the proper time.

Solution.
If we proceed with the same logic as in the previous problem, we have

$$u'^{\mu} = c\frac{\mathrm{d}x'^{\mu}}{\mathrm{d}x'^{0}} = c\frac{\Lambda^{\mu}{}_{\nu}\,\mathrm{d}x^{\nu}}{\Lambda^{0}{}_{\rho}\,\mathrm{d}x^{\rho}} \neq \Lambda^{\alpha}{}_{\beta}u^{\beta}$$

◀

Lec 05: Tensors

We have seen that differential displacements transform as follows:

$$\mathrm{d}\bar{\mathbf{x}}' = \Lambda\, \mathrm{d}\bar{\mathbf{x}}\,.$$

Assume that the coordinate transformation is invertible, *i.e.*, there exists a matrix Λ^{-1} such that

$$\mathrm{d}\bar{\mathbf{x}} = \Lambda^{-1}\, \mathrm{d}\bar{\mathbf{x}}'\,.$$

In component form, this is

$$\mathrm{d}x^\mu = (\Lambda^{-1})^\mu{}_\nu\, \mathrm{d}x'^\nu\,, \qquad \text{where } (\Lambda^{-1})^\mu{}_\nu = \frac{\partial x^\mu}{\partial x'^\nu}$$

§05.1 Gradient Vector

In 3D, the gradient vector is defined as

$$\vec{\boldsymbol{\nabla}} = \hat{\boldsymbol{\epsilon}}^i \frac{\partial}{\partial x^i} \equiv \hat{\boldsymbol{\epsilon}}^i \partial_i,$$

where $\hat{\boldsymbol{\epsilon}}^i$ is the unit vector for the i-th direction in Dual space. Define '0'-th component of the gradient vector as

$$\nabla_0 = \frac{1}{c}\frac{\partial}{\partial t} \equiv \frac{\partial}{\partial x^0} \equiv \partial_0.$$

Define "4-gradient" as

$$\bar{\boldsymbol{\nabla}} \equiv \bar{\boldsymbol{\partial}} := \begin{pmatrix} \partial_0 & \partial_1 & \partial_2 & \partial_3 \end{pmatrix}. \tag{05.1}$$

Under a coordinate transformation, we have

$$\partial'_\mu \equiv \frac{\partial}{\partial x'^\mu} = \frac{\partial x^\nu}{\partial x'^\mu}\frac{\partial}{\partial x^\nu} \equiv (\Lambda^{-1})^\nu{}_\mu \partial_\nu \quad \Rightarrow \quad \bar{\boldsymbol{\partial}}' = \Lambda^{-1}\bar{\boldsymbol{\partial}}. \tag{05.2}$$

This means that the '4-gradient' is not a contra-variant vector!!

Definition 05.1 (Covariant Vector):
If a mathematical object transforms under a coordinate transformation like the gradient $\bar{\boldsymbol{\partial}}$ (*i.e.*, $\bar{\boldsymbol{\partial}} \to \bar{\boldsymbol{\partial}}' = \Lambda^{-1}\bar{\boldsymbol{\partial}}$), then it is called a *covariant vector*, or a *co-vector*, or a *tensor of rank (0, 1)*.

§05.2 Tensors

With this, we can define a general tensor of rank (m, n). Consider the following object,

$$W = \mathrm{d}\bar{\mathbf{x}}\, \mathrm{d}\bar{\mathbf{x}}^\top = \begin{pmatrix} \mathrm{d}x^0 \\ \mathrm{d}x^1 \\ \mathrm{d}x^2 \\ \mathrm{d}x^3 \end{pmatrix} \begin{pmatrix} \mathrm{d}x^0 & \mathrm{d}x^1 & \mathrm{d}x^2 & \mathrm{d}x^3 \end{pmatrix} = \begin{pmatrix} \mathrm{d}x^0\,\mathrm{d}x^0 & \cdots & \mathrm{d}x^0\,\mathrm{d}x^3 \\ \vdots & \ddots & \vdots \\ \mathrm{d}x^3\,\mathrm{d}x^0 & \cdots & \mathrm{d}x^3\,\mathrm{d}x^3 \end{pmatrix}.$$

W is a square matrix, which is also written as

$$W = \mathrm{d}\bar{\mathbf{x}} \otimes \mathrm{d}\bar{\mathbf{x}}\,,$$

where $\otimes$ is the outer product. The components of this tensor is given by

$$W^{\mu\nu} = (\mathrm{d}\bar{\mathbf{x}} \otimes \mathrm{d}\bar{\mathbf{x}})^{\mu\nu} = \mathrm{d}x^\mu\, \mathrm{d}x^\nu\,.$$

Transformation of this object is given by

$$\begin{aligned} W'^{\mu\nu} &= (\mathrm{d}\bar{\mathbf{x}}' \otimes \mathrm{d}\bar{\mathbf{x}}')^{\mu\nu} = \mathrm{d}x'^{\mu}\,\mathrm{d}x'^{\nu} \\ &= \left(\frac{\partial x'^{\mu}}{\partial x^{\alpha}}\,\mathrm{d}x^{\alpha}\right)\left(\frac{\partial x'^{\nu}}{\partial x^{\beta}}\,\mathrm{d}x^{\beta}\right) \\ &= \frac{\partial x'^{\mu}}{\partial x^{\alpha}}\frac{\partial x'^{\nu}}{\partial x^{\beta}}\,\mathrm{d}x^{\alpha}\,\mathrm{d}x^{\beta} \\ &= \Lambda^{\mu}{}_{\alpha}\Lambda^{\nu}{}_{\beta}W^{\alpha\beta}. \end{aligned}$$

Thus, W is neither a contra-vector nor a co-vector, but a *tensor of rank (2, 0)*.

Definition 05.2 (Tensor):
If a mathematical object transforms under a coordinate transformation like $\mathrm{d}\bar{\mathbf{x}} \otimes \cdots \otimes \mathrm{d}\bar{\mathbf{x}}$ (m-copies) and $\bar{\boldsymbol{\partial}} \otimes \cdots \otimes \bar{\boldsymbol{\partial}}$ (n-copies), then it is called a *tensor of rank (m, n)*.

From classical mechanics, we know that the moment of inertia is a tensor of rank (2, 0), and stress is a tensor of rank (2, 0) as well.

Definition 05.3 (Scalar):
If a mathematical object doesn't transform under a coordinate transformation, then it is called a *scalar*, or a *tensor of rank (0, 0)*.

For example, $\mathrm{d}r^2 = (\mathrm{d}x^1)^2 + (\mathrm{d}x^2)^2 + (\mathrm{d}x^3)^2$ is a scalar under G.T. And $\mathrm{d}s^2 = -(\mathrm{d}x^0)^2 + \mathrm{d}r^2$ is a scalar under L.T.

§05.3 Inner Product

From basic linear algebra, we have defined the inner product of two vectors as

$$\vec{\mathbf{a}} \cdot \vec{\mathbf{b}} = \vec{\mathbf{a}}^{\top}\vec{\mathbf{b}}$$

Based on this, the invariant distance in 3D (under G.T.) is defined as

$$\mathrm{d}r^2 = \vec{\mathbf{dr}} \cdot \vec{\mathbf{dr}} = \vec{\mathbf{dr}}^{\top}\vec{\mathbf{dr}} = \begin{pmatrix}\mathrm{d}x^1 & \mathrm{d}x^2 & \mathrm{d}x^3\end{pmatrix}\begin{pmatrix}\mathrm{d}x^1 \\ \mathrm{d}x^2 \\ \mathrm{d}x^3\end{pmatrix}$$

In 4D (under L.T.), the invariant distance is defined as

$$\mathrm{d}s^2 = -(\mathrm{d}x^0)^2 + (\mathrm{d}x^1)^2 + (\mathrm{d}x^2)^2 + (\mathrm{d}x^3)^2 \neq \begin{pmatrix}\mathrm{d}x^0 & \mathrm{d}x^1 & \mathrm{d}x^2 & \mathrm{d}x^3\end{pmatrix}\begin{pmatrix}\mathrm{d}x^0 \\ \mathrm{d}x^1 \\ \mathrm{d}x^2 \\ \mathrm{d}x^3\end{pmatrix}$$

But if we introduce a matrix η such that

$$\mathrm{d}s^2 = \begin{pmatrix}\mathrm{d}x^0 & \mathrm{d}x^1 & \mathrm{d}x^2 & \mathrm{d}x^3\end{pmatrix}\underbrace{\begin{pmatrix}-1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1\end{pmatrix}}_{\eta}\begin{pmatrix}\mathrm{d}x^0 \\ \mathrm{d}x^1 \\ \mathrm{d}x^2 \\ \mathrm{d}x^3\end{pmatrix} = \mathrm{d}\bar{\mathbf{x}}^{\top}\,\eta\,\mathrm{d}\bar{\mathbf{x}}\,.$$

In component form, this is written as

$$\mathrm{d}s^2 = \eta_{\mu\nu}\,\mathrm{d}x^{\mu}\,\mathrm{d}x^{\nu}\,. \tag{05.3}$$

This matrix η is called the *metric tensor*. In Special Relativity, this is called the *Minkowski metric* which is diagonal with entries $(-1, 1, 1, 1)$.

In general, we have metric tensor g which is not necessarily diagonal, and the space-time interval is defined as

$$\mathrm{d}s^2 = g_{\mu\nu}\,\mathrm{d}x^{\mu}\,\mathrm{d}x^{\nu}\,. \tag{05.4}$$

Lec 06: Metric Tensor

In the last lecture, we have seen that the metric is like a map which takes a contra-vector and returns a number. Now the question is what is the rank of the metric tensor?

To answer this question, we expect the "norm" of a vector to be invariant under coordinate transformations *i.e.*, $\mathrm{d}s'^2 = \mathrm{d}s^2$.

$$\begin{aligned}\mathrm{d}s'^2 &= g'_{\mu\nu}\,\mathrm{d}x'^{\mu}\,\mathrm{d}x'^{\nu} \\ &= \left(\frac{\partial x'^{\mu}}{\partial x^{\alpha}}\frac{\partial x'^{\nu}}{\partial x^{\beta}}g'_{\mu\nu}\right)\mathrm{d}x^{\alpha}\,\mathrm{d}x^{\beta} = g_{\alpha\beta}\,\mathrm{d}x^{\alpha}\,\mathrm{d}x^{\beta}\,,\end{aligned}$$

since $\mathrm{d}\bar{\mathbf{x}}$ is arbitrary, we can write

$$\frac{\partial x'^{\mu}}{\partial x^{\alpha}}\frac{\partial x'^{\nu}}{\partial x^{\beta}}g'_{\mu\nu} = g_{\alpha\beta},$$

contracting both sides with $\frac{\partial x^{\alpha}}{\partial x'^{\rho}}$ gives

$$\delta^{\mu}{}_{\rho}\frac{\partial x'^{\nu}}{\partial x^{\beta}}g'_{\mu\nu} = \frac{\partial x^{\alpha}}{\partial x'^{\rho}}g_{\alpha\beta} \quad \Rightarrow \quad \frac{\partial x'^{\nu}}{\partial x^{\beta}}g'_{\rho\nu} = \frac{\partial x^{\alpha}}{\partial x'^{\rho}}g_{\alpha\beta},$$

contracting both sides with $\frac{\partial x^{\beta}}{\partial x'^{\sigma}}$ gives

$$\delta^{\nu}{}_{\sigma}g'_{\rho\nu} = \frac{\partial x^{\alpha}}{\partial x'^{\rho}}\frac{\partial x^{\beta}}{\partial x'^{\sigma}}g_{\alpha\beta} \quad \Rightarrow \quad g'_{\rho\sigma} = \frac{\partial x^{\alpha}}{\partial x'^{\rho}}\frac{\partial x^{\beta}}{\partial x'^{\sigma}}g_{\alpha\beta}.$$

This shows that the metric $g_{\mu\nu}$ is a rank (0, 2) tensor, and the transformation law for the metric tensor is given by

$$g'_{\mu\nu} = \frac{\partial x'^{\mu}}{\partial x^{\alpha}}\frac{\partial x'^{\nu}}{\partial x^{\beta}}g_{\alpha\beta}. \tag{06.1}$$

Assignment 02.1 (Norm of a contra-variant vector)**.**
Define the square of the norm of a contra-variant vector v^{μ} as

$$\|\bar{\mathbf{v}}\|^2 = g_{\mu\nu}v^{\mu}v^{\nu}. \tag{06.2}$$

Show that this is invariant under coordinate transformations if $g_{\mu\nu}$ is metric tensor.

Notice that the metric tensor is symmetric,

$$\mathrm{d}s^2 = g_{\mu\nu}\,\mathrm{d}x^{\mu}\,\mathrm{d}x^{\nu} = g_{\mu\nu}\,\mathrm{d}x^{\nu}\,\mathrm{d}x^{\mu} = g_{\nu\mu}\,\mathrm{d}x^{\mu}\,\mathrm{d}x^{\nu}$$

second equality follows the commutative property of multiplication, and the third equality follows from interchanging the dummy indices μ and ν. Since $\mathrm{d}\bar{\mathbf{x}}$ is arbitrary, we can write

$$g_{\mu\nu} = g_{\nu\mu}. \tag{06.3}$$

This means that we have 10 independent components in a metric tensor in a 4-dimensional space-time.

§06.1 Inverse Metric Tensor

If $\det\left(g_{\mu\nu}\right) \neq 0$, then the inverse metric tensor is defined as

$$gg^{-1} = g^{-1}g = I, \tag{06.4}$$

where g^{-1} is the inverse metric tensor, and I is the identity matrix. In terms of components, this is given by

$$g_{\nu\rho}(g^{-1})^{\rho\mu} = (g^{-1})^{\mu\rho}g_{\rho\nu} = \delta^{\mu}{}_{\nu}. \tag{06.5}$$

Assignment 02.2 (Inverse Metric Tensor)**.**
Show that the inverse metric tensor is a rank (2, 0) tensor, *i.e.*, it transforms as

$$(g'^{-1})^{\mu\nu} = \frac{\partial x'^{\mu}}{\partial x^{\alpha}} \frac{\partial x'^{\nu}}{\partial x^{\beta}} (g^{-1})^{\alpha\beta}. \tag{06.6}$$

From now on, we will the compact notation for the inverse metric tensor, *i.e.*, $(g^{-1})^{\mu\nu} = g^{\mu\nu}$, which means that if the indices are raised, then it is the inverse metric tensor, and if the indices are lowered, then it is the metric tensor.

Assignment 02.3 (Inner Product)**.**
Define the inner product of two contra-variant vectors A^{μ} and B^{ν} as

$$\bar{\mathbf{A}} \cdot \bar{\mathbf{B}} := g_{\mu\nu} A^{\mu} B^{\nu}. \tag{06.7}$$

We can use metric and inverse metric tensors to convert between contra-variant and covariant vectors.

- Contra-variant to Covariant: $A^{\mu} \xrightarrow{g_{\mu\nu}} A_{\mu} = g_{\mu\nu} A^{\nu}$.
- Covariant to Contra-variant: $A_{\mu} \xrightarrow{g^{\mu\nu}} A^{\mu} = g^{\mu\nu} A_{\nu}$.

With this, inner product can be written as

$$\bar{\mathbf{A}} \cdot \bar{\mathbf{B}} = g_{\mu\nu} A^{\mu} B^{\nu} = A^{\mu} B_{\mu} = A_{\nu} B^{\nu}. \tag{06.8}$$

Lec 07: Special Relativity in Tensor Notation

But before we proceed, let's quickly see how powerful the tensor notation is in coordinate transformations.

§07.1 Polar Coordinates to Cartesian Coordinates

Consider the coordinate transformation from polar coordinates (r, θ) to Cartesian coordinates (x, y) given by:

$$\begin{cases} x = r\cos\theta, \\ y = r\sin\theta. \end{cases}$$

Let's first compute the transformation matrix (or Jacobian) for this transformation:

$$\Lambda = \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{pmatrix} = \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix} \equiv \begin{pmatrix} \Lambda^x{}_r & \Lambda^x{}_\theta \\ \Lambda^y{}_r & \Lambda^y{}_\theta \end{pmatrix}.$$

Say we know the invariant distance in Cartesian coordinates, which is given by:

$$\mathrm{d}s'^2 = \mathrm{d}x^2 + \mathrm{d}y^2\,.$$

Thus, the metric tensor in Cartesian coordinates is:

$$g'_{\mu\nu} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \equiv \begin{pmatrix} g'_{xx} & g'_{xy} \\ g'_{yx} & g'_{yy} \end{pmatrix}.$$

Now, we can transform this metric tensor to polar coordinates using the transformation matrix:

$$g'_{\mu\nu} = (\Lambda^{-1})^\alpha{}_\mu (\Lambda^{-1})^\beta{}_\nu g_{\alpha\beta} \quad \Rightarrow \quad g_{\alpha\beta} = \Lambda^\alpha{}_\mu \Lambda^\beta{}_\nu g'_{\mu\nu}.$$

With this we can compute the metric tensor in polar coordinates:

$$\begin{aligned} g_{rr} &= \Lambda^x{}_r \Lambda^x{}_r g'_{xx} + \Lambda^y{}_r \Lambda^y{}_r g'_{yy} = \cos^2\theta + \sin^2\theta = 1, \\ g_{r\theta} &= \Lambda^x{}_r \Lambda^x{}_\theta g'_{xx} + \Lambda^y{}_r \Lambda^y{}_\theta g'_{yy} = \cos\theta(-r\sin\theta) + \sin\theta(r\cos\theta) = 0, \\ g_{\theta\theta} &= \Lambda^x{}_\theta \Lambda^x{}_\theta g'_{xx} + \Lambda^y{}_\theta \Lambda^y{}_\theta g'_{yy} = (-r\sin\theta)(-r\sin\theta) + (r\cos\theta)(r\cos\theta) = r^2. \end{aligned}$$

Since the metric tensor is symmetric, we have:

$$g_{\alpha\beta} = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix} \equiv \begin{pmatrix} g_{rr} & g_{r\theta} \\ g_{\theta r} & g_{\theta\theta} \end{pmatrix}.$$

Thus, the invariant distance in polar coordinates is given by:

$$\mathrm{d}s^2 = g_{\alpha\beta}\,\mathrm{d}x^\alpha\,\mathrm{d}x^\beta = \mathrm{d}r^2 + r^2\,\mathrm{d}\theta^2\,.$$

§07.2 Special Relativity

In Newtonian mechanics, we have the following physical quantities:

- Position: $\vec{\mathbf{r}}$,
- Velocity: $\vec{\mathbf{v}} = \frac{\mathrm{d}\vec{\mathbf{r}}}{\mathrm{d}t}$,
- Acceleration: $\vec{\mathbf{a}} = \frac{\mathrm{d}\vec{\mathbf{v}}}{\mathrm{d}t}$,
- Momentum: $\vec{\mathbf{p}} = m\vec{\mathbf{v}}$, where m is the 'rest' mass[1] of the particle.

We have seen that these quantities are not invariant under Lorentz transformations. Thus, we need to define notion of *4-vectors* and *proper time* to describe the physics in special relativity.

[1] However, in reality there is nothing known as 'relativistic mass'. It is just a convenient way to express the relativistic effects on momentum.

§07.2.1 Proper Time

We know that time is not invariant under Lorentz transformations. However, we know the speed of light (c), and space-time intervals are invariant. Using these, we can define the *proper time* as

$$\mathrm{d}\tau^2 = -\frac{\mathrm{d}s^2}{c^2}. \tag{07.1}$$

This is the time measured by a clock that is at rest, *i.e.*, a clock that is moving along the world line of the particle. Consider an experiment, we are the lab frame S and the particle is moving with velocity $\vec{\mathbf{v}}$ in the lab frame. At any instant, we attach a co-moving frame $\tilde{S}$ to the particle, which is at rest in the co-moving frame. The proper time is the time measured by a clock in the co-moving frame $\tilde{S}$. Since proper time is invariant, we can write:

$$\mathrm{d}\tau^2 = -\frac{\mathrm{d}s^2}{c^2} = \mathrm{d}t^2 - \frac{1}{c^2}|\vec{\mathbf{v}}|^2\,\mathrm{d}t^2 = \mathrm{d}t^2\left(1 - \frac{v^2}{c^2}\right).$$

Thus, we can write the proper time as:

$$\mathrm{d}\tau = \sqrt{1 - \frac{v^2}{c^2}}\,\mathrm{d}t = \frac{\mathrm{d}t}{\gamma}. \tag{07.2}$$

§07.2.2 4-Vectors

To start with, we define the *4-position* of a particle as:

$$\bar{\mathbf{x}} = \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix} = \begin{pmatrix} ct \\ \vec{\mathbf{r}} \end{pmatrix}.$$

With this, we can define the *4-velocity* as:

$$\bar{\mathbf{v}} = \frac{\mathrm{d}\bar{\mathbf{x}}}{\mathrm{d}\tau} = \frac{\mathrm{d}t}{\mathrm{d}\tau}\frac{\mathrm{d}\bar{\mathbf{x}}}{\mathrm{d}t} = \gamma\begin{pmatrix} c \\ \vec{\mathbf{v}} \end{pmatrix} = \begin{pmatrix} \gamma c \\ \gamma\vec{\mathbf{v}} \end{pmatrix},$$

where $\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$ is the Lorentz factor.

Recall, the invariant distance in Minkowski space-time is given by:

$$\mathrm{d}s^2 = -c^2\,\mathrm{d}t^2 + \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2 \equiv \eta_{\mu\nu}\,\mathrm{d}x^\mu\,\mathrm{d}x^\nu\,.$$

Thus, the metric tensor in Minkowski space-time is given by:

$$\eta_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Let's now compute the norm of the 4-velocity:

$$\|\bar{\mathbf{v}}\|^2 = \eta_{\mu\nu}v^\mu v^\nu = -\gamma^2c^2 + \gamma^2|\vec{\mathbf{v}}|^2 = -\gamma^2c^2\left(1 - \frac{v^2}{c^2}\right) = -c^2.$$

Define the *4-acceleration* as:

$$\bar{\mathbf{a}} = \frac{\mathrm{d}\bar{\mathbf{v}}}{\mathrm{d}\tau} = \frac{\mathrm{d}t}{\mathrm{d}\tau}\frac{\mathrm{d}\bar{\mathbf{v}}}{\mathrm{d}t} = \gamma\left[\frac{\mathrm{d}\gamma}{\mathrm{d}t}\begin{pmatrix} c \\ \vec{\mathbf{v}} \end{pmatrix} + \gamma\begin{pmatrix} 0 \\ \vec{\mathbf{a}} \end{pmatrix}\right]$$

where $\vec{\mathbf{a}} = \frac{\mathrm{d}\vec{\mathbf{v}}}{\mathrm{d}t}$ is the 3-acceleration of the particle. The first term is the time derivative of the Lorentz factor, which can be computed as:

$$\frac{\mathrm{d}\gamma}{\mathrm{d}t} = -\frac{1}{2}\gamma^3\left(-2\frac{\vec{\mathbf{v}}\cdot\vec{\mathbf{a}}}{c^2}\right) = \gamma^3\frac{\vec{\mathbf{v}}\cdot\vec{\mathbf{a}}}{c^2}.$$

So we can write the 4-acceleration as:

$$\bar{\mathbf{a}} = \begin{pmatrix} \gamma^4\frac{\vec{\mathbf{v}}\cdot\vec{\mathbf{a}}}{c} \\ \gamma^2\vec{\mathbf{a}} + \gamma^4\frac{\vec{\mathbf{v}}\cdot\vec{\mathbf{a}}}{c^2}\vec{\mathbf{v}} \end{pmatrix}.$$

Now, recall the square norm of the 4-velocity, and differentiate it with respect to proper time:

$$\frac{\mathrm{d}\bar{\mathbf{v}}\cdot\bar{\mathbf{v}}}{\mathrm{d}\tau} = 2\bar{\mathbf{v}}\cdot\bar{\mathbf{a}} = \frac{\mathrm{d}}{\mathrm{d}\tau}\left(-c^2\right) = 0 \quad \Rightarrow \quad \bar{\mathbf{v}}\cdot\bar{\mathbf{a}} = 0.$$

This means that the 4-acceleration is orthogonal to the 4-velocity (in Minkowski space-time).

Define the *4-momentum* as:

$$\bar{\mathbf{p}} = m\bar{\mathbf{v}} = \begin{pmatrix} \gamma mc \\ \gamma m\vec{\mathbf{v}} \end{pmatrix}$$

where m is the rest mass of the particle. The norm of the 4-momentum is given by:

$$\|\bar{\mathbf{p}}\|^2 = m^2\|\bar{\mathbf{v}}\|^2 = -m^2c^2.$$

Lec 08: 4-Force and First Attempt at General Relativity

Recall the 4-momentum of a particle is defined as:

$$\bar{\mathbf{p}} = m\bar{\mathbf{v}} = \begin{pmatrix} \gamma mc \\ \gamma m\vec{\mathbf{v}} \end{pmatrix},$$

where m is the rest mass of the particle. Define the *4-force* as:

$$\bar{\mathbf{K}} = \frac{\mathrm{d}\bar{\mathbf{p}}}{\mathrm{d}\tau} = \frac{\mathrm{d}t}{\mathrm{d}\tau}\frac{\mathrm{d}\bar{\mathbf{p}}}{\mathrm{d}t} = \gamma m\left[\frac{\mathrm{d}\gamma}{\mathrm{d}t}\begin{pmatrix} c \\ \vec{\mathbf{v}} \end{pmatrix} + \gamma\begin{pmatrix} 0 \\ \vec{\mathbf{a}} \end{pmatrix}\right] = m\bar{\mathbf{a}}. \tag{08.1}$$

This is the relativistic version of Newton's second law. Say we observe the same particle from a different inertial frame S' moving with a constant velocity $\vec{\mathbf{u}}$ with respect to the lab frame S. Let τ' be the proper time in the frame S', and $\Lambda(\vec{\mathbf{u}})$ be the Lorentz transformation from S to S'. Then the 4-force in the frame S' is given by:

$$\bar{\mathbf{K}}' = \frac{\mathrm{d}\bar{\mathbf{p}}'}{\mathrm{d}\tau'} = \frac{\mathrm{d}\tau}{\mathrm{d}\tau'}\frac{\mathrm{d}}{\mathrm{d}\tau}(\Lambda(\vec{\mathbf{u}})\bar{\mathbf{p}}) = 1\cdot\Lambda(\vec{\mathbf{u}})\frac{\mathrm{d}\bar{\mathbf{p}}}{\mathrm{d}\tau} = \Lambda(\vec{\mathbf{u}})\bar{\mathbf{K}}.$$

This shows that the 2nd law of motion is *covariant* under Lorentz transformations.

> Remark 08.1 (Invariance vs Covariance). Notice the non-relativistic version of Newton's second law is invariant under Galilean transformations, *i.e.*, both sides of the equation does not change under a Galilean transformation. In contrast, the relativistic version of Newton's second law is covariant, meaning that the quantities on both sides of the equation transform under Lorentz transformations, but the form of the equation remains the same.

§08.1 Interpretation of 4-Momentum: Energy-Momentum Relation

Similar, to Newtonian mechanics, if 4-force $\bar{\mathbf{K}} = 0$, then the 4-momentum is conserved, *i.e.*,

$$\frac{\mathrm{d}\bar{\mathbf{p}}}{\mathrm{d}\tau} = 0 \quad \Rightarrow \quad \bar{\mathbf{p}} = \text{constant}. \tag{08.2}$$

This means that each component of the 4-momentum is conserved $(\gamma mc, \gamma m\vec{\mathbf{v}})$. We can interpret the spatial components of the 4-momentum as 3-momentum $\vec{\mathbf{p}} = \gamma m\vec{\mathbf{v}}$, but interpretation for the time component γmc is not very clear. Looking at the time component of the 4-momentum, we have

$$p^0 = mv^0 = \gamma mc = mc\left(1 - \frac{v^2}{c^2}\right)^{-1/2}$$

Consider the Newtonian limit, where $v/c \ll 1$, then in the leading order, we have

$$p^0 = mc\left(1 + \frac{1}{2}\frac{v^2}{c^2}\right) \xrightarrow{\text{multiply by } c} p^0 c = mc^2 + \underbrace{\frac{1}{2}mv^2}_{\text{kinetic energy}}$$

Since in the Newtonian limit, the time component of the 4-momentum contains the kinetic energy, we can interpret the time component of the 4-momentum as the total energy of the particle, *i.e.*,

$$E = p^0 c = \gamma mc^2.$$

With 3-momentum $\vec{\mathbf{p}} = \gamma m\vec{\mathbf{v}}$, we can write the 4-momentum as:

$$\bar{\mathbf{p}} = \begin{pmatrix} E/c \\ \vec{\mathbf{p}} \end{pmatrix}. \tag{08.3}$$

Thus, the square norm of the 4-momentum is:

$$-m^2c^2 = \bar{\mathbf{p}} \cdot \bar{\mathbf{p}} = -\frac{E^2}{c^2} + \vec{\mathbf{p}} \cdot \vec{\mathbf{p}} \quad \Rightarrow \quad E^2 = |\vec{\mathbf{p}}|^2c^2 + m^2c^4. \tag{08.4}$$

Assuming the total energy E to be +ve, we can write the energy-momentum relation as:

$$E = \sqrt{|\vec{\mathbf{p}}|^2c^2 + m^2c^4}. \tag{08.5}$$

For massive particles, $m > 0$ at rest, the particle still has a rest energy $E_0 = mc^2$. For massless particles, $m = 0$, the energy-momentum relation reduces to $E = |\vec{\mathbf{p}}|c$.

§08.2 First Attempt at General Relativity

We know the 3-force of gravity (Newton's law of gravitation) is given by:

$$\vec{\mathbf{F}}_G = -\frac{Gm_1m_2}{r^2}\hat{\mathbf{r}}.$$

We also know the general structure of the 4-force, so the only part Einstein needs to figure out is the time component of the 4-force of gravity. As we expect the 4-force of gravity to be of the form:

$$\bar{\mathbf{K}}_G = \begin{pmatrix} K_G^0 \\ (_)\vec{\mathbf{F}}_G \end{pmatrix},$$

where K_G^0 is the time component of the 4-force of gravity, which we don't know yet.

Before going further, we need to understand the term '*inertial mass*'.

§08.2.1 Inertial Mass

From Newton's second law, we know

$$\vec{\mathbf{F}} \propto \vec{\mathbf{a}}$$

where $\vec{\mathbf{F}}$ is applied force, $\vec{\mathbf{a}}$ is acceleration of the object. Experimentally, we know that same force $\vec{\mathbf{F}}$ applied to different objects may result in different accelerations. We define the *inertial mass* m^I of an object as the proportionality constant between the applied force and the acceleration:

$$\vec{\mathbf{F}} = m^I\vec{\mathbf{a}} \quad \Rightarrow \quad m^I = \frac{|\vec{\mathbf{F}}|}{|\vec{\mathbf{a}}|}.$$

§08.2.2 Drawing Parallels with Electrodynamics

Einstein knew that the Electrodynamics is consistent with Special Relativity, so he tried to find a similar structure for the 4-force of gravity. Recall the electrostatic force (Coulomb's law),

$$\vec{\mathbf{F}}_E = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r^2}\hat{\mathbf{r}},$$

where $1/(4\pi\varepsilon_0)$ is the electrostatic coupling constant, and q_1 and q_2 are the electric charges.

Einstein noticed that the Newton's formula for gravitational force is similar to the Coulomb's law, and he thought that why should the "mass" appearing in the gravitational force is same as the "inertial mass" of the object? As the inertial mass is the property of the object that how much it resists the change in its motion, Einstein thought that why this property is resulting in the attractive force between two objects. He introduced the concept of *gravitational mass* m^G of an object, which is the property of the object that how much it attracts other objects. So the gravitational force can be written as:

$$\vec{\mathbf{F}}_G = -\frac{Gm_1^Gm_2^G}{r^2}\hat{\mathbf{r}},$$

where m_1^G and m_2^G are the gravitational masses of the two objects, and G is the gravitational coupling constant.

Object under Static Field

Consider an object with electric charge Q creates an electric field $\vec{\mathbf{E}}$ around it, which is given by:

$$\vec{\mathbf{E}} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\hat{\mathbf{r}}.$$

Now, if we place a test charge q in this electric field, it experiences a force:

$$\vec{\mathbf{F}} = q\vec{\mathbf{E}} = \frac{1}{4\pi\varepsilon_0}\frac{Qq}{r^2}\hat{\mathbf{r}}.$$

The acceleration of the test charge is given by:

$$\vec{\mathbf{a}} = \frac{\vec{\mathbf{F}}}{m^I} = \frac{q}{m^I}\vec{\mathbf{E}}$$

where m^I is the inertial mass of the test charge. Thus, the acceleration of the test charge is proportional to the electric field, but the coupling is controlled by the inertial mass as well as the charge of the test charge.

Similarly, we can think of the gravitational field created by an object with gravitational mass M^G as:

$$\vec{\mathbf{g}} = -\frac{GM^G}{r^2}\hat{\mathbf{r}}.$$

Now, if we place a test object with inertial mass m^I and gravitational mass m^G in this gravitational field, it experiences a force:

$$\vec{\mathbf{F}} = m^I\vec{\mathbf{g}} = -\frac{GM^Gm^I}{r^2}\hat{\mathbf{r}}.$$

The acceleration of the test mass is given by:

$$\vec{\mathbf{a}} = \frac{\vec{\mathbf{F}}}{m^I} = -\frac{m^G}{m^I}\vec{\mathbf{g}}.$$

Now the question arises, is the gravitational mass m^G same as the inertial mass m^I? In 1885, in famous experiment by Eötvös, it was shown that

$$n = \left|\frac{m_1^G/m_1^I - m_2^G/m_2^I}{1/2(m_1^G/m_1^I + m_2^G/m_2^I)}\right| \approx 10^{-9},$$

which is very small, in the recent experiments, it is shown that $n \approx 10^{-13}$. But experimentally, we can never for sure say that $m^G = m^I$. So, Einstein assumed that $m^G = m^I$, which is known as the *weak equivalence principle*.

As electrostatic force is not the complete story, we also have the magnetic force, and combined, we have the Lorentz force. Similarly, Einstein thought that the Newtonian gravitational force is not the complete story, and there must be a magnetic-gravitational force (velocity dependent term), which will be the complete story of gravity. But we know that this is not the case, and we'll see later what goes wrong with this assumption.

Lec 09: Equivalence Principle

A simple experiment to test whether $m^G = m^I$ or not is the following: Consider a body with gravitational mass M^G creates a gravitational field $\vec{\mathbf{g}}$ around it, which is given by:

$$\vec{\mathbf{g}} = -\frac{GM^G}{r^2}\hat{\mathbf{r}}.$$

Now, consider two test objects with inertial masses m_1^I and m_2^I and gravitational masses m_1^G and m_2^G respectively, placed as a satellite in this gravitational field. Thus, the force balance equation for the two test objects can be written as:

$$m_i^I \frac{v_i^2}{r} = \frac{GM^G m_i^G}{r^2} \quad \Rightarrow \quad \omega_i^2 = \left(\frac{m_i^G}{m_i^I}\right)\frac{GM^G}{r^3} \quad (i = 1, 2),$$

where v_i is the tangential velocity of the i-th test object, and $\omega_i = v_i/r$ is the angular velocity of the i-th test object. So, if m_i^G/m_i^I are different for different objects, then they will have different angular velocities, and after some time, their angular positions will be different from initial configuration. But experimentally, it is found that their angular positions remain same (within experimental accuracy) even after long time.

Principle 09.1 (Weak Equivalence Principle)
The gravitational mass m^G and inertial mass m^I are same for all objects, *i.e.*,

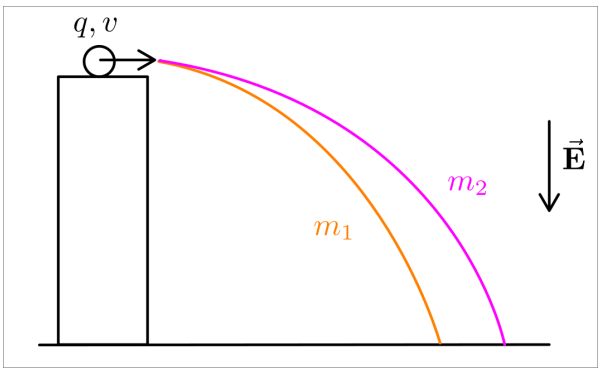
$$m^G = m^I. \tag{09.1}$$

A direct consequence of the weak equivalence principle is that acceleration of an object in a gravitational force is independent of its mass, *i.e.*,

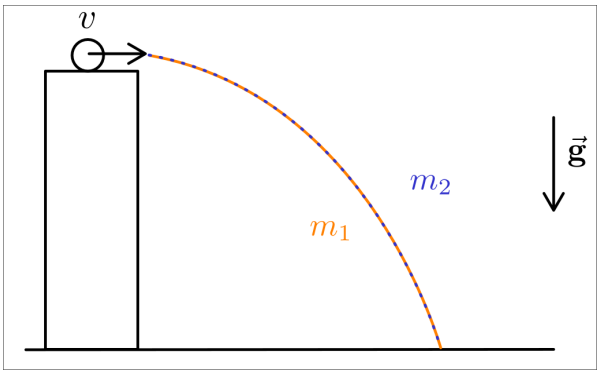
$$m^I\vec{\mathbf{a}} = -G\frac{M^G m^G}{r^2}\hat{\mathbf{r}} \quad \Rightarrow \quad \vec{\mathbf{a}} = -\frac{GM^G}{r^2}\hat{\mathbf{r}}.$$

§09.1 Unique Trajectory under Gravity

First consider two objects with different masses m_1 and m_2 but same electric charge q, placed in an electric field $\vec{\mathbf{E}}$ (see fig. 09.1a). We give them same initial horizontal velocity v, and let them fall under the electric field. We have seen that their accelerations depend on their charge-to-mass ratio q/m, so they will follow different trajectories, and after some time, they will be at different horizontal positions.



(a) Trajectories of objects in an electric field.



(b) Trajectories of objects in a gravitational field.

Figure 09.1: Objects falling under different fields.

Now, consider similar experiment under gravity (see fig. 09.1b). We take two objects with different masses m_1 and m_2. We give them same initial horizontal velocity v, and let them fall under the gravitational

field. From the weak equivalence principle, we know that their accelerations are independent of their masses, so they will follow same trajectory, and after some time, they will be at same horizontal position.

This shows that under gravity, all objects follow the same trajectory irrespective of their masses. This is a unique property of gravity, and it is not true for any other force.

§09.2 Einstein's Thought Experiment

Let's now look at a thought experiment by Einstein which led him to so-called (strong-)*equivalence principle*. Consider a long lift, we attach a reference frame S' to the lift, and an inertial reference frame S outside the lift is observing the lift. For simplicity, we assume all motions are in vertical direction. At time $t = 0$, we release a ball from the ceiling of the lift. Let's analyze the motion of the ball from both reference frames under different scenarios.

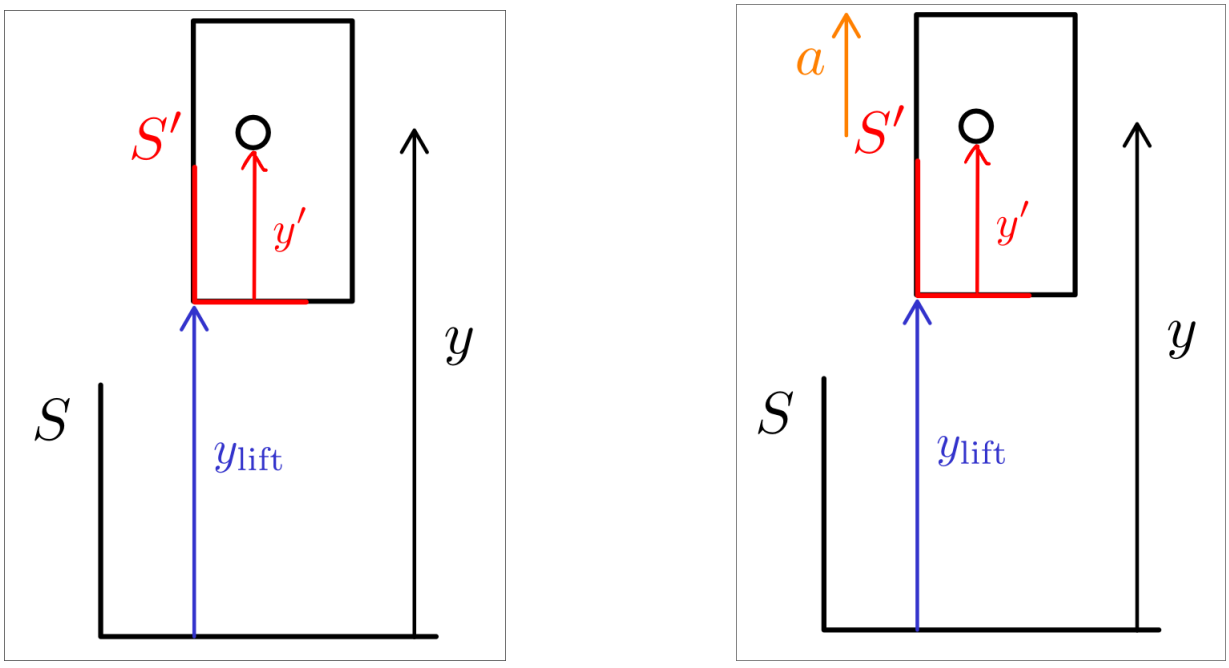
Scenario 1: Lift at Rest, No Gravity.

Say the lift is at rest, and there is no gravity. Since the lift is at rest, both reference frames are inertial (see fig. 09.2a). We have the following relation between position of the ball in both reference frames:

$$y = y' + y_{\text{lift}}.$$

Since the lift is at rest, $\ddot{y}_{\text{lift}} = 0$. Also, as there is no gravity, the ball will remain at rest after being released, *i.e.* $\ddot{y} = 0$. Thus, we have

$$\ddot{y}' = \ddot{y} - \ddot{y}_{\text{lift}} = 0 - 0 = 0.$$



(a) Lift at rest, no gravity.

(b) Lift accelerating upwards, no gravity.

Figure 09.2: Einstein's thought experiment scenarios 1 and 2.

Scenario 2: Lift Accelerating Upwards, No Gravity.

Now, say the lift is accelerating upwards with constant acceleration a, and there is no gravity (see fig. 09.2b). The reference frame S' attached to the lift is no longer inertial, but the reference frame S outside the lift is still inertial. From the reference frame S, the ball will remain at rest after being released, *i.e.* $\ddot{y} = 0$. Since the lift is accelerating upwards with acceleration a, we have $\ddot{y}_{\text{lift}} = a$. Thus, we have

$$\ddot{y}' = \ddot{y} - \ddot{y}_{\text{lift}} = 0 - a = -a.$$

So, from the reference frame S', the ball is accelerating downwards with acceleration a, which is referred as a *pseudo-force*. But in reality, there is no force acting on the ball, it is the result of applying Newton's second law in a non-inertial reference frame.

Now let's switch on a homogeneous gravitational field $\vec{\mathbf{g}} = -g\hat{y}$ (pointing downwards).

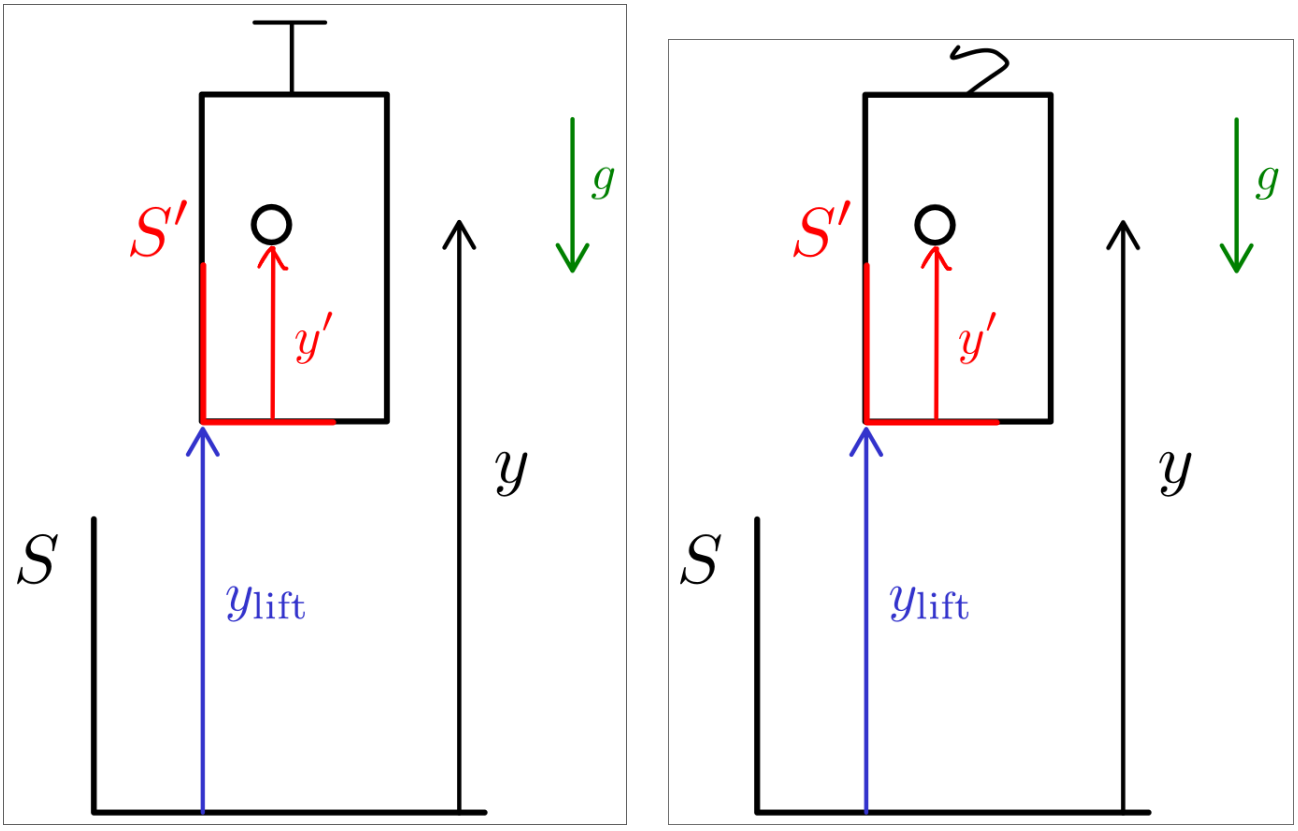
Scenario 3: Lift at Rest (hanging from a rope) in Gravity.

Now, say the lift is at rest (hanging from a rope), and there is a homogeneous gravitational field $\vec{\mathbf{g}} = -g\hat{y}$ (see fig. 09.3a). The reference frame S' attached to the lift is no longer inertial. From the reference frame S, the ball will accelerate downwards with acceleration g after being released, from Newton's second law, we have

$$m^I \ddot{y} = -m^G g \quad \Rightarrow \quad \ddot{y} = -g,$$

where we have used the weak equivalence principle $m^I = m^G$. Since the lift is at rest, we have $\ddot{y}_{\text{lift}} = 0$. Thus, we have

$$\ddot{y}' = \ddot{y} - \ddot{y}_{\text{lift}} = -g - 0 = -g.$$



(a) Lift at rest (hanging from a rope) in gravity. **(b)** Free-falling lift in gravity.

Figure 09.3: Einstein's thought experiment scenarios 3 and 4.

Scenario 4: Lift in Free Fall in Gravity.

Now, say the lift is in free fall, and there is a homogeneous gravitational field $\vec{\mathbf{g}} = -g\hat{y}$ (see fig. 09.3b). The reference frame S' attached to the lift is no longer inertial. From the reference frame S, the ball will accelerate downwards with acceleration g after being released, from Newton's second law, we have

$$m^I \ddot{y} = -m^G g \quad \Rightarrow \quad \ddot{y} = -g,$$

where we have used the weak equivalence principle $m^I = m^G$. Since the lift is in free fall, we have $m^I_{\text{lift}} \ddot{y}_{\text{lift}} = -m^G g$, which gives us $\ddot{y}_{\text{lift}} = -g$. Thus, we have

$$\ddot{y}' = \ddot{y} - \ddot{y}_{\text{lift}} = -g - (-g) = 0.$$

So, from the reference frame S', the ball remains at rest after being released.

Let's discuss the implications of these scenarios.

- Look at the scenarios 2 and 3, in both cases, the lift frame S' see the ball accelerating downwards, thus these scenarios are indistinguishable from each other.
- In scenario 1 and 4, the lift frame S' see the ball at rest, thus these scenarios are also indistinguishable from each other, *i.e.*, a free-falling observer is same as an observer at rest.

With this thought experiment, we have argued that there exists at least one reference frame (the free-falling frame) with respect to which the equations of motion take the same form as in absence of gravity. In other words, there exists a reference frame in which the laws of physics are same as if there is no gravity. This is the essence of the *strong equivalence principle*.

Another important implication of this thought experiment is that we have found a frame in which the force of gravity vanishes, but in other frames the force of gravity is present[1]. This suggests that gravity is not a force (vector field) like other forces.

§09.2.1 A deeper look at the thought experiment

In the scenarios 3 and 4, we have assumed that the gravitational field is homogeneous, which means that the gravitational acceleration g is same everywhere in the lift. But it will not be true in general, consider the gravitational field of the Earth, which is not homogeneous. Consider we have two balls attached to the ceiling of the lift with a separation d (large enough to see the curvature of the Earth) between them. The lift is free-falling in the gravitational field of the Earth, and at time $t = 0$, we release the two balls. As the gravitational field is directed towards the center of the Earth, the separation between the two balls will decrease as they fall. But if we consider the same experiment in scenario 1, the separation between the two balls will remain same.

So in case of a non-homogeneous gravitational field, the free-falling frame can actually figure out that it is in a gravitational field, and it is not same as an inertial frame. This effect is known as the *tidal force*, which is a result of the non-homogeneity of the gravitational field.

To overcome this issue, we must consider lift to be very small, such that the gravitational field can be considered homogeneous inside the lift. Thus, the strong equivalence principle can be stated as follows:

Principle 09.2 (Strong Equivalence Principle)
In a sufficiently small region of spacetime ("local" region), there exists a reference frame in which the laws of physics are same as if there is no gravity.

This special reference frame is known as the *local inertial frame*.

[1] We know if all component of a tensor is zero in some frame, then it is zero in all frames.

Lec 10: Affine Connection and Geodesics

In the previous lecture, we have discussed that there exists an observer such that it doesn't experience any gravitational effects locally. This is the essence of the *Equivalence Principle*. We have also seen that weak equivalence principle (WEP) leads to the idea of a unique path followed by a free-falling particle in a gravitational field.

§10.1 Equivalence Principle and Local Inertial Frame

The *Einstein's Equivalence Principle* states that:

Principle 10.1 (Einstein's Equivalence Principle)
All freely falling frames under the influence of gravity are locally equivalent to inertial frames (*i.e.* Lorentz frames where special relativity is valid and gravitational effects are absent).

Recall the scenario 4, where we had a free-falling lift in a uniform gravitational field. For an observer outside the lift (inertial frame S), the instantaneous position of the released ball is given by:

$$\frac{\mathrm{d}^2\vec{\mathbf{r}}}{\mathrm{d}t^2} = -\vec{\mathbf{g}} \quad \Rightarrow \quad \vec{\mathbf{r}}(t) = \vec{\mathbf{r}}_0 - \frac{1}{2}\vec{\mathbf{g}}t^2.$$

Since the motion is only in the vertical direction, we can write:

$$y(t) = y_0 - \frac{1}{2}gt^2.$$

Now, for an observer inside the lift (free-falling frame S'), the position of the ball is given by:

$$y'(t) = y'_0,$$

where y'_0 is the initial position of the ball in the lift frame. Say initially, the lift is at height $y_0^{(\text{lift})}$ in the inertial frame S. Then, the relation between the two frames is given by:

$$y_0 = y'_0 + y_0^{(\text{lift})}.$$

The position of the ball in the inertial frame S can be written as:

$$y(t) = \underbrace{y'_0}_{=y'(t)} + y_0^{(\text{lift})} - \frac{1}{2}gt^2 \quad \Rightarrow \quad y'(t) = y(t) - y_0^{(\text{lift})} + \frac{1}{2}gt^2.$$

So we see that physics in two frames can be related by a coordinate transformation. In this case the transformation is given by:

$$y' = y - y_0^{(\text{lift})} + \frac{1}{2}gt^2, \quad t' = t.$$

This is a non-linear transformation. The transformation matrix is given by:

$$\Lambda = \begin{pmatrix} 1 & 0 \\ gt & 1 \end{pmatrix}.$$

Till now, we have only seen transformation matrices which are coordinate independent (like Lorentz transformations). But here, the transformation matrix depends on time t. This is a general feature of transformations between inertial and non-inertial frames.

In relativistic mechanics (in presence of gravity), the equations of motion for a free-falling particle in a gravitational field observed in a local inertial frame S' are given by:

$$\frac{\mathrm{d}^2x'^{\mu}}{\mathrm{d}\tau^2} = 0,$$

where τ is the proper time. Now we want to find the equations of motion in a general frame S. Let $\Lambda^{\mu}{}_{\nu}$ be the transformation matrix from S to S'. Then, we have:

$$\mathrm{d}x'^{\mu} = \Lambda^{\mu}{}_{\nu}\,\mathrm{d}x^{\nu}\,.$$

Recall that $u^{\mu} = \mathrm{d}x^{\mu}/\mathrm{d}\tau$ is the 4-velocity, and transformation for 4-velocity is given by:

$$u'^{\mu} = \Lambda^{\mu}{}_{\nu}u^{\nu}.$$

Differentiating w.r.t. τ, we get:

$$\frac{\mathrm{d}u'^{\mu}}{\mathrm{d}\tau} = 0 \quad \Rightarrow \quad \frac{\mathrm{d}\Lambda^{\mu}{}_{\nu}}{\mathrm{d}\tau}u^{\nu} + \Lambda^{\mu}{}_{\nu}\frac{\mathrm{d}u^{\nu}}{\mathrm{d}\tau} = \partial_{\rho}(\Lambda^{\mu}{}_{\nu})u^{\rho}u^{\nu} + \Lambda^{\mu}{}_{\nu}\frac{\mathrm{d}^2x^{\nu}}{\mathrm{d}\tau^2} = 0.$$

Multiplying by $(\Lambda^{-1})^{\sigma}{}_{\mu}$, we get:

$$\frac{\mathrm{d}^2x^{\sigma}}{\mathrm{d}\tau^2} + (\Lambda^{-1})^{\sigma}{}_{\mu}\partial_{\rho}(\Lambda^{\mu}{}_{\nu})\frac{\mathrm{d}x^{\rho}}{\mathrm{d}\tau}\frac{\mathrm{d}x^{\nu}}{\mathrm{d}\tau} = 0.$$

We define the *affine connection* as:

$$\Gamma^{\alpha}_{\mu\nu} = (\Lambda^{-1})^{\alpha}{}_{\rho}\partial_{\nu}(\Lambda^{\rho}{}_{\mu}) \equiv \frac{\partial x^{\alpha}}{\partial x'^{\rho}}\partial_{\nu}\left(\frac{\partial x'^{\rho}}{\partial x^{\mu}}\right) = \frac{\partial x^{\alpha}}{\partial x'^{\rho}}\frac{\partial^2 x'^{\rho}}{\partial x^{\nu}\partial x^{\mu}}. \tag{10.1}$$

Assignment 03.1 (Affine Connection Symmetry)**.**
Show that $\Gamma^{\alpha}_{\mu\nu} = \Gamma^{\alpha}_{\nu\mu}$, *i.e.*, the affine connection is symmetric in the lower indices.

The equations of motion in a general frame S are given by:

$$\frac{\mathrm{d}^2x^{\mu}}{\mathrm{d}\tau^2} + \Gamma^{\mu}_{\alpha\beta}\frac{\mathrm{d}x^{\alpha}}{\mathrm{d}\tau}\frac{\mathrm{d}x^{\beta}}{\mathrm{d}\tau} = 0. \tag{10.2}$$

This is known as the *geodesic equation*, it describes the motion of a free-falling particle in a gravitational field.

§10.2 Affine Connection

The affine connection $\Gamma^{\alpha}_{\mu\nu}$ in its current form depends on both the coordinate of S and S', so it is impossible to determine since we don't know S'. In the local inertial frame (LIF) S', the metric is given by the Minkowski metric $\eta_{\mu\nu}$. We can write the metric in the general frame S as:

$$g_{\mu\nu} = \Lambda^{\alpha}{}_{\mu}\Lambda^{\beta}{}_{\nu}\eta_{\alpha\beta}.$$

Differentiating w.r.t. x^{ρ}, we get:

$$\partial_{\rho}(g_{\mu\nu}) = \partial_{\rho}(\Lambda^{\alpha}{}_{\mu})\Lambda^{\beta}{}_{\nu}\eta_{\alpha\beta} + \Lambda^{\alpha}{}_{\mu}\partial_{\rho}(\Lambda^{\beta}{}_{\nu})\eta_{\alpha\beta}.$$

Recall that $\Lambda^{\gamma}{}_{\alpha}\Gamma^{\alpha}_{\mu\nu} = \partial_{\nu}(\Lambda^{\gamma}{}_{\mu})$. Using this, we can write:

$$\partial_{\rho}(g_{\mu\nu}) = \Gamma^{\sigma}_{\mu\rho}\underbrace{\Lambda^{\alpha}{}_{\sigma}\Lambda^{\beta}{}_{\nu}\eta_{\alpha\beta}}_{=g_{\sigma\nu}} + \Gamma^{\sigma}_{\nu\rho}\underbrace{\Lambda^{\alpha}{}_{\mu}\Lambda^{\beta}{}_{\sigma}\eta_{\alpha\beta}}_{=g_{\mu\sigma}}.$$

So we have:

$$\partial_{\rho}(g_{\mu\nu}) = \Gamma^{\sigma}_{\mu\rho}g_{\sigma\nu} + \Gamma^{\sigma}_{\nu\rho}g_{\mu\sigma}.$$

By cyclically permuting the indices, we get:

$$\partial_{\rho}(g_{\mu\nu}) = \Gamma^{\sigma}_{\mu\rho}g_{\sigma\nu} + \Gamma^{\sigma}_{\nu\rho}g_{\mu\sigma},$$
$$\partial_{\mu}(g_{\nu\rho}) = \Gamma^{\sigma}_{\nu\mu}g_{\sigma\rho} + \Gamma^{\sigma}_{\rho\mu}g_{\nu\sigma},$$

$$\partial_\nu(g_{\rho\mu}) = \Gamma^\sigma_{\rho\nu} g_{\sigma\mu} + \Gamma^\sigma_{\mu\nu} g_{\rho\sigma}.$$

Adding the last two equations and subtracting the first, we get:

$$\partial_\mu(g_{\nu\rho}) + \partial_\nu(g_{\rho\mu}) - \partial_\rho(g_{\mu\nu}) = 2\Gamma^\sigma_{\mu\nu} g_{\rho\sigma}.$$

Multiplying by $g^{\alpha\rho}$, we get:

$$\Gamma^\alpha_{\mu\nu} = \frac{1}{2} g^{\alpha\rho}\Big[\partial_\mu(g_{\nu\rho}) + \partial_\nu(g_{\rho\mu}) - \partial_\rho(g_{\mu\nu})\Big]. \tag{10.3}$$

This is known as the *Christoffel symbol*. We see that the Christoffel symbol depends only on the metric and its derivatives. So if we know the metric in a general frame S, we can determine the Christoffel symbols and hence the equations of motion of a free-falling particle in a gravitational field.

The Christoffel symbols are not tensors. To see this, consider a transformation from frame S to S' given by $\frac{\partial x'^\mu}{\partial x^\nu}$. The Christoffel symbols in the two frames are related by:

$$\Gamma'^\alpha_{\mu\nu} = \frac{\partial x'^\alpha}{\partial x^\lambda}\frac{\partial x^\rho}{\partial x'^\mu}\frac{\partial x^\sigma}{\partial x'^\nu}\Gamma^\lambda_{\rho\sigma} + \frac{\partial x'^\alpha}{\partial x^\lambda}\frac{\partial^2 x^\lambda}{\partial x'^\mu \partial x'^\nu}. \tag{10.4}$$

The second term is not tensorial, hence the Christoffel symbols are not tensors.

We can derive the geodesic equation from a variational principle as well.

§10.3 Geodesic Equation from Action Principle

From classic mechanics, we know that a symmetry of the action leads to a symmetry of the equations of motion. For example, invariance of the action under 3-rotations leads to conservation of angular momentum. Similarly, we want an action which is invariant under general coordinate transformations.

We have to construct a scalar from the coordinates, their derivatives, the metric, and speed of light c. In special relativity, we have

$$\eta_{\mu\nu}\frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}\frac{\mathrm{d}x^\nu}{\mathrm{d}\tau} = -c^2.$$

In presence of gravity, we replace the Minkowski metric by the general metric $g_{\mu\nu}$. So we have:

$$g_{\mu\nu}\frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}\frac{\mathrm{d}x^\nu}{\mathrm{d}\tau} = -c^2.$$

This is a scalar. We want action to have the dimension $[S] = [\hbar] = [\mathrm{ML^2T^{-1}}]$. The proper time τ has the dimension of time [T]. Consider a massive particle of mass m. The quantity mc^2 has the dimension of energy $[\mathrm{ML^2T^{-2}}]$. So the action can be written as:

$$S = \int \mathrm{d}\tau\,(-mc^2) = m\int \mathrm{d}\tau\, g_{\mu\nu}\frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}\frac{\mathrm{d}x^\nu}{\mathrm{d}\tau}. \tag{10.5}$$

This is invariant under general coordinate transformations. We can also see that in the Newtonian limit, this action reduces to the action of a non-relativistic particle in a potential, recall that for Minkowski metric, we have:

$$\mathrm{d}\tau = \frac{\mathrm{d}t}{\gamma} = \mathrm{d}t\sqrt{1-\frac{v^2}{c^2}} \approx \mathrm{d}t\left(1-\frac{v^2}{2c^2}\right),$$

where $v \ll c$. So we have:

$$S = -\int \mathrm{d}t\, mc^2\left(1-\frac{v^2}{2c^2}\right) = \int \mathrm{d}t\left(\frac{1}{2}mv^2 - mc^2\right) \equiv \int \mathrm{d}t\,(T-V),$$

where $T = \frac{1}{2}mv^2$ is the kinetic energy and $V = mc^2$ is the potential energy (constant).

The Lagrangian is given by:

$$L = L\left(x^\mu, u^\mu = \frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}\right) = m g_{\mu\nu} \frac{\mathrm{d}x^\mu}{\mathrm{d}\tau} \frac{\mathrm{d}x^\nu}{\mathrm{d}\tau} = m g_{\mu\nu} u^\mu u^\nu.$$

The Euler-Lagrange equations are given by:

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\left(\frac{\partial L}{\partial u^\alpha}\right) - \frac{\partial L}{\partial x^\alpha} = 0. \tag{10.6}$$

Calculating the derivatives, we get:

$$m\left[\frac{\mathrm{d}}{\mathrm{d}\tau}\left(g_{\alpha\nu}u^\nu + g_{\mu\alpha}u^\mu\right) - \partial_\alpha(g_{\mu\nu})u^\mu u^\nu\right] = 0.$$

Using chain rule, we can write:

$$\partial_\mu(g_{\alpha\nu})u^\mu u^\nu + \partial_\nu(g_{\mu\alpha})u^\mu u^\nu + g_{\alpha\nu}\frac{\mathrm{d}u^\nu}{\mathrm{d}\tau} + g_{\mu\alpha}\frac{\mathrm{d}u^\mu}{\mathrm{d}\tau} - \partial_\alpha(g_{\mu\nu})u^\mu u^\nu = 0.$$

Combining the terms, we get:

$$g_{\alpha\mu}\frac{\mathrm{d}^2x^\mu}{\mathrm{d}\tau^2} + \frac{1}{2}\left[\partial_\mu(g_{\alpha\nu}) + \partial_\nu(g_{\mu\alpha}) - \partial_\alpha(g_{\mu\nu})\right]\frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}\frac{\mathrm{d}x^\nu}{\mathrm{d}\tau} = 0.$$

Multiplying by $g^{\sigma\alpha}$, we get:

$$\frac{\mathrm{d}^2x^\sigma}{\mathrm{d}\tau^2} + \Gamma^\sigma_{\mu\nu}\frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}\frac{\mathrm{d}x^\nu}{\mathrm{d}\tau} = 0.$$

This is the geodesic equation, which we had derived earlier using the equivalence principle.

In this derivation, we have assumed that the particle is massive. For a massless particle, we cannot use the proper time τ as a parameter. We can use an affine parameter λ instead. The action can be written as:

$$S = \int \mathrm{d}\lambda\, g_{\mu\nu}\frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda}.$$

The equations of motion can be derived similarly, and we get the same geodesic equation:

$$\frac{\mathrm{d}^2x^\sigma}{\mathrm{d}\lambda^2} + \Gamma^\sigma_{\mu\nu}\frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda} = 0.$$

Lec 11: Geodesics on Curved Surfaces

In the last lecture, we introduced the concept of geodesics as the path that extremizes the proper time between two events in spacetime. We derived the geodesic equation using the coordinate-transformation properties of the metric tensor and the principle of extremal action. Now, let's use the geodesic equation to find the extremal distance paths between two points in different geometries.

§11.1 Geodesics on a 2-Sphere S^2

Consider a 2-sphere of radius R embedded in $\mathbb{R}^3$. The metric on the sphere can be expressed in spherical coordinates (θ, ϕ) as:

$$\mathrm{d}s^2 = R^2\Big(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2\Big). \tag{11.1}$$

The metric tensor is given by:

$$g_{\mu\nu} = \begin{pmatrix} R^2 & 0 \\ 0 & R^2\sin^2\theta \end{pmatrix} \equiv \begin{pmatrix} g_{\theta\theta} & g_{\theta\phi} \\ g_{\phi\theta} & g_{\phi\phi} \end{pmatrix}. \tag{11.2}$$

To find the geodesics on the sphere, we need to compute the Christoffel symbols. For 2-dimensional manifolds, we have total of $2^3 = 8$ Christoffel symbols, but due to the symmetry (assignment 03.1) we only need to compute $n \times \frac{n(n+1)}{2} = 6$ of them. For a diagonal metric, we have the following simplifications:

$$\begin{aligned}
&\Gamma^\alpha_{\mu\nu} = 0 && \text{if } \alpha \neq \mu \neq \nu, \\
&\Gamma^\alpha_{\mu\mu} = \tfrac{1}{2}g^{\alpha\alpha}\partial_\mu(g_{\mu\mu}) && \text{if } \alpha \neq \mu, \text{ no sum over } \mu, \\
&\Gamma^\mu_{\mu\nu} = \tfrac{1}{2}g^{\mu\mu}\partial_\nu(g_{\mu\mu}) && \text{if } \mu \neq \nu, \text{ no sum over } \mu, \\
&\Gamma^\mu_{\mu\mu} = \tfrac{1}{2}g^{\mu\mu}\partial_\mu(g_{\mu\mu}) && \text{no sum over } \mu.
\end{aligned} \tag{11.3}$$

Also note that metric components are only functions of θ and not ϕ. Using these simplifications, we can compute the Christoffel symbols for the 2-sphere:

$$\Gamma^\theta_{\theta\theta} = 0, \qquad \Gamma^\phi_{\phi\phi} = 0, \tag{11.4}$$

$$\Gamma^\theta_{\theta\phi} = 0, \qquad \Gamma^\phi_{\phi\theta} = \cot\theta, \tag{11.5}$$

$$\Gamma^\theta_{\phi\phi} = -\sin\theta\cos\theta, \qquad \Gamma^\phi_{\theta\theta} = 0. \tag{11.6}$$

This gives us total of 3 non-zero Christoffel symbols. Now, we can write down the geodesic equations for θ and ϕ:

$$\frac{\mathrm{d}^2\theta}{\mathrm{d}\tau^2} - \sin\theta\cos\theta\left(\frac{\mathrm{d}\phi}{\mathrm{d}\tau}\right)^2 = 0, \qquad \frac{\mathrm{d}^2\phi}{\mathrm{d}\tau^2} + 2\cot\theta\frac{\mathrm{d}\theta}{\mathrm{d}\tau}\frac{\mathrm{d}\phi}{\mathrm{d}\tau} = 0.$$

These are coupled second-order differential equations. Before solving them, we need to look at the equation of great circles on a sphere.

Consider the center of the 2-sphere at the origin of $\mathbb{R}^3$, thus equation of a plain passing through the origin is given by:

$$ax + by + cz = 0. \tag{11.7}$$

The coordinates (x, y, z) of any point on the sphere can be expressed in terms of spherical coordinates as:

$$\begin{cases} x = R\sin\theta\cos\phi, \\ y = R\sin\theta\sin\phi, \\ z = R\cos\theta. \end{cases} \tag{11.8}$$

To find the intersection of the plane with the sphere, we substitute these coordinates into the plane equation:

$$aR\sin\theta\cos\phi + bR\sin\theta\sin\phi + cR\cos\theta = 0.$$

Dividing through by R and rearranging, we get:

$$\cot\theta = c_1\cos\phi + c_2\sin\phi, \tag{11.9}$$

where $c_1 = -\frac{a}{c}$ and $c_2 = -\frac{b}{c}$ are constants. This is the equation of a great circle on the sphere.

Now, let's integrate the geodesic equations. Starting with the ϕ equation, multiply both sides by $\sin^2\theta$:

$$\sin^2\theta\frac{\mathrm{d}^2\phi}{\mathrm{d}\tau^2} + 2\sin\theta\cos\theta\frac{\mathrm{d}\theta}{\mathrm{d}\tau}\frac{\mathrm{d}\phi}{\mathrm{d}\tau} = 0.$$

This can be rewritten as:

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\left(\sin^2\theta\frac{\mathrm{d}\phi}{\mathrm{d}\tau}\right) = 0 \quad \Rightarrow \quad \sin^2\theta\frac{\mathrm{d}\phi}{\mathrm{d}\tau} = l,$$

where l is a constant of integration. This gives us:

$$\frac{\mathrm{d}\phi}{\mathrm{d}\tau} = \frac{l}{\sin^2\theta}. \tag{11.10}$$

Next, we substitute this into the θ equation:

$$\frac{\mathrm{d}^2\theta}{\mathrm{d}\tau^2} - \sin\theta\cos\theta\left(\frac{l^2}{\sin^4\theta}\right) = 0,$$

this equation gives us $\theta(\tau)$, but to find the shape of the geodesic, we need to eliminate τ and find $\theta(\phi)$. Using the chain rule, we have:

$$\frac{\mathrm{d}}{\mathrm{d}\tau} = \frac{\mathrm{d}\phi}{\mathrm{d}\tau}\frac{\mathrm{d}}{\mathrm{d}\phi} = \frac{l}{\sin^2\theta}\frac{\mathrm{d}}{\mathrm{d}\phi} \qquad \Rightarrow \qquad \frac{l}{\sin^2\theta}\frac{\mathrm{d}}{\mathrm{d}\phi}\left(\frac{l}{\sin^2\theta}\frac{\mathrm{d}\theta}{\mathrm{d}\phi}\right) - \sin\theta\cos\theta\left(\frac{l^2}{\sin^4\theta}\right) = 0.$$

Simplifying, we get:

$$\frac{\mathrm{d}}{\mathrm{d}\phi}\left(\frac{1}{\sin^2\theta}\frac{\mathrm{d}\theta}{\mathrm{d}\phi}\right) - \cot\theta = 0.$$

Recall that $\frac{\mathrm{d}}{\mathrm{d}\phi}(\cot\theta) = -\csc^2\theta\frac{\mathrm{d}\theta}{\mathrm{d}\phi}$, thus we can rewrite the above equation as:

$$-\frac{\mathrm{d}}{\mathrm{d}\phi}\left(\frac{\mathrm{d}}{\mathrm{d}\phi}(\cot\theta)\right) - \cot\theta = 0 \quad \Rightarrow \quad \frac{\mathrm{d}^2(\cot\theta)}{\mathrm{d}\phi^2} + \cot\theta = 0.$$

This is a simple harmonic oscillator equation with general solution:

$$\cot\theta = c_1\cos\phi + c_2\sin\phi, \tag{11.11}$$

which is exactly the equation of a great circle on the sphere. Thus, we have shown that the geodesics on a 2-sphere are great circles.

§11.2 Geodesics on a Cylinder $S^1 \times \mathbb{R}$

Next, let's consider a cylinder, which can be represented as $S^1 \times \mathbb{R}$. Let the radius of the cylinder be a. The metric on the cylinder in coordinates (φ, z) is given by:

$$\mathrm{d}s^2 = a^2\,\mathrm{d}\varphi^2 + \mathrm{d}z^2\,. \tag{11.12}$$

The metric tensor is:

$$g_{\mu\nu} = \begin{pmatrix} a^2 & 0 \\ 0 & 1 \end{pmatrix} \equiv \begin{pmatrix} g_{\varphi\varphi} & g_{\varphi z} \\ g_{z\varphi} & g_{zz} \end{pmatrix}. \tag{11.13}$$

Notice that the metric components are constants, thus all Christoffel symbols vanish:

$$\Gamma^{\alpha}_{\mu\nu} = 0 \quad \forall \alpha, \mu, \nu. \tag{11.14}$$

The geodesic equations simplify to:

$$\frac{\mathrm{d}^2\varphi}{\mathrm{d}\tau^2} = 0, \qquad \frac{\mathrm{d}^2 z}{\mathrm{d}\tau^2} = 0. \tag{11.15}$$

The general solutions are:

$$\varphi(\tau) = c_1\tau + c_2, \qquad z(\tau) = c_3\tau + c_4,$$

where c_1, c_2, c_3, c_4 are constants. Since we can be always affine transform τ, we can set $c_2 = c_4 = 0$ without loss of generality. Thus, the geodesics on a cylinder are given by:

$$\varphi(\tau) = c_1\tau, \qquad z(\tau) = c_3\tau. \tag{11.16}$$

Eliminating τ, we get:

$$z = m\varphi, \quad \text{where } m = \frac{c_3}{c_1}. \tag{11.17}$$

This is the equation of a helix on the cylinder with pitch $2\pi m$. Thus, the geodesics on a cylinder are helices.

Remark 11.1 (Geodesics in Flat Spacetime). In flat spacetime with the Minkowski metric $\eta_{\mu\nu} = \mathrm{diag}(-1, 1, 1, 1)$, all Christoffel symbols vanish. Thus, the geodesic equations reduce to:

$$\frac{\mathrm{d}^2 x^{\mu}}{\mathrm{d}\tau^2} = 0.$$

The general solution is:

$$x^{\mu}(\tau) = u^{\mu}\tau + x^{\mu}_0,$$

where u^{μ} is a constant four-velocity vector and x^{μ}_0 is the initial position. This describes straight-line motion in spacetime, consistent with the principle of inertia in special relativity.

Remark 11.2 (Geodesics in Flat Space). In flat space with the Euclidean metric $g_{ij} = \delta_{ij}$, all Christoffel symbols vanish. Thus, the geodesic equations reduce to:

$$\frac{\mathrm{d}^2 x^{i}}{\mathrm{d}\tau^2} = 0.$$

The general solution is:

$$x^{i}(\tau) = v^{i}\tau + x^{i}_0,$$

where v^{i} is a constant velocity vector and x^{i}_0 is the initial position. This describes straight-line motion in space, consistent with the principle of inertia in classical mechanics.

Lec 12: Invariant Measure under General Coordinate Transformations

In the last few lectures, we have discussed the concept of geodesics, which are the path taken by massive particles under the influence of gravity alone. We also discussed the geodesic equation, describes the path of the shortest distance between two points in a curved spacetime. In order to derive the geodesic equation, we have used following action:

$$S = m \int \mathrm{d}\tau \, g_{\mu\nu} \frac{\mathrm{d}x^\mu}{\mathrm{d}\tau} \frac{\mathrm{d}x^\nu}{\mathrm{d}\tau}. \tag{12.1}$$

Now, we want to find the action for the metric tensor $g_{\mu\nu}$ itself. But before that, let's recall a well known Field Theory which follows Special Relativity from an action principle, the Electromagnetic Field Theory. The action for the electromagnetic field is given by:

$$S_{\mathrm{EM}} = \int \mathrm{d}^4x \left[-\frac{1}{4\mu_\circ} F_{\mu\nu} F^{\mu\nu} \right], \tag{12.2}$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is the electromagnetic field tensor, and A_μ is the four-potential. Let's note some important points about this action:

- It is a functional of the four-potential $A_\mu(x)$, *i.e.*, $S_{\mathrm{EM}} = S_{\mathrm{EM}}[A_\mu(x)]$.
- The action is a scalar under Lorentz transformations.
- The action contains at most two derivatives of the field A_μ.

Einstein was very much inspired from the Maxwell's theory of electromagnetism, so he thought that the action for the metric tensor should be similar to the action for the electromagnetic field. So, he extended the scalar property of the action to general coordinate transformations, and proposed the action for the metric tensor should be of the form:

$$S_{\mathrm{G}} \approx \int \mathrm{d}^4x \, f(g_{\mu\nu}, \partial_\rho g_{\mu\nu}, \partial_\rho \partial_\sigma g_{\mu\nu}), \tag{12.3}$$

where f is some scalar function of the metric tensor and its derivatives. We want this action to be invariant under general coordinate transformations. But the measure d^4x is not invariant under general coordinate transformations. An easy way to see this is to consider the Cartesian to plain polar transformation in flat space, $\{x, y\} \to \{r, \theta\}$. The elemental area transforms as:

$$\mathrm{d}x \, \mathrm{d}y \to r \, \mathrm{d}r \, \mathrm{d}\theta \equiv J(r, \theta) \, \mathrm{d}r \, \mathrm{d}\theta \,,$$

where $J(r, \theta) = r$ is the Jacobian of the transformation.

So, the measure d^4x is not invariant under general coordinate transformations. We need to find an invariant measure under general coordinate transformations. But before that, let's recall the definition of determinant of a matrix.

Determinant of a Matrix

Consider a square matrix A of size $n \times n$. The determinant of the matrix A is defined as:

$$\det(A) = \det(a_{ij}) = \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) \prod_{i=1}^{n} a_{i\sigma(i)}, \tag{12.4}$$

where S_n is the set of all permutations of n elements, and $\mathrm{sgn}(\sigma)$ is the sign of the permutation σ, which is $+1$ for even permutations and -1 for odd permutations.

We can rewrite the determinant using the Levi-Civita symbol, which is defined as:

$$\varepsilon_{i_1 i_2 \cdots i_n} = \begin{cases} +1 & \text{if } (i_1, i_2, \ldots, i_n) \text{ is an even permutation of } (1, 2, \ldots, n), \\ -1 & \text{if } (i_1, i_2, \ldots, i_n) \text{ is an odd permutation of } (1, 2, \ldots, n), \\ 0 & \text{if any two indices are equal.} \end{cases} \tag{12.5}$$

Using the Levi-Civita symbol, we can express the determinant as:

$$\det(A) = \varepsilon^{j_1 j_2 \cdots j_n} a_{1 j_1} a_{2 j_2} \cdots a_{n j_n}. \tag{12.6}$$

Using this definition, we can also write another useful identity:

$$\det(A) = \frac{1}{n!} \varepsilon^{i_1 i_2 \cdots i_n} \varepsilon^{j_1 j_2 \cdots j_n} a_{i_1 j_1} a_{i_2 j_2} \cdots a_{i_n j_n}. \tag{12.7}$$

§12.1 Invariant Measure

Consider an n-dimensional Riemannian manifold with coordinates x^i and metric tensor g_{ij}. We assume the invariant measure to be:

$$\mathrm{d}V = J(x) \frac{1}{n!} \varepsilon_{i_1 i_2 \cdots i_n} \Big[\mathrm{sgn}(\sigma)\, \mathrm{d}x^{i_1}\, \mathrm{d}x^{i_2} \cdots \mathrm{d}x^{i_n} \Big], \tag{12.8}$$

where $J(x)$ is some scalar function of the coordinates, $\varepsilon_{i_1 i_2 \cdots i_n}$ is the Levi-Civita symbol in n-dimensions, and σ is the permutation such that $\sigma(k) = i_k$.[1] We want this ansatz to be invariant under general coordinate transformations. So we have:

$$\mathrm{d}V = \mathrm{d}V'$$

$$J(x) \frac{1}{n!} \varepsilon_{i_1 i_2 \cdots i_n} \Big[\mathrm{sgn}(\sigma)\, \mathrm{d}x^{i_1}\, \mathrm{d}x^{i_2} \cdots \mathrm{d}x^{i_n} \Big] = J'(x') \frac{1}{n!} \varepsilon_{j_1 j_2 \cdots j_n} \Big[\mathrm{sgn}(\sigma)\, \mathrm{d}x'^{j_1}\, \mathrm{d}x'^{j_2} \cdots \mathrm{d}x'^{j_n} \Big]$$

Let x'^i be the local inertial coordinates at a point P on the manifold, *i.e.*, $g'_{ij}(P) = \delta_{ij}$, and we want $J'(P) = 1$. Now, we have:

$$J(x) \frac{1}{n!} \varepsilon_{i_1 i_2 \cdots i_n} \Big[\mathrm{sgn}(\sigma)\, \mathrm{d}x^{i_1}\, \mathrm{d}x^{i_2} \cdots \mathrm{d}x^{i_n} \Big] = \frac{1}{n!} \varepsilon_{j_1 j_2 \cdots j_n} \Big[\mathrm{sgn}(\sigma)\, \mathrm{d}x'^{j_1}\, \mathrm{d}x'^{j_2} \cdots \mathrm{d}x'^{j_n} \Big]$$

Using the transformation $\mathrm{d}x'^j = \frac{\partial x'^j}{\partial x^i}\, \mathrm{d}x^i$, we get:

$$J(x) \frac{1}{n!} \varepsilon_{i_1 i_2 \cdots i_n} \Big[\mathrm{sgn}(\sigma)\, \mathrm{d}x^{i_1}\, \mathrm{d}x^{i_2} \cdots \mathrm{d}x^{i_n} \Big] = \frac{1}{n!} \varepsilon_{j_1 j_2 \cdots j_n} \left[\mathrm{sgn}(\sigma) \frac{\partial x'^{j_1}}{\partial x^{i_1}}\, \mathrm{d}x^{i_1} \frac{\partial x'^{j_2}}{\partial x^{i_2}}\, \mathrm{d}x^{i_2} \cdots \frac{\partial x'^{j_n}}{\partial x^{i_n}}\, \mathrm{d}x^{i_n} \right]$$

Since $\mathrm{d}x^i$ are linearly independent, we can equate the coefficients of $\mathrm{d}x^1\, \mathrm{d}x^2 \cdots \mathrm{d}x^n$ on both sides to get:

$$J(x) \varepsilon_{i_1 i_2 \cdots i_n} = \varepsilon_{j_1 j_2 \cdots j_n} \frac{\partial x'^{j_1}}{\partial x^{i_1}} \frac{\partial x'^{j_2}}{\partial x^{i_2}} \cdots \frac{\partial x'^{j_n}}{\partial x^{i_n}}$$

Contracting both sides with $\varepsilon^{i_1 i_2 \cdots i_n}$, we get:

$$J(x) \varepsilon_{i_1 i_2 \cdots i_n} \varepsilon^{i_1 i_2 \cdots i_n} = \varepsilon_{j_1 j_2 \cdots j_n} \frac{\partial x'^{j_1}}{\partial x^{i_1}} \frac{\partial x'^{j_2}}{\partial x^{i_2}} \cdots \frac{\partial x'^{j_n}}{\partial x^{i_n}} \varepsilon^{i_1 i_2 \cdots i_n}$$

Using the identity $\varepsilon_{i_1 i_2 \cdots i_n} \varepsilon^{i_1 i_2 \cdots i_n} = n!$, and the definition of determinant, we get:

$$J(x) n! = n! \det \left(\frac{\partial x'^j}{\partial x^i} \right)$$

$$J(x) = \det \left(\frac{\partial x'^j}{\partial x^i} \right) = \det \Big(\Lambda^j{}_i \Big).$$

[1] The term in the square brackets is just $\mathrm{d}x^{i_1} \wedge \mathrm{d}x^{i_2} \wedge \cdots \wedge \mathrm{d}x^{i_n}$.

This scalar function $J(x)$ is the determinant of the transformation matrix $\Lambda^{j}{}_{i}$, and known as the Jacobian of the transformation. Now, we can find the Jacobian in terms of the metric tensor. We know that the metric tensor transforms as:

$$g'_{ij} = (\Lambda^{-1})^{k}{}_{i}(\Lambda^{-1})^{l}{}_{j} g_{kl} \quad \Rightarrow \quad g_{kl} = \Lambda_{k}{}^{i}\Lambda_{l}{}^{j} g'_{ij}.$$

Taking the determinant on both sides, we get:

$$\det(g_{kl}) = \det\left(\Lambda_{k}{}^{i}\right)\det\left(\Lambda_{l}{}^{j}\right)\det\left(g'_{ij}\right) = \left[\det\left(\Lambda_{k}{}^{i}\right)\right]^{2}\det\left(g'_{ij}\right).$$

Since $g'_{ij} = \delta_{ij}$, we have $\det\left(g'_{ij}\right) = 1$. So, we have:

$$\det(g_{kl}) = \left[\det\left(\Lambda_{k}{}^{i}\right)\right]^{2} \quad \Rightarrow \quad \det\left(\Lambda_{k}{}^{i}\right) = \sqrt{\det(g_{kl})}.$$

Define $g = \det\left(g_{ij}\right)$, then we have:

$$J(x) = \sqrt{g}. \tag{12.9}$$

So, the invariant measure on an n-dimensional Riemannian manifold is

$$\mathrm{d}V = \sqrt{g}\frac{1}{n!}\varepsilon_{i_1 i_2 \cdots i_n}\left[\mathrm{sgn}(\sigma)\,\mathrm{d}x^{i_1}\,\mathrm{d}x^{i_2}\cdots\mathrm{d}x^{i_n}\right]. \tag{12.10}$$

In case of a 4-dimensional Lorentzian manifold (spacetime), the local inertial coordinates x'^{μ} are such that $g'_{\mu\nu} = \eta_{\mu\nu}$. So, we have $\det\left(g'_{\mu\nu}\right) = \det\left(\eta_{\mu\nu}\right) = -1$. Thus, the Jacobian is given by:

$$J(x) = \sqrt{-g}, \tag{12.11}$$

where $g = \det\left(g_{\mu\nu}\right)$. So, the invariant measure on a 4-dimensional Lorentzian manifold is given by:

$$\mathrm{d}V = \sqrt{-g}\frac{1}{4!}\varepsilon_{\mu_1\mu_2\mu_3\mu_4}[\mathrm{sgn}(\sigma)\,\mathrm{d}x^{\mu_1}\,\mathrm{d}x^{\mu_2}\,\mathrm{d}x^{\mu_3}\,\mathrm{d}x^{\mu_4}]. \tag{12.12}$$

This measure is also known as the volume element in curved spacetime.

Let's look at some known spaces to see how this invariant measure works.

Example 12.1 (Plane Polar Coordinates)
In 2-dimensional flat space, the metric tensor in plain polar coordinates $\{r, \theta\}$ is given by:

$$g_{ij} = \mathrm{diag}(1, r^2) \quad \Rightarrow \quad g = \det\left(g_{ij}\right) = r^2.$$

So, the invariant measure is given by:

$$\mathrm{d}V = \sqrt{g}\frac{1}{2!}\varepsilon_{i_1 i_2}\left[\mathrm{sgn}(\sigma)\,\mathrm{d}x^{i_1}\,\mathrm{d}x^{i_2}\right] = r\,\mathrm{d}r\,\mathrm{d}\theta\,.$$

Example 12.2 (Spherical Coordinates)
In 3-dimensional flat space, the metric tensor in spherical coordinates $\{r, \theta, \phi\}$ is given by:

$$g_{ij} = \mathrm{diag}(1, r^2, r^2\sin^2\theta) \quad \Rightarrow \quad g = \det\left(g_{ij}\right) = r^4\sin^2\theta.$$

So, the invariant measure is given by:

$$\mathrm{d}V = \sqrt{g}\frac{1}{3!}\varepsilon_{i_1 i_2 i_3}\left[\mathrm{sgn}(\sigma)\,\mathrm{d}x^{i_1}\,\mathrm{d}x^{i_2}\,\mathrm{d}x^{i_3}\right] = r^2\sin\theta\,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}\phi\,.$$

We can simplify the expression for the invariant measure using the antisymmetric property of the Levi-Civita symbol. We have:

$$\frac{1}{n!}\varepsilon_{i_1 i_2 \cdots i_n}\Big[\mathrm{sgn}(\sigma)\,\mathrm{d}x^{i_1}\,\mathrm{d}x^{i_2}\cdots\mathrm{d}x^{i_n}\Big] = \mathrm{d}x^1\,\mathrm{d}x^2\cdots\mathrm{d}x^n\,.$$

So, the invariant measure on an n-dimensional Riemannian manifold is

$$\mathrm{d}V = \sqrt{g}\,\mathrm{d}^n x\,, \tag{12.13}$$

and on a 4-dimensional Lorentzian manifold (spacetime), the invariant measure is

$$\mathrm{d}V = \sqrt{-g}\,\mathrm{d}^4 x\,. \tag{12.14}$$

Lec 13: Covariant Derivative

We have argued that the action for the metric field have the following form

$$S_{\mathrm{G}} \sim \int \mathrm{d}^4x \sqrt{-g} f(g_{\mu\nu}, \cdots, \partial_\alpha \partial_\beta g_{\mu\nu})$$

where f is a scalar function of the metric and its derivatives (up to second order). But before we move forward, is the quantity $\partial_\alpha \partial_\beta g_{\mu\nu}$ a tensor of any kind? Spoiler alert: it is not.

§13.1 Covariant Derivative of Scalar Field

Consider a scalar field $\phi(x)$, which means that it is invariant under coordinate transformations

$$\phi'(x') = \phi(x).$$

Look at the quantity $\partial_\mu \phi(x)$. Under a coordinate transformation, we have

$$\partial'_\mu \phi'(x') = \partial'_\mu \phi(x) = \frac{\partial x^\alpha}{\partial x'^\mu} \partial_\alpha \phi(x). \tag{13.1}$$

This means that $\partial_\mu \phi$ transforms like a covariant vector, and hence is a tensor of rank $(0,1)$. We call $\partial_\mu \phi$ the *covariant derivative* of the scalar field ϕ, and denote it by $\nabla_\mu \phi$:

$$\nabla_\mu \phi \equiv \partial_\mu \phi. \tag{13.2}$$

§13.2 Covariant Derivative of Vector Field

Now consider a contravariant vector field $V^\rho(x)$, so that it transforms as

$$V'^\rho(x') = \frac{\partial x'^\rho}{\partial x^\beta} V^\beta(x).$$

Consider the quantity $\partial_\mu V^\rho(x)$. Under a general coordinate transformation, we have

$$\begin{aligned} \partial'_\mu V'^\rho(x') &= \frac{\partial x^\alpha}{\partial x'^\mu} \partial_\alpha \left[\frac{\partial x'^\rho}{\partial x^\beta} V^\beta(x) \right] \\ &= \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x'^\rho}{\partial x^\beta} \partial_\alpha V^\beta(x) + \underbrace{\frac{\partial x^\alpha}{\partial x'^\mu} \partial_\alpha \left[\frac{\partial x'^\rho}{\partial x^\beta} \right]}_{\neq 0} V^\beta(x). \end{aligned}$$

The first term on the right-hand side is the tensorial transformation we would expect for a rank $(1,1)$ tensor. However, the second term is an additional term that prevents $\partial_\mu V^\rho$ from being a tensor.

To fix this, we define the *covariant derivative* of the vector field V^ρ such that $\nabla_\mu V^\rho$ is a tensor of rank $(1,1)$. Thus, we have

$$\nabla'_\mu V'^\rho(x') = \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x'^\rho}{\partial x^\beta} \nabla_\alpha V^\beta(x). \tag{13.3}$$

So inverting the transformation, we have

$$\nabla_\alpha V^\beta(x) = \frac{\partial x^\beta}{\partial x'^\rho} \frac{\partial x'^\mu}{\partial x^\alpha} \nabla'_\mu V'^\rho(x').$$

In local inertial frame, we want the covariant derivative to reduce to the partial derivative, *i.e.*, $\nabla'_\mu = \partial'_\mu$. Thus, we have

$$\begin{aligned}
\nabla_\alpha V^\beta &= \frac{\partial x^\beta}{\partial x'^\rho} \underbrace{\frac{\partial x'^\mu}{\partial x^\alpha}\partial'_\mu}_{=\partial_\alpha} \left[\frac{\partial x'^\rho}{\partial x^\nu} V^\nu\right] \\
&= \frac{\partial x^\beta}{\partial x'^\rho}\partial_\alpha \left[\frac{\partial x'^\rho}{\partial x^\nu} V^\nu\right] \\
&= \underbrace{\frac{\partial x^\beta}{\partial x'^\rho}\frac{\partial x'^\rho}{\partial x^\nu}}_{\delta^\beta{}_\nu}\partial_\alpha V^\nu + \underbrace{\frac{\partial x^\beta}{\partial x'^\rho}\partial_\alpha\left[\frac{\partial x'^\rho}{\partial x^\nu}\right]}_{\text{affine connection}} V^\nu \\
&= \partial_\alpha V^\beta + \Gamma^\beta_{\alpha\nu} V^\nu,
\end{aligned}$$

where we have used the definition of the Christoffel symbols (see eqn. (10.1)). Thus, we have the *covariant derivative* of a contravariant vector field V^β as

$$\nabla_\alpha V^\beta = \partial_\alpha V^\beta + \Gamma^\beta_{\alpha\nu} V^\nu. \tag{13.4}$$

Assignment 04.1 (Covariant Derivative of Covariant Vector Field)**.**
Show that the covariant derivative of a covariant vector field W_β is given by

$$\nabla_\alpha W_\beta = \partial_\alpha W_\beta - \Gamma^\nu_{\alpha\beta} W_\nu .$$

Solution.
We have

$$\nabla'_\mu W'_\rho(x') = \frac{\partial x^\alpha}{\partial x'^\mu}\frac{\partial x^\beta}{\partial x'^\rho}\nabla_\alpha W_\beta(x).$$

Inverting the transformation, we have

$$\nabla_\alpha W_\beta(x) = \frac{\partial x'^\rho}{\partial x^\beta}\frac{\partial x'^\mu}{\partial x^\alpha}\ \nabla'_\mu W'_\rho(x').$$

In local inertial frame, we want the covariant derivative to reduce to the partial derivative, *i.e.*, $\nabla'_\mu = \partial'_\mu$. Thus, we have

$$\begin{aligned}
\nabla_\alpha W_\beta &= \frac{\partial x'^\rho}{\partial x^\beta} \underbrace{\frac{\partial x'^\mu}{\partial x^\alpha}\partial'_\mu}_{=\partial_\alpha} \left[\frac{\partial x^\nu}{\partial x'^\rho} W_\nu\right] \\
&= \frac{\partial x'^\rho}{\partial x^\beta}\partial_\alpha \left[\frac{\partial x^\nu}{\partial x'^\rho} W_\nu\right] \\
&= \underbrace{\frac{\partial x'^\rho}{\partial x^\beta}\frac{\partial x^\nu}{\partial x'^\rho}}_{\delta_\beta{}^\nu}\partial_\alpha W_\nu + \frac{\partial x'^\rho}{\partial x^\beta}\partial_\alpha\left[\frac{\partial x^\nu}{\partial x'^\rho}\right] W_\nu
\end{aligned}$$

Using chain rule for the second term $\partial_\alpha = \frac{\partial x'^\mu}{\partial x^\alpha}\partial'_\mu$, we have

$$= \partial_\alpha W_\beta + \frac{\partial x'^\rho}{\partial x^\beta}\frac{\partial x'^\mu}{\partial x^\alpha}\frac{\partial^2 x^\nu}{\partial x'^\mu \partial x'^\rho} W_\nu$$

For the second term, we need to convert it into the standard form of the affine connection. Since the Kronecker delta is invariant, we have

$$0 = \partial_\alpha(\delta_\beta{}^\nu) = \partial_\alpha\left(\frac{\partial x'^\rho}{\partial x^\beta}\frac{\partial x^\nu}{\partial x'^\rho}\right) = \partial_\alpha\left(\frac{\partial x'^\rho}{\partial x^\beta}\right)\frac{\partial x^\nu}{\partial x'^\rho} + \frac{\partial x'^\rho}{\partial x^\beta}\partial_\alpha\left(\frac{\partial x^\nu}{\partial x'^\rho}\right).$$

In the last term, x^ν is a function of x'^μ, so we can use the chain rule to write

$$0 = \frac{\partial x'^\mu}{\partial x^\alpha}\frac{\partial x'^\rho}{\partial x^\beta}\frac{\partial^2 x^\nu}{\partial x'^\mu \partial x'^\rho} + \frac{\partial x^\nu}{\partial x'^\rho}\frac{\partial^2 x'^\rho}{\partial x^\alpha \partial x^\beta}.$$

Rearranging, we have

$$\frac{\partial x'^{\rho}}{\partial x^{\beta}}\frac{\partial x'^{\mu}}{\partial x^{\alpha}}\frac{\partial^2 x^{\nu}}{\partial x'^{\mu}\partial x'^{\rho}} = -\frac{\partial x^{\nu}}{\partial x'^{\rho}}\partial_{\alpha}\left(\frac{\partial x'^{\rho}}{\partial x^{\beta}}\right) = -\Gamma^{\nu}_{\alpha\beta}.$$

Thus, we have

$$\nabla_{\alpha}W_{\beta} = \partial_{\alpha}W_{\beta} - \Gamma^{\nu}_{\alpha\beta}W_{\nu}\,.$$

◀

Summary

We have seen that partial derivatives of tensor fields do not transform like tensors. To fix this, we have defined the covariant derivative ∇_{μ} such that the covariant derivative of a tensor field is also a tensor field. The covariant derivative of various tensor fields are given by

$$\begin{aligned}
\nabla_{\mu}\phi &= \partial_{\mu}\phi, \\
\nabla_{\mu}V^{\nu} &= \partial_{\mu}V^{\nu} + \Gamma^{\nu}_{\mu\rho}V^{\rho}, \\
\nabla_{\mu}W_{\nu} &= \partial_{\mu}W_{\nu} - \Gamma^{\rho}_{\mu\nu}W_{\rho}\,.
\end{aligned}$$

Lec 14: Riemann & Ricci Tensors

In the last lecture, we have defined the covariant derivative for scalar, and vector fields. Let's now extend this definition to higher rank tensors. For a tensor of rank $(1,1)$, the covariant derivative is given by

$$\nabla_\lambda T^\mu{}_\nu = \partial_\lambda T^\mu{}_\nu + \Gamma^\mu_{\lambda\sigma} T^\sigma{}_\nu - \Gamma^\sigma_{\lambda\nu} T^\mu{}_\sigma \,, \tag{14.1}$$

a direct way to remember this is to note that each contravariant index (upper index) contributes a positive Christoffel symbol term, while each covariant index (lower index) contributes a negative Christoffel symbol term.

For a general tensor of rank (m,n), the covariant derivative is given by

$$\nabla_\lambda T^{\mu_1\mu_2\ldots\mu_m}{}_{\nu_1\nu_2\ldots\nu_n} = \partial_\lambda T^{\mu_1\mu_2\ldots\mu_m}{}_{\nu_1\nu_2\ldots\nu_n} + \sum_{i=1}^{m} \Gamma^{\mu_i}_{\lambda\sigma} T^{\mu_1\ldots\sigma\ldots\mu_m}{}_{\nu_1\nu_2\ldots\nu_n} - \sum_{j=1}^{n} \Gamma^\sigma_{\lambda\nu_j} T^{\mu_1\mu_2\ldots\mu_m}{}_{\nu_1\ldots\sigma\ldots\nu_n} \,.$$

§14.1 Covariant Derivative of Metric Tensor

We started our discussion by noting that we want a scalar quantity with at most two derivatives of the metric tensor. So let's first compute the covariant derivative of the metric tensor. We have

$$\nabla_\lambda g_{\mu\nu} = \partial_\lambda g_{\mu\nu} - \Gamma^\sigma_{\lambda\mu} g_{\sigma\nu} - \Gamma^\sigma_{\lambda\nu} g_{\mu\sigma} . \tag{14.2}$$

Using the expression for the Christoffel symbols in terms of the metric tensor, we have

$$\begin{aligned} g_{\sigma\nu}\Gamma^\sigma_{\lambda\mu} &= \frac{1}{2} g_{\sigma\nu} g^{\sigma\rho}\Big(\partial_\lambda g_{\rho\mu} + \partial_\mu g_{\lambda\rho} - \partial_\rho g_{\lambda\mu}\Big); \\ &= \frac{1}{2}\Big(\partial_\lambda g_{\nu\mu} + \partial_\mu g_{\lambda\nu} - \partial_\nu g_{\lambda\mu}\Big), \end{aligned} \qquad \begin{aligned} g_{\mu\sigma}\Gamma^\sigma_{\lambda\nu} &= \frac{1}{2} g_{\mu\sigma} g^{\sigma\rho}\Big(\partial_\lambda g_{\rho\nu} + \partial_\nu g_{\lambda\rho} - \partial_\rho g_{\lambda\nu}\Big) \\ &= \frac{1}{2}\Big(\partial_\lambda g_{\mu\nu} + \partial_\nu g_{\lambda\mu} - \partial_\mu g_{\lambda\nu}\Big). \end{aligned}$$

Substituting these back, we get

$$\nabla_\lambda g_{\mu\nu} = \partial_\lambda g_{\mu\nu} - \frac{1}{2}\Big(\partial_\lambda g_{\nu\mu} + \partial_\mu g_{\lambda\nu} - \partial_\nu g_{\lambda\mu}\Big) - \frac{1}{2}\Big(\partial_\lambda g_{\mu\nu} + \partial_\nu g_{\lambda\mu} - \partial_\mu g_{\lambda\nu}\Big) = 0. \tag{14.3}$$

This is a very important result, which states that the covariant derivative of the metric tensor is zero. This is also known as the *metric compatibility condition*. So, this means that the metric tensor is covariantly conserved.

Assignment 04.2 (Christoffel Symbols in terms of determinant of Metric Tensor)**.**
Show that the Christoffel symbols can also be written in terms of the determinant of the metric tensor as

$$\Gamma^\alpha_{\alpha\mu} = \frac{1}{\sqrt{-g}} \partial_\mu \sqrt{-g}, \tag{14.4}$$

where $g = \det\Big(g_{\alpha\beta}\Big)$.

Solution.
We have

$$\Gamma^\alpha_{\alpha\mu} = \frac{1}{2} g^{\alpha\beta}\Big(\partial_\alpha g_{\beta\mu} + \partial_\mu g_{\alpha\beta} - \partial_\beta g_{\alpha\mu}\Big) = \frac{1}{2}\Big(g^{\alpha\beta}\partial_\alpha g_{\beta\mu} + g^{\alpha\beta}\partial_\mu g_{\alpha\beta} - g^{\alpha\beta}\partial_\beta g_{\alpha\mu}\Big).$$

In the last term, we can rename the dummy indices $\alpha \leftrightarrow \beta$ to get

$$\Gamma^\alpha_{\alpha\mu} = \frac{1}{2}\Big(g^{\alpha\beta}\partial_\alpha g_{\beta\mu} + g^{\alpha\beta}\partial_\mu g_{\alpha\beta} - g^{\beta\alpha}\partial_\alpha g_{\beta\mu}\Big) = \frac{1}{2} g^{\alpha\beta}\partial_\mu g_{\alpha\beta}.$$

Now, consider a square matrix invertible matrix $M(x)$ whose elements depend on coordinates x. Then, we have the following identity[1]

$$\operatorname{tr}\left(M^{-1}\partial_\mu M\right) = \frac{1}{\det(M)}\partial_\mu(\det(M)).$$

Applying this to the metric tensor $g_{\alpha\beta}$, we have

$$\operatorname{tr}\left(g^{\alpha\beta}\partial_\mu g_{\beta\gamma}\right) = g^{\alpha\beta}\partial_\mu g_{\beta\alpha} = \frac{1}{\det\left(g_{\alpha\beta}\right)}\partial_\mu(\det\left(g_{\alpha\beta}\right)) = \frac{1}{g}\partial_\mu(g).$$

Massaging the last term, we get

$$\frac{1}{g}\partial_\mu g = \frac{1}{-g}\partial_\mu(-g) = \frac{1}{-g}\partial_\mu(\sqrt{-g}\sqrt{-g}) = \frac{2}{-g}\left(\sqrt{-g}\partial_\mu\sqrt{-g}\right) = \frac{2}{\sqrt{-g}}\partial_\mu\sqrt{-g}.$$

Thus, we have

$$g^{\alpha\beta}\partial_\mu g_{\alpha\beta} = \frac{2}{\sqrt{-g}}\partial_\mu\sqrt{-g}.$$

Substituting this back, we get

$$\Gamma^\alpha_{\alpha\mu} = \frac{1}{2}g^{\alpha\beta}\partial_\mu g_{\alpha\beta} = \frac{1}{\sqrt{-g}}\partial_\mu\sqrt{-g}.$$

◀

Assignment 04.3 (4-Gradient).
Show that the 4-gradient of a contravariant vector field V^μ is given by

$$\nabla_\mu V^\mu = \frac{1}{\sqrt{-g}}\partial_\mu(\sqrt{-g}V^\mu). \tag{14.5}$$

Solution.
We have

$$\nabla_\mu V^\mu = \partial_\mu V^\mu + \Gamma^\mu_{\mu\sigma}V^\sigma.$$

Using the previous result, we have

$$\nabla_\mu V^\mu = \partial_\mu V^\mu + \frac{1}{\sqrt{-g}}\left(\partial_\mu\sqrt{-g}\right)V^\mu = \frac{1}{\sqrt{-g}}\left(\sqrt{-g}\partial_\mu V^\mu + \left(\partial_\mu\sqrt{-g}\right)V^\mu\right) = \frac{1}{\sqrt{-g}}\partial_\mu(\sqrt{-g}V^\mu),$$

where we have used the product rule of differentiation in the last step. ◀

Assignment 04.4 (Leibniz Product Rule for Covariant Derivative).
Show that the covariant derivative satisfies the Leibniz product rule, *i.e.*,

(i) $\nabla_\mu(\phi V^\nu) = (\nabla_\mu\phi)V^\nu + \phi(\nabla_\mu V^\nu)$, where ϕ is a scalar field.

(ii) $\nabla_\mu(V^\alpha W_\beta) = (\nabla_\mu V^\alpha)W_\beta + V^\alpha(\nabla_\mu W_\beta)$, where V^α is a contravariant vector field and W_β is a covariant vector field.

Solution. (i) We have

$$\begin{aligned}\nabla_\mu(\phi V^\nu) &= \partial_\mu(\phi V^\nu) + \Gamma^\nu_{\mu\sigma}(\phi V^\sigma)\\ &= (\partial_\mu\phi)V^\nu + \phi(\partial_\mu V^\nu) + \Gamma^\nu_{\mu\sigma}\phi V^\sigma\\ &= (\nabla_\mu\phi)V^\nu + \phi(\nabla_\mu V^\nu),\end{aligned}$$

where we have used the fact that the covariant derivative of a scalar field is equal to its partial derivative.

[1] For a proof, see section B.1.

(ii) We have

$$\begin{aligned}\nabla_\mu(V^\alpha W_\beta) &= \partial_\mu(V^\alpha W_\beta) + \Gamma^\alpha_{\mu\sigma}(V^\sigma W_\beta) - \Gamma^\sigma_{\mu\beta}(V^\alpha W_\sigma)\\ &= (\partial_\mu V^\alpha)W_\beta + V^\alpha(\partial_\mu W_\beta) + \Gamma^\alpha_{\mu\sigma}V^\sigma W_\beta - \Gamma^\sigma_{\mu\beta}V^\alpha W_\sigma\\ &= (\nabla_\mu V^\alpha)W_\beta + V^\alpha(\nabla_\mu W_\beta),\end{aligned}$$

where we have used the definition of the covariant derivative for contravariant and covariant vector fields.

◀

§14.2 Riemann Curvature Tensor

Consider V^ρ be a contravariant vector field. Let's compute the second covariant derivative $\nabla_\mu(\nabla_\nu V^\rho)$. We have

$$\begin{aligned}\nabla_\mu(\nabla_\nu V^\rho) &= \partial_\mu(\nabla_\nu V^\rho) + \Gamma^\rho_{\mu\sigma}(\nabla_\nu V^\sigma) - \Gamma^\sigma_{\mu\nu}(\nabla_\sigma V^\rho)\\ &= \partial_\mu[\partial_\nu V^\rho + \Gamma^\rho_{\nu\sigma}V^\sigma] + \Gamma^\rho_{\mu\sigma}\left[\partial_\nu V^\sigma + \Gamma^\sigma_{\nu\lambda}V^\lambda\right] - \Gamma^\lambda_{\mu\nu}(\nabla_\lambda V^\rho)\\ &= \partial_\mu\partial_\nu V^\rho + \partial_\mu(\Gamma^\rho_{\nu\sigma})V^\sigma + \Gamma^\rho_{\nu\sigma}\partial_\mu V^\sigma + \Gamma^\rho_{\mu\sigma}\partial_\nu V^\sigma + \Gamma^\rho_{\mu\sigma}\Gamma^\sigma_{\nu\lambda}V^\lambda - \Gamma^\lambda_{\mu\nu}(\nabla_\lambda V^\rho).\end{aligned}$$

Notice that the terms in blue and green are symmetric in μ and ν. Let us now compute $\nabla_\nu\Big(\nabla_\mu V^\rho\Big)$. We have

$$\nabla_\nu\Big(\nabla_\mu V^\rho\Big) = \partial_\nu\partial_\mu V^\rho + \partial_\nu(\Gamma^\rho_{\mu\sigma})V^\sigma + \Gamma^\rho_{\mu\sigma}\partial_\nu V^\sigma + \Gamma^\rho_{\nu\sigma}\partial_\mu V^\sigma + \Gamma^\rho_{\nu\sigma}\Gamma^\sigma_{\mu\lambda}V^\lambda - \Gamma^\lambda_{\nu\mu}(\nabla_\lambda V^\rho).$$

Taking the difference, the terms in blue and green cancel out, and we are left with

$$\begin{aligned}\Big[\nabla_\mu,\nabla_\nu\Big]V^\rho &= \nabla_\mu(\nabla_\nu V^\rho) - \nabla_\nu\Big(\nabla_\mu V^\rho\Big)\\ &= \Big(\partial_\mu(\Gamma^\rho_{\nu\sigma}) - \partial_\nu(\Gamma^\rho_{\mu\sigma})\Big)V^\sigma + \Big(\Gamma^\rho_{\mu\sigma}\Gamma^\sigma_{\nu\lambda} - \Gamma^\rho_{\nu\sigma}\Gamma^\sigma_{\mu\lambda}\Big)V^\lambda.\end{aligned}$$

In the last line for the second term, we have renamed the dummy index $\lambda \leftrightarrow \sigma$. We can write this as

$$\Big[\nabla_\mu,\nabla_\nu\Big]V^\rho = \Big(\partial_\mu\Gamma^\rho_{\nu\sigma} - \partial_\nu\Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma}\Big)V^\sigma. \tag{14.6}$$

Since left-hand side is a rank $(1,2)$ tensor, and V^σ is a rank $(1,0)$ tensor, the quantity in the parentheses must be a rank $(1,3)$ tensor. This is known as the *Riemann curvature tensor*, and is denoted by $R^\rho{}_{\sigma\mu\nu}$. Thus, we have

$$R^\rho{}_{\sigma\mu\nu} = \partial_\mu\Gamma^\rho_{\sigma\nu} - \partial_\nu\Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\sigma\nu} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma}. \tag{14.7}$$

Consider a covariant vector field W_ρ. We can similarly compute $\Big[\nabla_\mu,\nabla_\nu\Big]W_\rho$. We know $W_\rho = g_{\rho\lambda}W^\lambda$. Using the Leibniz product rule, we have

$$\Big[\nabla_\mu,\nabla_\nu\Big]W_\rho = \Big[\nabla_\mu,\nabla_\nu\Big](g_{\rho\lambda}W^\lambda) = \Big(\Big[\nabla_\mu,\nabla_\nu\Big]g_{\rho\lambda}\Big)W^\lambda + g_{\rho\lambda}\Big(\Big[\nabla_\mu,\nabla_\nu\Big]W^\lambda\Big).$$

Using the metric compatibility condition, the first term vanishes, and we are left with

$$\Big[\nabla_\mu,\nabla_\nu\Big]W_\rho = g_{\rho\lambda}\Big(\Big[\nabla_\mu,\nabla_\nu\Big]W^\lambda\Big) = g_{\rho\lambda}\Big(R^\lambda{}_{\sigma\mu\nu}W^\sigma\Big) = R_{\rho\sigma\mu\nu}W^\sigma,$$

using all lower indices for the Riemann tensor. Thus, we have

$$\Big[\nabla_\mu,\nabla_\nu\Big]W_\rho = R_{\rho\lambda\mu\nu}W^\lambda = R_{\rho\lambda\mu\nu}g^{\lambda\sigma}W_\sigma = R_\rho{}^\sigma{}_{\mu\nu}W_\sigma\,, \tag{14.8}$$

where we have raised the second index of the Riemann tensor using the metric tensor.

In this derivation, we have seen all lowered indices Riemann tensor $R_{\rho\sigma\mu\nu}$, which is related to the mixed index Riemann tensor $R^\rho{}_{\sigma\mu\nu}$ by

$$R_{\rho\sigma\mu\nu} = g_{\rho\lambda}R^\lambda{}_{\sigma\mu\nu}. \tag{14.9}$$

From definition eqn. (14.7), it is evident that for the flat spacetime (*i.e.*, Minkowski spacetime), the Riemann tensor vanishes, $R^\rho{}_{\sigma\mu\nu} = 0$, as all the Christoffel symbols vanish in this case.

The Riemann tensor has the following properties:

1. From the definition, we can see that the Riemann tensor is antisymmetric in the last two indices μ and ν, *i.e.*,
$$R_{\rho\sigma\mu\nu} = -R_{\rho\sigma\nu\mu}.$$

2. The Riemann tensor is also antisymmetric in the first two indices ρ and σ, *i.e.*,
$$R_{\rho\sigma\mu\nu} = -R_{\sigma\rho\mu\nu}.$$
This is not immediately obvious from the definition, since the covariant derivative of the metric tensor is zero, we have
$$0 = \left[\nabla_\mu, \nabla_\nu\right] g_{\alpha\beta} = R_\alpha{}^\lambda{}_{\mu\nu} g_{\lambda\beta} + R_\beta{}^\lambda{}_{\mu\nu} g_{\alpha\lambda} = R_{\alpha\beta\mu\nu} + R_{\beta\alpha\mu\nu}.$$
In the last step, we have used metric tensor to lower the second index of the Riemann tensor.

However, the rank of $(0,4)$ is very large, and it is useful to contract some indices to get lower rank tensors. Since $g^{\mu\nu}$ is a symmetric tensor, so we have
$$g^{\mu\nu} R_{\rho\sigma\mu\nu} = 0, \qquad g^{\rho\sigma} R_{\rho\sigma\mu\nu} = 0.$$
Thus, we can only contract the first and third indices or the second and fourth indices.

§14.3 Ricci Tensor and Ricci Scalar

Contracting the first and third indices of the Riemann tensor, we get the *Ricci tensor*, denoted by $R_{\mu\nu}$, given by
$$R_{\mu\nu} = g^{\alpha\beta} R_{\alpha\mu\beta\nu} \equiv R^\alpha{}_{\mu\alpha\nu}. \tag{14.10}$$
The Ricci tensor is a rank $(0,2)$ tensor. We can further contract the Ricci tensor using the metric tensor to get a scalar quantity known as the *Ricci scalar*, denoted by R, given by
$$R = g^{\mu\nu} R_{\mu\nu} \equiv R^\mu{}_\mu. \tag{14.11}$$
The Ricci scalar is a scalar quantity constructed from the metric tensor and its first and second derivatives. This is the quantity we were looking for to construct the action for General Relativity.

Even if we go to a local inertial frame at a point P, where the Christoffel symbols vanish, the Riemann tensor does not vanish, since it depends on the derivatives of the Christoffel symbols. Thus, the Ricci tensor and Ricci scalar do not vanish in a local inertial frame. This is in contrast to Special Relativity, as in that case, the first derivatives of the metric tensor vanish at all points in spacetime, and thus the Riemann tensor, Ricci tensor, and Ricci scalar all vanish.

§14.4 Einstein-Hilbert Action

We are now ready to write down the action for General Relativity. We want a scalar quantity constructed from the metric tensor and at most two derivatives of the metric tensor. The only such quantity is the Ricci scalar R. So we want to write down an action of the form
$$S_{\mathrm{G}} \sim \int \mathrm{d}^4x \sqrt{-g} f(R), \tag{14.12}$$
where $f(R)$ is some function of the Ricci scalar. Keeping the simplicity in mind, we take $f(R) = c_0 + c_1 R$, where c_0 and c_1 are some constants. Thus, we have
$$f(R) = \alpha[R - 2\Lambda], \tag{14.13}$$
where α and Λ are some constants. The constant Λ is known as the *cosmological constant*. Thus, we have
$$S_{\mathrm{G}} = \alpha \int \mathrm{d}^4x \sqrt{-g}(R - 2\Lambda). \tag{14.14}$$

We want action to have the dimension of $\hbar$. So we can fix the dimension of α. We have

$$[\alpha] = \frac{[\hbar]}{[\mathrm{d}^4x][\sqrt{-g}][R]} = \frac{ML^2T^{-1}}{L^4 \cdot 1 \cdot L^{-2}} = M \cdot T^{-1} = \frac{L^3T^{-3}}{[G]},$$

since Ricci scalar contains two derivatives of the metric tensor, and the metric tensor is dimensionless. Thus, we have

$$\alpha \sim \frac{c^3}{G}. \tag{14.15}$$

To fix the constant of proportionality, we want the action to reduce to the Newtonian limit in the weak gravity regime (non-relativistic limit). This fixes the constant of proportionality to be $1/(16\pi)$. Thus, we have the Einstein-Hilbert action

$$S_{\mathrm{EH}} = \frac{c^3}{16\pi G}\int \mathrm{d}^4x\,\sqrt{-g}(R - 2\Lambda) = \frac{c^4}{16\pi G}\int \mathrm{d}t\,\mathrm{d}^3\mathbf{x}\,\sqrt{-g}(R - 2\Lambda). \tag{14.16}$$

Lec 15: Ricci Curvature Example

In the last lecture, we defined the Riemann curvature tensor, Ricci curvature tensor, and Ricci scalar. We have also noticed that the Ricci curvature tensor contains at most two derivatives of the metric. Let's now first see how second derivatives is related to curvature.

Let us consider a function $f : \mathbb{R} \to \mathbb{R}$. The first derivative f' gives the slope of the function, and the second derivative f'' gives the rate of change of the slope. Let's now relate second derivatives to curvature. Say we want to approximate the function f around a point x_0 with a circle such that the circle has the same first and second derivatives as f at x_0 (see Figure 15.1).

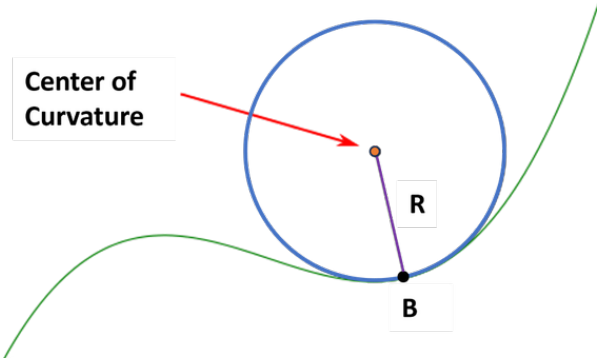


Figure 15.1: Approximating a 'smooth' function f around a point x_0 with a circle. The radius of the circle is called the radius of curvature.

Let us move the coordinate system such that center of the circle is at the origin, so the equation of the circle is $x^2 + y^2 = R^2$, where R is the radius of the circle. We can write the first derivative at x_0 as

$$y\frac{\mathrm{d}y}{\mathrm{d}x} + x = 0 \qquad \Rightarrow \qquad f'(x_0) = \left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x_0} = -\frac{x_0}{y_0}, \tag{15.1}$$

where $y_0 = f(x_0)$. Similarly, differentiating again, we get

$$y\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 + 1 = 0 \qquad \Rightarrow \qquad y_0 = \frac{1 + (f'(x_0))^2}{-f''(x_0)}. \tag{15.2}$$

Taking square root of the equation of the circle, we have

$$R = \left| y_0\sqrt{1 + \left(\frac{x_0}{y_0}\right)^2} \right| = \left| \frac{(1 + (f'(x_0))^2)^{3/2}}{f''(x_0)} \right|. \tag{15.3}$$

By choosing appropriate coordinates, we can set $f'(x_0) = 0$, and hence the radius of curvature is given by

$$R = \frac{1}{|f''(x_0)|}. \tag{15.4}$$

This shows that the second derivative of a function is related to the curvature of the function.

§15.1 Example: Ricci Curvature of 2-Sphere

Previously, we had computed the geodesics of a 2-sphere. Let us now compute the curvature of a 2-sphere. The metric of a 2-sphere is given by

$$\mathrm{d}s^2 = r_\circ^2\,\mathrm{d}\theta^2 + r_\circ^2 \sin^2\theta\,\mathrm{d}\phi^2\,, \tag{15.5}$$

where $r_\circ$ is the radius of the sphere. We can write the metric as

$$g_{\mu\nu} = \begin{pmatrix} r_\circ^2 & 0 \\ 0 & r_\circ^2 \sin^2\theta \end{pmatrix}, \qquad g^{\mu\nu} = \begin{pmatrix} \frac{1}{r_\circ^2} & 0 \\ 0 & \frac{1}{r_\circ^2 \sin^2\theta} \end{pmatrix}. \tag{15.6}$$

See eqn. (11.4), we have already computed the Christoffel symbols for the 2-sphere, the only non-zero Christoffel symbols are

$$\Gamma^\theta_{\phi\phi} = -\sin\theta\cos\theta, \qquad \Gamma^\phi_{\phi\theta} = \cot\theta = \Gamma^\phi_{\theta\phi}. \tag{15.7}$$

Now, we can compute the Riemann curvature tensor using eqn. (14.7), due to anti-symmetry property of the Riemann tensor, only non-zero components is $R^\theta{}_{\phi\theta\phi}$:

$$\begin{aligned} R^\theta{}_{\phi\theta\phi} &= \partial_\theta\Gamma^\theta_{\phi\phi} - \partial_\phi\Gamma^\theta_{\theta\phi} + \Gamma^\theta_{\theta\lambda}\Gamma^\lambda_{\phi\phi} - \Gamma^\theta_{\phi\lambda}\Gamma^\lambda_{\theta\phi} \\ &= \partial_\theta(-\sin\theta\cos\theta) - 0 + 0 - (-\sin\theta\cos\theta)(\cot\theta) \\ &= -\cos^2\theta + \sin^2\theta + \cos^2\theta \\ R^\theta{}_{\phi\theta\phi} &= \sin^2\theta. \end{aligned} \tag{15.8}$$

Now, we can compute the Ricci tensor using eqn. (14.10). Since the Ricci tensor is symmetric, we only need to compute two components, $R_{\theta\theta}$, $R_{\theta\phi}$, and $R_{\phi\phi}$:

$$R_{\theta\theta} = R^\lambda{}_{\theta\lambda\theta} \xrightarrow[\lambda=\phi]{\text{non-zero when}} R^\phi{}_{\theta\phi\theta} = g^{\phi\alpha}R_{\alpha\theta\phi\theta}$$

since the metric is diagonal, only non-zero contribution is when $\alpha = \phi$, thus

$$\begin{aligned} R_{\theta\theta} &= g^{\phi\phi}R_{\phi\theta\phi\theta} \\ &= \frac{1}{r_\circ^2\sin^2\theta}R_{\theta\phi\theta\phi} \\ &= \frac{1}{r_\circ^2\sin^2\theta}g_{\theta\beta}R^\beta{}_{\phi\theta\phi} \end{aligned}$$

again, only non-zero contribution is when $\beta = \theta$, thus

$$\begin{aligned} &= \frac{1}{r_\circ^2\sin^2\theta}g_{\theta\theta}R^\theta{}_{\phi\theta\phi} \\ &= \frac{1}{r_\circ^2\sin^2\theta}(r_\circ^2)(\sin^2\theta) \\ R_{\theta\theta} &= 1. \end{aligned}$$

Due to anti-symmetry property of the Riemann tensor, $R_{\theta\phi} = R^\lambda{}_{\theta\lambda\phi} = 0$. Finally, we compute $R_{\phi\phi}$:

$$R_{\phi\phi} = R^\lambda{}_{\phi\lambda\phi} \xrightarrow[\lambda=\theta]{\text{non-zero when}} R^\theta{}_{\phi\theta\phi} = \sin^2\theta$$

So summarizing, we have

$$R_{\mu\nu} = \begin{pmatrix} 1 & 0 \\ 0 & \sin^2\theta \end{pmatrix}. \tag{15.9}$$

Now, we can compute the Ricci scalar using eqn. (14.11):

$$\begin{aligned} R = g^{\mu\nu}R_{\mu\nu} &= g^{\theta\theta}R_{\theta\theta} + g^{\phi\phi}R_{\phi\phi} \\ &= \frac{1}{r_\circ^2}(1) + \frac{1}{r_\circ^2\sin^2\theta}(\sin^2\theta) \\ R &= \frac{2}{r_\circ^2}. \end{aligned} \tag{15.10}$$

We see that the Ricci scalar is constant everywhere on the sphere, and is inversely proportional to the square of the radius of the sphere. Thus, as the radius of the sphere increases, the curvature decreases, and in the limit $r_\circ \to \infty$, we get a flat surface with zero curvature.

This is consistent with our intuition that a sphere with larger radius is 'less curved' than a sphere with smaller radius. Also note that the Ricci scalar has dimensions of inverse length squared, which is consistent with our earlier observation that curvature is related to second derivatives of the metric.

Lec 16: Natural Units and Bianchi Identities

§16.1 Natural Units

Till now, we have been using the SI units. However, these choices of units are arbitrary, and we can choose other units as well. In theoretical physics, it is often convenient to use the so-called natural units, where we set the speed of light c and the reduced Planck's constant $\hbar$ to unity:

$$c = 1, \qquad \hbar = 1.$$

So in natural units, the dimensions of length and time are the same, and we can measure length in terms of kg^{-1}. If we set $c = 1$, then we have

$$1\,\mathrm{s} = 3 \times 10^8\,\mathrm{m}.$$

Similarly, if we set $\hbar = 1$, then we have

$$1 = \frac{1 \times 10^{-34}\,\mathrm{kg}\cdot\mathrm{m}^2}{\mathrm{s}} = \frac{1}{3} \times 10^{-42}\,\mathrm{kg}\cdot\mathrm{m} \quad \Rightarrow \quad 1\,\mathrm{m} = 3 \times 10^{42}\,\mathrm{kg}^{-1}.$$

To get a feel of these units, compute the following quantities as part of your assignment.

Assignment 05.1.
Using natural units, express the distance between the Earth and the Sun (approximately 1.5×10^{11} m) in the units of s and kg^{-1}.

Assignment 05.2.
Using natural units, express the mass of the Sun (approximately 2×10^{30} kg) in the units of km^{-1} and yr^{-1}.

Assignment 05.3.
Using natural units, express 1 Earth-day in the units of km and kg^{-1}.

Since we have set $\hbar = 1$, thus in natural units, the action S is dimensionless.

§16.2 Bianchi Identities

The space of tensors of type (p, q) at a point P in a manifold $\mathcal{M}$ is denoted by $\mathcal{T}^p_q(P)$. This space forms a vector space over the field of real numbers $\mathbb{R}$ with the following operations:

- Addition: If $T_1, T_2 \in \mathcal{T}^p_q(P)$, then $T_1 + T_2 \in \mathcal{T}^p_q(P)$.
- Scalar Multiplication: If $T \in \mathcal{T}^p_q(P)$ and $\alpha \in \mathbb{R}$, then $\alpha T \in \mathcal{T}^p_q(P)$.

The dimension of this vector space is given by

$$\dim(\mathcal{T}^p_q(P)) = n^{p+q},$$

where n is the dimension of the manifold $\mathcal{M}$.

The covariant derivative operator ∇ is a linear map from $\mathcal{T}^p_q(P)$ to $\mathcal{T}^p_{q+1}(P)$:

$$\nabla : \mathcal{T}^p_q(P) \to \mathcal{T}^p_{q+1}(P).$$

§16.2.1 Jacobi Identity

Define the commutator of two linear operators A and B as

$$[A, B] = AB - BA. \tag{16.1}$$

The commutator satisfies the Jacobi identity:

$$[A, [B, C]] + [B, [C, A]] + [C, [A, B]] = 0. \tag{16.2}$$

In shorthand, we can write this as

$$[A, [B, C]] + \text{cyclic}(A, B, C) = 0. \tag{16.3}$$

In particular, if we take $A = \nabla_\sigma$, $B = \nabla_\mu$, and $C = \nabla_\nu$, then we have

$$\Big[\nabla_\sigma, \Big[\nabla_\mu, \nabla_\nu\Big]\Big] + \text{cyclic}(\sigma, \mu, \nu) = 0. \tag{16.4}$$

§16.2.2 First Bianchi Identity

In last lecture, we have computed the commutator of two covariant derivatives acting on a contra-variant vector field V^ρ as well as on a co-variant vector field W_ρ. Let's now compute the commutator of two covariant derivatives acting on a scalar field ϕ,

$$\Big[\nabla_\mu, \nabla_\nu\Big]\phi = \nabla_\mu(\nabla_\nu\phi) - \nabla_\nu(\nabla_\mu\phi).$$

Since the scalar field is a $(0, 0)$ tensor, thus the outer covariant derivative acts on a tensor of rank $(0, 1)$. Therefore, we have

$$\Big[\nabla_\mu, \nabla_\nu\Big]\phi = \Big\{\partial_\mu(\nabla_\nu\phi) - \Gamma^\lambda_{\mu\nu}(\nabla_\lambda\phi)\Big\} - \Big\{\partial_\nu(\nabla_\mu\phi) - \Gamma^\lambda_{\nu\mu}(\nabla_\lambda\phi)\Big\}$$

The action of the covariant derivative on a scalar field is the same as that of the partial derivative, *i.e.*, $\nabla_\mu\phi = \partial_\mu\phi$. Thus, we have

$$= \Big\{\partial_\mu\partial_\nu\phi - \Gamma^\lambda_{\mu\nu}\partial_\lambda\phi\Big\} - \Big\{\partial_\nu\partial_\mu\phi - \Gamma^\lambda_{\nu\mu}\partial_\lambda\phi\Big\}$$

the partial derivatives commute for appropriately smooth functions, and the Christoffel symbols are symmetric in the lower two indices, we get

$$= 0.$$

Thus, the commutator of two covariant derivatives acting on a scalar field vanishes.

> Remark 16.1 (Torsion-Free Connection). The above result holds true for a torsion-free connection, where the Christoffel symbols are symmetric in the lower two indices. If the connection has torsion, then the Christoffel symbols are not symmetric in the lower two indices, and we have
>
> $$\big[\nabla_\mu, \nabla_\nu\big]\phi = -T^\lambda{}_{\mu\nu}\,\partial_\lambda\phi,$$
>
> where $T^\lambda{}_{\mu\nu}$ is the torsion tensor define as
>
> $$T^\lambda{}_{\mu\nu} = \Gamma^\lambda_{\mu\nu} - \Gamma^\lambda_{\nu\mu}.$$
>
> In this course, we will only consider torsion-free connections.

The left-hand side of the Jacobi identity is still a linear operator, so it's action on a scalar field must also vanish:

$$\Big[\nabla_\sigma, \Big[\nabla_\mu, \nabla_\nu\Big]\Big]\phi + \text{cyclic}(\sigma, \mu, \nu) = 0.$$

Expanding the first term, we have

$$\nabla_\sigma\Big(\Big[\nabla_\mu, \nabla_\nu\Big]\phi\Big) - \Big[\nabla_\mu, \nabla_\nu\Big](\nabla_\sigma\phi) + \text{cyclic}(\sigma, \mu, \nu) = 0.$$

The first term vanishes, as we have just shown that the commutator of two covariant derivatives acting on a scalar field vanishes. We have

$$-\Big[\nabla_\mu, \nabla_\nu\Big](\nabla_\sigma\phi) + \text{cyclic}(\sigma, \mu, \nu) = 0.$$

The term $\nabla_\sigma\phi$ is a $(0, 1)$ tensor, so we can use the result we have derived in last lecture for the commutator of two covariant derivatives acting on a co-variant vector field:

$$-R_{\sigma\rho\mu\nu}(\nabla^\rho\phi) + \text{cyclic}(\sigma, \mu, \nu) = 0.$$

Exchanging the first two indices to absorb the negative sign, and taking the factor $\nabla^\rho\phi$ out, we have

$$\Big\{R_{\rho\sigma\mu\nu} + R_{\rho\mu\nu\sigma} + R_{\rho\nu\sigma\mu}\Big\}\nabla^\rho\phi = 0.$$

Since this must hold for any scalar field ϕ, thus the quantity in the curly brackets must vanish. So we have

$$R_{\rho\sigma\mu\nu} + R_{\rho\mu\nu\sigma} + R_{\rho\nu\sigma\mu} = 0. \tag{16.5}$$

This is known as the first Bianchi identity (or algebraic Bianchi identity).

With the help of the first Bianchi identity, we can prove one more symmetry property of the Riemann curvature tensor. We have already shown that the Riemann tensor is antisymmetric in the first two indices as well as in the last two indices. Let's now show that it is symmetric under the exchange of the first pair of indices with the second pair of indices, *i.e.*,

$$R_{\rho\sigma\mu\nu} = R_{\mu\nu\rho\sigma}. \tag{16.6}$$

Starting from the first Bianchi identity, we have

$$R_{\rho\sigma\mu\nu} = -R_{\rho\mu\nu\sigma} - R_{\rho\nu\sigma\mu}$$

Using the anti-symmetry in the last two indices, we have

$$= R_{\mu\rho\nu\sigma} + R_{\nu\rho\sigma\mu}$$

Using the 1st Bianchi identity again for fixed indices μ and ν, we have

$$= \Big\{-R_{\mu\nu\sigma\rho} - R_{\mu\sigma\rho\nu}\Big\} + \Big\{-R_{\nu\sigma\mu\rho} - R_{\nu\mu\rho\sigma}\Big\}$$

Using the anti-symmetry property on the pair of indices highlighted in red and blue, we get

$$= 2R_{\mu\nu\rho\sigma} + \Big\{R_{\sigma\mu\rho\nu} + R_{\sigma\nu\mu\rho}\Big\}$$

Using the 1st Bianchi identity again for fixed indices σ, we have

$$= 2R_{\mu\nu\rho\sigma} - R_{\rho\sigma\nu\mu}$$

Using the anti-symmetry in the first two indices as well as in the last two indices for the second term, we have

$$= 2R_{\mu\nu\rho\sigma} - R_{\rho\sigma\mu\nu}$$

Taking the second term to the left-hand side, we have

$$2R_{\rho\sigma\mu\nu} = 2R_{\mu\nu\rho\sigma}$$

$$\boxed{R_{\rho\sigma\mu\nu} = R_{\mu\nu\rho\sigma}}. \tag{16.7}$$

□

A more brute-force way to prove this is sum all the four 1st Bianchi identities obtained by fixing different first indices $\{\rho, \sigma, \mu, \nu\}$ and then use the anti-symmetry properties to simplify the resulting expression.

With this additional symmetry property, we can now prove that the Ricci Curvature tensor is symmetric, we have

$$R_{\alpha\mu\beta\nu} = R_{\beta\nu\alpha\mu}$$

Contracting the indices α and β with $g^{\alpha\beta}$, we have

$$R_{\mu\nu} = R_{\nu\mu}.$$

Thus, the Ricci tensor is symmetric.

Remark 16.2 (Number of Independent Components of Riemann Tensor). Since the Riemann tensor is a rank $(0, 4)$ tensor in d-dimensional spacetime, it has d^4 components. However, due to the symmetries and anti-symmetries mentioned above, the number of independent components is reduced.

To start with, the anti-symmetry in the first two indices and the last two indices means that those pair of indices can each take $\binom{d}{2} = \frac{d(d-1)}{2}$ independent values. Thus, we can think of the Riemann tensor as a matrix with dimension $\binom{d}{2} \times \binom{d}{2}$. The pair exchange symmetry means that this matrix is symmetric. Thus, the number of independent components of the Riemann tensor is given by the number of independent components of a symmetric matrix of dimension $\binom{d}{2}$, which is given by

$$\frac{1}{2}\binom{d}{2}\left[\binom{d}{2} + 1\right] = \frac{d^4 - 2d^3 + 3d^2 - 2d}{8}.$$

Finally, for the cyclic identity (algebraic Bianchi identity), if any of the indices are same then the identity reduces to any of the previous symmetry or anti-symmetry. Thus, the cyclic identity gives non-trivial constraints only when all the indices are different. So this imposes $\binom{d}{4}$ independent constraints. Thus, the total number of independent components of the Riemann tensor is given by

$$\frac{1}{2}\binom{d}{2}\left[\binom{d}{2} + 1\right] - \binom{d}{4} = \frac{d^2(d^2-1)}{12}.$$

In 4-dimensional spacetime, the Riemann tensor has 20 independent components.

§16.3 Second Bianchi Identity

To prove the 1st Bianchi identity, we have used the Jacobi identity on a scalar field. Now, applying the Jacobi identity on a contra-variant vector field V^ρ, we have

$$\Big[\nabla_\sigma, \Big[\nabla_\mu, \nabla_\nu\Big]\Big]V^\rho + \text{cyclic}(\sigma, \mu, \nu) = 0,$$
$$\nabla_\sigma\Big(\Big[\nabla_\mu, \nabla_\nu\Big]V^\rho\Big) - \Big[\nabla_\mu, \nabla_\nu\Big](\nabla_\sigma V^\rho) + \text{cyclic}(\sigma, \mu, \nu) = 0.$$

Using the result we have derived in last lecture for the commutator of two covariant derivatives acting on a tensor, we have

$$\nabla_\sigma\Big(R^\rho{}_{\lambda\mu\nu}V^\lambda\Big) - \Big(R^\rho{}_{\lambda\mu\nu}\nabla_\sigma V^\lambda + R_\sigma{}^\lambda{}_{\mu\nu}\nabla_\lambda V^\rho\Big) + \text{cyclic}(\sigma, \mu, \nu) = 0.$$

Using Leibniz rule for the covariant derivative on the first term, we have

$$\Big\{\nabla_\sigma\Big(R^\rho{}_{\lambda\mu\nu}\Big)V^\lambda + R^\rho{}_{\lambda\mu\nu}\nabla_\sigma(V^\lambda)\Big\} - \Big\{R^\rho{}_{\lambda\mu\nu}\nabla_\sigma(V^\lambda) + R_\sigma{}^\lambda{}_{\mu\nu}\nabla_\lambda(V^\rho)\Big\} + \text{cyclic}(\sigma, \mu, \nu) = 0.$$

The second term in the first curly brackets and the first term in the second curly brackets cancel out, and we have

$$\Rightarrow \quad \nabla_\sigma\Big(R^\rho{}_{\lambda\mu\nu}\Big)V^\lambda - R_\sigma{}^\lambda{}_{\mu\nu}\nabla_\lambda(V^\rho) + \text{cyclic}(\sigma, \mu, \nu) = 0,$$

$$\Rightarrow \quad \nabla_\sigma\Big(R^\rho{}_{\lambda\mu\nu}\Big)V^\lambda - R_{\sigma\lambda\mu\nu}\nabla^\lambda(V^\rho) + \text{cyclic}(\sigma,\mu,\nu) = 0,$$
$$\Rightarrow \quad \nabla_\sigma\Big(R^\rho{}_{\lambda\mu\nu}\Big)V^\lambda + R_{\lambda\sigma\mu\nu}\nabla^\lambda(V^\rho) + \text{cyclic}(\sigma,\mu,\nu) = 0.$$

Now we can cluster the terms with V^λ and $\nabla^\lambda(V^\rho)$ separately:

$$\Big\{\nabla_\sigma\Big(R^\rho{}_{\lambda\mu\nu}\Big) + \nabla_\mu(R^\rho{}_{\lambda\nu\sigma}) + \nabla_\nu\Big(R^\rho{}_{\lambda\sigma\mu}\Big)\Big\}V^\lambda + \Big\{R_{\lambda\sigma\mu\nu} + \text{cyclic}(\sigma,\mu,\nu)\Big\}\nabla^\lambda(V^\rho) = 0.$$

The term in the second curly brackets vanishes due to the 1st Bianchi identity. Since this must hold for any vector field V^ρ, thus the term in the first curly brackets must also vanish. Thus, we have

$$\nabla_\sigma(R^\rho{}_{\lambda\mu\nu}) + \nabla_\mu(R^\rho{}_{\lambda\nu\sigma}) + \nabla_\nu(R^\rho{}_{\lambda\sigma\mu}) = 0. \tag{16.8}$$

This is known as the second Bianchi identity (or differential Bianchi identity). Since the metric tensor is covariantly constant, $\nabla_\alpha g_{\mu\nu} = 0$, thus we can lower the first index of the Riemann tensor in the above equation to get

$$\nabla_\sigma(R_{\rho\lambda\mu\nu}) + \nabla_\mu(R_{\rho\lambda\nu\sigma}) + \nabla_\nu(R_{\rho\lambda\sigma\mu}) = 0. \tag{16.9}$$

§16.4 Einstein Tensor

Contract the 2nd Bianchi identity with $g^{\rho\mu}$, we have

$$\nabla_\sigma(R_{\lambda\nu}) + \nabla^\rho(R_{\rho\lambda\nu\sigma}) - \nabla_\nu(R_{\lambda\sigma}) = 0.$$

Contracting further with $g^{\lambda\sigma}$, we have

$$\nabla^\lambda(R_{\lambda\nu}) + \nabla^\rho(R_{\rho\nu}) - \nabla_\nu(R) = 0.$$

In the first two terms, the dummy indices λ and ρ can be replaced with μ, and in the second term, we can raise the index ν to get

$$2\nabla^\mu(R_{\mu\nu}) - g_{\nu\mu}\nabla^\mu(R) = 0.$$

Rearranging, we have

$$\nabla^\mu\Big(R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R\Big) = 0.$$

Define the tensor in the parentheses as the Einstein tensor:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R. \tag{16.10}$$

Thus, the Einstein tensor is divergence-free:

$$\nabla^\mu G_{\mu\nu} = 0. \tag{16.11}$$

Lec 17: Einstein's Field Equations

We have argued that the action governing the dynamics of the metric tensor should be the Einstein-Hilbert action

$$S_{\mathrm{EH}} = \int \mathrm{d}^4x \sqrt{-g}R, \tag{17.1}$$

without the cosmological constant term. The total action of the universe is then given by

$$S = S_{\mathrm{EH}} + S_{\mathrm{M}}, \tag{17.2}$$

where S_{M} is the action for the matter fields, it contains all the matter fields and their interactions, like the Electromagnetic field, Dirac field, etc. So we have

$$S[g_{\mu\nu}, \psi] = \int \mathrm{d}^4x \sqrt{-g}R + S_{\mathrm{M}}[g_{\mu\nu}, \psi], \tag{17.3}$$

where ψ represents all the matter fields. The equations of motion for the metric tensor are obtained by varying the action with respect to the metric tensor and setting it to zero, *i.e.*,

$$\delta S = \delta S_{\mathrm{EH}} + \delta S_{\mathrm{M}} = 0. \tag{17.4}$$

Assignment 06.1.
Show that the variation of the determinant of the metric tensor under an arbitrary variation of the metric tensor is given by

$$\delta g = g g^{\mu\nu}\, \delta g_{\mu\nu} = -g g_{\mu\nu}\, \delta g^{\mu\nu}\,. \tag{17.5}$$

Solution.
Recall Jacobi's formula eqn. (B.5) from section B.1, which states that for a square invertible matrix M,

$$\delta(\det(M)) = \det(M)\, \mathrm{tr}\Big(M^{-1}\, \delta M\Big).$$

Applying this to the metric tensor $g_{\mu\nu}$, we have

$$\delta g = g\, \mathrm{tr}(g^{\mu\sigma}\, \delta g_{\sigma\nu}) = g g^{\mu\nu}\, \delta g_{\mu\nu}\,.$$

To show the second equality, we start from the identity

$$g^{\mu\sigma} g_{\sigma\nu} = \delta^{\mu}{}_{\nu}.$$

Varying both sides, we have

$$\delta(g^{\mu\sigma} g_{\sigma\nu}) = \delta(\delta^{\mu}{}_{\nu}) = 0 \quad \Rightarrow \quad g^{\mu\sigma}(\delta g_{\sigma\nu}) + (\delta g^{\mu\sigma}) g_{\sigma\nu} = 0.$$

Take the trace on both sides, we have

$$g^{\mu\nu}\, \delta g_{\mu\nu} + g_{\mu\nu}\, \delta g^{\mu\nu} = 0 \quad \Rightarrow \quad g^{\mu\nu}\, \delta g_{\mu\nu} = -g_{\mu\nu}\, \delta g^{\mu\nu}\,.$$

So we have

$$\delta g = g g^{\mu\nu}\, \delta g_{\mu\nu} = -g g_{\mu\nu}\, \delta g^{\mu\nu}\,.$$

◀

Assignment 06.2.
Show that

$$\delta(\sqrt{-g}) = -\frac{1}{2}\sqrt{-g} g_{\mu\nu}\, \delta g^{\mu\nu}\,. \tag{17.6}$$

Solution.
We have

$$\delta(\sqrt{-g}) = \frac{1}{2\sqrt{-g}}\,\delta(-g) = -\frac{1}{2\sqrt{-g}}\,\delta g\,.$$

Using the result from the previous problem, we have

$$\delta(\sqrt{-g}) = -\frac{1}{2\sqrt{-g}}(-g g_{\mu\nu}\,\delta g^{\mu\nu}) = -\frac{1}{2}\sqrt{-g} g_{\mu\nu}\,\delta g^{\mu\nu}\,.$$

◀

Assignment 06.3.
Show that

$$g^{\mu\nu}\,\delta R_{\mu\nu} = \nabla_\mu A^\mu, \tag{17.7}$$

where

$$A^\mu = g^{\alpha\beta}\,\delta\Gamma^\mu_{\alpha\beta} - g^{\mu\beta}\,\delta\Gamma^\alpha_{\alpha\beta}\,.$$

Solution.
Recall the expression for the Ricci tensor in terms of the Christoffel symbols,

$$R_{\mu\nu} = R^\alpha{}_{\mu\alpha\nu} = \partial_\alpha\Gamma^\alpha_{\mu\nu} - \partial_\nu\Gamma^\alpha_{\mu\alpha} + \Gamma^\alpha_{\alpha\lambda}\Gamma^\lambda_{\mu\nu} - \Gamma^\alpha_{\nu\lambda}\Gamma^\lambda_{\mu\alpha}.$$

Varying both sides, we have

$$\delta R_{\mu\nu} = \partial_\alpha\,\delta\Gamma^\alpha_{\mu\nu} - \partial_\nu\,\delta\Gamma^\alpha_{\mu\alpha} + \delta\Gamma^\alpha_{\alpha\lambda}\,\Gamma^\lambda_{\mu\nu} + \Gamma^\alpha_{\alpha\lambda}\,\delta\Gamma^\lambda_{\mu\nu} - \delta\Gamma^\alpha_{\nu\lambda}\,\Gamma^\lambda_{\mu\alpha} - \Gamma^\alpha_{\nu\lambda}\,\delta\Gamma^\lambda_{\mu\alpha}\,.$$

Add and subtract the terms $\Gamma^\lambda_{\nu\alpha}\,\delta\Gamma^\alpha_{\mu\lambda}$, we have

$$\begin{aligned}\delta R_{\mu\nu} = {}& \partial_\alpha\,\delta\Gamma^\alpha_{\mu\nu} + \Gamma^\alpha_{\alpha\lambda}\,\delta\Gamma^\lambda_{\mu\nu} - \Gamma^\lambda_{\alpha\mu}\,\delta\Gamma^\alpha_{\lambda\nu} - \Gamma^\lambda_{\alpha\nu}\,\delta\Gamma^\alpha_{\lambda\mu} \\ & - \partial_\nu\,\delta\Gamma^\alpha_{\mu\alpha} - \Gamma^\alpha_{\nu\lambda}\,\delta\Gamma^\lambda_{\mu\alpha} + \Gamma^\lambda_{\nu\mu}\,\delta\Gamma^\alpha_{\lambda\alpha} + \Gamma^\lambda_{\nu\alpha}\,\delta\Gamma^\alpha_{\mu\lambda}.\end{aligned}$$

The terms in red are extra terms that we added and subtracted. Notice that the terms in separate lines can be written as covariant derivatives, we have

$$\delta R_{\mu\nu} = \nabla_\alpha\,\delta\Gamma^\alpha_{\mu\nu} - \nabla_\nu\,\delta\Gamma^\alpha_{\mu\alpha}\,.$$

Contracting with $g^{\mu\nu}$, we have

$$g^{\mu\nu}\,\delta R_{\mu\nu} = g^{\mu\nu}\nabla_\alpha\,\delta\Gamma^\alpha_{\mu\nu} - g^{\mu\nu}\nabla_\nu\,\delta\Gamma^\alpha_{\mu\alpha}\,.$$

Using the metric compatibility condition $\nabla_\alpha g^{\mu\nu} = 0$, we have

$$g^{\mu\nu}\,\delta R_{\mu\nu} = \underbrace{\nabla_\alpha(g^{\mu\nu}\,\delta\Gamma^\alpha_{\mu\nu})}_{\substack{\alpha\leftrightarrow\mu\\ \nu\to\beta}} - \underbrace{\nabla_\nu(g^{\mu\nu}\,\delta\Gamma^\alpha_{\mu\alpha})}_{\substack{\mu\to\beta\\ \nu\to\mu}},$$

which can be written as

$$g^{\mu\nu}\,\delta R_{\mu\nu} = \nabla_\mu A^\mu,$$

where

$$A^\mu = g^{\alpha\beta}\,\delta\Gamma^\mu_{\alpha\beta} - g^{\mu\beta}\,\delta\Gamma^\alpha_{\alpha\beta}\,.$$

◀

Using the results from the above problems, we can now vary the full action, we have

$$\delta S = 0 \quad \Rightarrow \quad \int \mathrm{d}^4x\left[\delta(\sqrt{-g})\,R + \sqrt{-g}(\delta g^{\mu\nu})R_{\mu\nu} + \sqrt{-g}g^{\mu\nu}\,\delta R_{\mu\nu}\right] + 16\pi G\,\delta S_{\mathrm{M}} = 0.$$

Substituting the results from the above problems, we have

$$\delta S = \int \mathrm{d}^4x\,\sqrt{-g}\left[-\frac{1}{2}g_{\mu\nu}R + R_{\mu\nu}\right]\delta g^{\mu\nu} + \int \mathrm{d}^4x\,\sqrt{-g}(\nabla_\mu A^\mu) + 16\pi G\,\delta S_{\mathrm{M}} = 0.$$

The second term is a total derivative, using the divergence theorem, it can be converted to a surface integral at infinity, which vanishes assuming appropriate boundary conditions. So we have

$$\delta S = \int \mathrm{d}^4x\, \sqrt{-g} G_{\mu\nu}\, \delta g^{\mu\nu} + 16\pi G\, \delta S_{\mathrm{M}} = 0.$$

Defining the *energy-momentum tensor* (or *stress-energy tensor*) as

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta S_{\mathrm{M}}}{\delta g^{\mu\nu}}, \tag{17.8}$$

we have the variation of the matter action as

$$\delta S_{\mathrm{M}} = -\frac{1}{2} \int \mathrm{d}^4x\, \sqrt{-g} T_{\mu\nu}\, \delta g^{\mu\nu}\,.$$

Substituting this back, we have

$$\delta S = \int \mathrm{d}^4x\, \sqrt{-g} \Big[G_{\mu\nu} - 8\pi G T_{\mu\nu} \Big]\, \delta g^{\mu\nu} = 0.$$

Since $\delta g^{\mu\nu}$ is arbitrary, the integrand must vanish, we have the Einstein's field equations

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = 8\pi G T_{\mu\nu}\,. \tag{17.9}$$

Recall that the Einstein tensor is covariantly conserved, $\nabla^\mu G_{\mu\nu} = 0$, (see section 16.4) which implies that the energy-momentum tensor is also covariantly conserved, $\nabla^\mu T_{\mu\nu} = 0$. We will see later that this is "equivalent" to the local conservation of energy and momentum in curved spacetime.

In the Einstein-Hilbert action, we had not justified the normalization constant $16\pi G$ in front of the matter action. This is fixed by requiring that in the non-relativistic limit, the Einstein's field equations reduces to the Poisson equation of Newtonian gravity, $\nabla^2\Phi = 4\pi G\rho$, where Φ is the Newtonian gravitational potential and ρ is the mass density. We will see this in detail later.

In the derivation of the Einstein's field equations, we had not included the cosmological constant term in the Einstein-Hilbert action. If we include it, the equations of motion becomes

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}\,. \tag{17.10}$$

The cosmological constant term does not affect the conservation of the energy-momentum tensor.

First thing we should ask is that, can we recover Newtonian's law of gravity in an appropriate domain?

§17.1 Motion of Planets around a massive Star

Consider a massive star of mass M at the origin, and a planet of mass m orbiting around it ($M \gg m$). In Newtonian gravity, the equation of motion of the planet is given by

$$m\ddot{\vec{\mathbf{r}}} = -\frac{GMm}{r^2}\hat{\mathbf{r}}, \tag{17.11}$$

which can be written in terms of the gravitational potential $\Phi = -GM/r$ as

$$\ddot{\vec{\mathbf{r}}} = -\vec{\boldsymbol{\nabla}}\Phi. \tag{17.12}$$

Just from these equations, we can observe following properties of the Newtonian gravity:

- Time Reversal Symmetry: The equations of motion are invariant under time reversal $t \to -t$.
- Spherical Symmetry: The gravitational potential is spherically symmetric, $\Phi = \Phi(r)$.
- Time Independence (Static Field): The gravitational potential is time independent, $\partial_t \Phi = 0$.

- Motion in Matter Free Region: The planet is moving in a region where there is no matter, $\rho = 0$, so the gravitational potential satisfies the Laplace equation, $\nabla^2\Phi = 0$. In terms of stress-energy tensor, this means that $T_{\mu\nu} = 0$ in that region.

So we should look for a solution of the Einstein's field equations which satisfies these properties. Precisely, we are looking for a *vacuum solution* of the Einstein's field equations, which is *spherically symmetric*, *static*, and *time reversal symmetric*.

From the description of the properties, it is evident that spherical polar coordinates (t, r, θ, ϕ) are the most natural choice of coordinates to describe the spacetime around the star. The most general metric is given by

$$\begin{aligned} \mathrm{d}s^2 = g_{tt}\,\mathrm{d}t^2 + 2g_{tr}\,\mathrm{d}t\,\mathrm{d}r + 2g_{t\theta}\,\mathrm{d}t\,\mathrm{d}\theta + 2g_{t\phi}\,\mathrm{d}t\,\mathrm{d}\phi \\ + g_{rr}\,\mathrm{d}r^2 + 2g_{r\theta}\,\mathrm{d}r\,\mathrm{d}\theta + 2g_{r\phi}\,\mathrm{d}r\,\mathrm{d}\phi \\ + g_{\theta\theta}\,\mathrm{d}\theta^2 + 2g_{\theta\phi}\,\mathrm{d}\theta\,\mathrm{d}\phi + g_{\phi\phi}\,\mathrm{d}\phi^2\,. \end{aligned}$$

Now we will impose the symmetries one by one to simplify the metric. First, spherical symmetry, till now we haven't studied how to impose spherical symmetry on a metric, but intuitively, for a fixed radius r and time t, the metric should reduce to the metric on a 2-sphere, which is given by

$$r^2\,\mathrm{d}\Omega^2 = r^2(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2).$$

So we must have

$$g_{\theta\theta} = r^2, \quad g_{\phi\phi} = r^2\sin^2\theta, \quad g_{\theta\phi} = 0.$$

As well as it should be independent of orientation of (θ, ϕ), *i.e.*, it should be invariant under $\theta \to -\theta$ and $\phi \to -\phi$. This implies that

$$g_{t\theta} = 0, \quad g_{t\phi} = 0, \quad g_{r\theta} = 0, \quad g_{r\phi} = 0.$$

So the metric reduces to

$$\mathrm{d}s^2 = g_{tt}\,\mathrm{d}t^2 + 2g_{tr}\,\mathrm{d}t\,\mathrm{d}r + g_{rr}\,\mathrm{d}r^2 + r^2\,\mathrm{d}\Omega^2\,.$$

Next, time independence, this implies that the metric components should be independent of time t, so

$$g_{tt} = g_{tt}(r) = f(r), \qquad g_{rr} = g_{rr}(r) = h(r), \qquad g_{tr} = g_{tr}(r).$$

Finally, time reversal symmetry, this implies that the metric should be invariant under $t \to -t$. Under this transformation, $\mathrm{d}t^2$ is invariant, but $\mathrm{d}t\,\mathrm{d}r \to -\,\mathrm{d}t\,\mathrm{d}r$, so for the metric to be invariant, we must have

$$g_{tr} = 0.$$

So the metric reduces to

$$\mathrm{d}s^2 = f(r)\,\mathrm{d}t^2 + h(r)\,\mathrm{d}r^2 + r^2\,\mathrm{d}\Omega^2\,.$$

This is the most general form of a spherically symmetric, static, and time reversal symmetric metric. Now we need to solve the vacuum Einstein's field equations, $R_{\mu\nu} = 0$, for this metric to find the functions $f(r)$ and $h(r)$. This solution is known as the *Schwarzschild solution*, which we will study in detail in the next lecture.

Lec 18: Christoffel Symbols of Spherically Symmetric Static Metric

In the last lecture, we have derived the most general form of a spherically symmetric, static, and time reversal symmetric metric. The metric is given by

$$\mathrm{d}s^2 = f(r)\,\mathrm{d}t^2 + h(r)\,\mathrm{d}r^2 + r^2\,\mathrm{d}\Omega^2\,,$$

where $f(r)$ and $h(r)$ are two arbitrary functions of the radial coordinate r, and $\mathrm{d}\Omega^2 = \mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2$ is the metric on a unit 2-sphere.

If a metric is time-independent, *i.e.*, $\partial_t g_{\mu\nu} = 0$, then the metric is said to be *stationary*. If a metric is stationary and also invariant under time reversal, *i.e.*, $t \to -t$, then the metric is said to be *static*.

Theorem 18.1 (Birkhoff-Jebsen's Theorem)
Any spherically symmetric solution of the vacuum Einstein's field equations must be static and asymptotically flat.

Converse of this theorem is also true, and it is known as the *Israel's Theorem*.

Theorem 18.2 (Israel's Theorem)
Any static and asymptotically flat solution of the vacuum Einstein's field equations that has a regular event horizon must be spherically symmetric.

So these two theorems together imply the uniqueness of the Schwarzschild solution in vacuum.

To fix the functions $f(r)$ and $h(r)$, we need to solve the vacuum Einstein's field equations, *i.e.*, $R_{\mu\nu} = 0$. To do this, we first need to compute the Christoffel symbols for the given metric.

§18.1 Christoffel Symbols for Schwarzschild-like Metric

Before we compute the Christoffel symbols, let us write down the metric components explicitly.

$$g_{\mu\nu} = \begin{pmatrix} f(r) & 0 & 0 & 0 \\ 0 & h(r) & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2\sin^2\theta \end{pmatrix}; \qquad g^{\mu\nu} = \begin{pmatrix} \frac{1}{f(r)} & 0 & 0 & 0 \\ 0 & \frac{1}{h(r)} & 0 & 0 \\ 0 & 0 & \frac{1}{r^2} & 0 \\ 0 & 0 & 0 & \frac{1}{r^2\sin^2\theta} \end{pmatrix}.$$

Let's introduce the comma and semicolon notation for partial and covariant derivatives, as follows:

$$A_{\mu,\nu} \coloneqq \partial_\nu A_\mu = \frac{\partial A_\mu}{\partial x^\nu} \qquad A_{\mu;\nu} \coloneqq \nabla_\nu A_\mu. \tag{18.1}$$

Since $f(r)$ and $h(r)$ are functions of r only, we will use the following notation for their derivatives:

$$f'(r) \coloneqq \frac{\mathrm{d}f(r)}{\mathrm{d}r} \qquad h'(r) \coloneqq \frac{\mathrm{d}h(r)}{\mathrm{d}r}.$$

Recall that the Christoffel symbols are given by

$$\Gamma^\mu_{\alpha\beta} = \frac{1}{2}g^{\mu\rho}\Big[g_{\rho\beta,\alpha} + g_{\alpha\rho,\beta} - g_{\alpha\beta,\rho}\Big].$$

Since the Christoffel symbols are symmetric in the lower two indices, we have a total of 40 independent components. However, many of these components are zero due to the diagonal nature of the metric and the fact that metric is independent of t and ϕ.

- $\mu = t$: So the formula reduces to

$$\Gamma^t_{\alpha\beta} = \frac{1}{2}g^{tt}\Big[g_{t\beta,\alpha} + g_{\alpha t,\beta} - g_{\alpha\beta,t}\Big] = \frac{1}{2f(r)}\Big[g_{t\beta,\alpha} + g_{\alpha t,\beta}\Big],$$

since the metric is independent of t. Now, since the metric is diagonal, the only non-zero components are when either $\alpha = t$ or $\beta = t$. Thus, we have

$$\Gamma^t_{tr} = \Gamma^t_{rt} = \frac{f'(r)}{2f(r)} \qquad \Gamma^t_{t\theta} = \Gamma^t_{\theta t} = 0 \qquad \Gamma^t_{t\phi} = \Gamma^t_{\phi t} = 0.$$

All other components are zero.

$\alpha\backslash\beta$	t	r	θ	ϕ
t	0	$\frac{f'(r)}{2f(r)}$	0	0
r	$\frac{f'(r)}{2f(r)}$	0	0	0
θ	0	0	0	0
ϕ	0	0	0	0

Table 18.1: Christoffel symbols of form $\Gamma^t_{\alpha\beta}$.

- $\mu = r$: The formula reduces to

$$\Gamma^r_{\alpha\beta} = \frac{1}{2}g^{rr}\Big[g_{r\beta,\alpha} + g_{\alpha r,\beta} - g_{\alpha\beta,r}\Big] = \frac{1}{2h(r)}\Big[g_{r\beta,\alpha} + g_{\alpha r,\beta} - g_{\alpha\beta,r}\Big].$$

Since the metric is diagonal, the only non-zero components are when either $\alpha = r$ or $\beta = r$ or $\alpha = \beta$. Thus, we have

$$\Gamma^r_{tt} = \frac{f'(r)}{2h(r)} \qquad \Gamma^r_{rr} = \frac{h'(r)}{2h(r)} \qquad \Gamma^r_{\theta\theta} = -\frac{r}{h(r)} \qquad \Gamma^r_{\phi\phi} = -\frac{r\sin^2\theta}{h(r)}.$$

All other components are zero.

$\alpha\backslash\beta$	t	r	θ	ϕ
t	$-\frac{f'(r)}{2h(r)}$	0	0	0
r	0	$\frac{h'(r)}{2h(r)}$	0	0
θ	0	0	$-\frac{r}{h(r)}$	0
ϕ	0	0	0	$-\frac{r\sin^2\theta}{h(r)}$

Table 18.2: Christoffel symbols of form $\Gamma^r_{\alpha\beta}$.

- $\mu = \theta$: The formula reduces to

$$\Gamma^\theta_{\alpha\beta} = \frac{1}{2}g^{\theta\theta}\Big[g_{\theta\beta,\alpha} + g_{\alpha\theta,\beta} - g_{\alpha\beta,\theta}\Big] = \frac{1}{2r^2}\Big[g_{\theta\beta,\alpha} + g_{\alpha\theta,\beta} - g_{\alpha\beta,\theta}\Big].$$

From the symmetry of the metric, we see that the only non-zero components are when either $\alpha = \theta, \beta = r$ or $\alpha = r, \beta = \theta$ or $\alpha = \beta = \phi$. Thus, we have

$$\Gamma^\theta_{\theta r} = \Gamma^\theta_{r\theta} = \frac{1}{r} \qquad \Gamma^\theta_{\phi\phi} = -\sin\theta\cos\theta.$$

$\alpha\backslash\beta$	t	r	θ	ϕ
t	0	0	0	0
r	0	0	r^{-1}	0
θ	0	r^{-1}	0	$-\sin\theta\cos\theta$
ϕ	0	0	0	0

Table 18.3: Christoffel symbols of form $\Gamma^{\theta}_{\alpha\beta}$.

- $\mu = \phi$: The formula reduces to

$$\Gamma^{\phi}_{\alpha\beta} = \frac{1}{2}g^{\phi\phi}\Big[g_{\phi\beta,\alpha} + g_{\alpha\phi,\beta} - g_{\alpha\beta,\phi}\Big] = \frac{1}{2r^2\sin^2\theta}\Big[g_{\phi\beta,\alpha} + g_{\alpha\phi,\beta}\Big].$$

  From the symmetry of the metric, we see that the only non-zero components are when either $\alpha = \phi, \beta = r$ or $\alpha = \phi, \beta = \theta$ or $\alpha = r, \beta = \phi$ or $\alpha = \theta, \beta = \phi$. Thus, we have

$$\Gamma^{\phi}_{\phi r} = \Gamma^{\phi}_{r\phi} = \frac{1}{r} \qquad\qquad \Gamma^{\phi}_{\phi\theta} = \Gamma^{\phi}_{\theta\phi} = \cot\theta.$$

$\alpha\backslash\beta$	t	r	θ	ϕ
t	0	0	0	0
r	0	0	0	r^{-1}
θ	0	0	0	$\cot\theta$
ϕ	0	r^{-1}	$\cot\theta$	0

Table 18.4: Christoffel symbols of form $\Gamma^{\phi}_{\alpha\beta}$.

Thus, the non-zero Christoffel symbols for the spherically symmetric static metric are:

$$\Gamma^{t}_{tr} = \frac{f'(r)}{2f(r)} = \Gamma^{t}_{rt}, \qquad \Gamma^{r}_{tt} = \frac{f'(r)}{2h(r)}, \qquad \Gamma^{r}_{rr} = \frac{h'(r)}{2h(r)},$$
$$\Gamma^{r}_{\theta\theta} = -\frac{r}{h(r)}, \qquad \Gamma^{r}_{\phi\phi} = -\frac{r\sin^2\theta}{h(r)}, \qquad \Gamma^{\theta}_{\theta r} = r^{-1} = \Gamma^{\theta}_{r\theta},$$
$$\Gamma^{\theta}_{\phi\phi} = -\sin\theta\cos\theta, \qquad \Gamma^{\phi}_{\phi r} = r^{-1} = \Gamma^{\phi}_{r\phi}, \qquad \Gamma^{\phi}_{\phi\theta} = \cot\theta = \Gamma^{\phi}_{\theta\phi}.$$

In the next lecture, we will use these Christoffel symbols to compute the Ricci tensor components and solve the vacuum Einstein's field equations to determine the functions $f(r)$ and $h(r)$.

Lec 19: Ricci Tensor for Spherically Symmetric Static Metric

In the previous lecture, we computed all the Christoffel symbols for the Schwarzschild metric. Today, we will use these Christoffel symbols to compute the Ricci tensor components and then solve the vacuum Einstein equations to find the arbitrary functions in the metric.

The Ricci tensor is given by:

$$R_{\mu\nu} = R^{\rho}{}_{\mu\rho\nu} = \Gamma^{\rho}_{\mu\nu,\rho} - \Gamma^{\rho}_{\mu\rho,\nu} + \Gamma^{\rho}_{\rho\sigma}\Gamma^{\sigma}_{\mu\nu} - \Gamma^{\rho}_{\nu\sigma}\Gamma^{\sigma}_{\mu\rho}. \tag{19.1}$$

We will compute each component of the Ricci tensor step by step. Let's start with the $R_{00} = R_{tt}$ component.

$$R_{tt} = R^{\rho}{}_{t\rho t} = R^{r}{}_{trt} + R^{\theta}{}_{t\theta t} + R^{\phi}{}_{t\phi t}$$

since $\Gamma^{\mu}_{\alpha\beta,t} = 0$ for all μ, α, β, we have

$$\begin{aligned} &= \Gamma^{r}_{tt,r} + \Gamma^{r}_{r\sigma}\Gamma^{\sigma}_{tt} - \Gamma^{r}_{t\sigma}\Gamma^{\sigma}_{tr} \\ &\quad + \Gamma^{\theta}_{tt,\theta} + \Gamma^{\theta}_{\theta\sigma}\Gamma^{\sigma}_{tt} - \Gamma^{\theta}_{t\sigma}\Gamma^{\sigma}_{t\theta} \\ &\quad + \Gamma^{\phi}_{tt,\phi} + \Gamma^{\phi}_{\phi\sigma}\Gamma^{\sigma}_{tt} - \Gamma^{\phi}_{t\sigma}\Gamma^{\sigma}_{t\phi} \end{aligned}$$

since $\Gamma^{\theta}_{tt}, \Gamma^{\phi}_{tt}$ are independent of t, and $\Gamma^{\theta}_{t\sigma} = 0 = \Gamma^{\phi}_{t\sigma}$ for all σ, we have

$$\begin{aligned} &= \Gamma^{r}_{tt,r} + \Gamma^{r}_{r\sigma}\Gamma^{\sigma}_{tt} - \Gamma^{r}_{t\sigma}\Gamma^{\sigma}_{tr} \\ &\quad + \Gamma^{\theta}_{\theta\sigma}\Gamma^{\sigma}_{tt} + \Gamma^{\phi}_{\phi\sigma}\Gamma^{\sigma}_{tt} \end{aligned}$$

now, each sum over σ only has one non-zero term, so we have

$$\begin{aligned} &= \Gamma^{r}_{tt,r} + \Gamma^{r}_{rr}\Gamma^{r}_{tt} - \Gamma^{r}_{tt}\Gamma^{t}_{tr} \\ &\quad + \Gamma^{\theta}_{\theta r}\Gamma^{r}_{tt} + \Gamma^{\phi}_{\phi r}\Gamma^{r}_{tt} \end{aligned}$$

substituting the values of the Christoffel symbols, we get

$$\begin{aligned} &= \left(-\frac{f'}{2h}\right)' + \left(\frac{h'}{2h}\right)\left(-\frac{f'}{2h}\right) - \left(-\frac{f'}{2h}\right)\left(\frac{f'}{2f}\right) \\ &\quad + \left(\frac{1}{r}\right)\left(-\frac{f'}{2h}\right) + \left(\frac{1}{r}\right)\left(-\frac{f'}{2h}\right) \\ &= -\frac{f''}{2h} + \frac{f'h'}{2h^2} - \frac{f'h'}{4h^2} + \frac{(f')^2}{4fh} - \frac{f'}{rh} \\ &= -\frac{f''}{2h} + \frac{f'h'}{4h^2} + \frac{(f')^2}{4fh} - \frac{f'}{rh} \\ &= -\frac{1}{2h}\left[f'' + \frac{2f'}{r} - \frac{f'h'}{2h} - \frac{(f')^2}{2f}\right]. \end{aligned} \tag{19.2}$$

We can see that to compute R_{tt}, we had to do so many steps. So, to avoid repetition, we will directly write down the expressions for the other components of the Ricci tensor [1]:

$$R_{rr} = -\frac{1}{2f}\left[f'' - \frac{(f')^2}{2f} - \frac{f'h'}{2h} + \frac{2fh'}{rh}\right], \tag{19.3}$$

$$R_{\theta\theta} = -\frac{r}{h^2 f}(fh)' - \left(\frac{r}{h}\right)' + 1, \tag{19.4}$$

$$R_{\phi\phi} = \sin^2\theta R_{\theta\theta}, \tag{19.5}$$

all other components (off-diagonal) are zero. Now, let's look at the vacuum Einstein equations:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 0.$$

[1] However, we can compute these components using any symbolic computation software, *e.g.*, `Mathmatica` (see section B.2).

Contracting both sides with $g^{\mu\nu}$, we get:

$$g^{\mu\nu}R_{\mu\nu} - \frac{1}{2}g^{\mu\nu}g_{\mu\nu}R = 0 \quad \Rightarrow \quad R - 2R = 0 \quad \Rightarrow \quad R = 0.$$

Since $R = g^{\mu\nu}R_{\mu\nu}$, the vacuum Einstein equations reduce to:

$$R_{\mu\nu} = 0. \tag{19.6}$$

Since off-diagonal components of the Ricci tensor are already zero, and $R_{\phi\phi} = \sin^2\theta R_{\theta\theta}$, we only need to solve the following three equations:

$$R_{tt} = 0, \qquad R_{rr} = 0, \qquad R_{\theta\theta} = 0.$$

To solve these equations, we will first manipulate R_{tt} and R_{rr} to get a simpler form. Multiplying eqn. (19.2) by h and eqn. (19.3) by f, and then subtracting the two equations ($h \times R_{tt} - f \times R_{rr} = 0$), we get:

$$\frac{2fh'}{rh} + \frac{2f'}{r} = 0 \quad \Rightarrow \quad \frac{h'}{h} + \frac{f'}{f} = 0 \quad \Rightarrow \quad (fh)' = 0. \tag{19.7}$$

Integrating, we get:

$$f(r)h(r) = c_1, \tag{19.8}$$

where c_1 is an integration constant. In the asymptotic limit $r \to \infty$, we expect the metric to be flat (*i.e.*, $g_{tt} \to -1$ and $g_{rr} \to 1$), so we have

$$\lim_{r\to\infty} f(r) = -1, \quad \lim_{r\to\infty} h(r) = 1 \quad \Rightarrow \quad \lim_{r\to\infty} f(r)h(r) = -1. \tag{19.9}$$

Thus, we get $c_1 = -1$, and hence:

$$f(r)h(r) = -1 \quad \Rightarrow \quad h(r) = -\frac{1}{f(r)}. \tag{19.10}$$

Now, substituting this result into eqn. (19.4) and setting $R_{\theta\theta} = 0$, we get:

$$0 - \left(\frac{r}{h}\right)' + 1 = 0 \quad \Rightarrow \quad \left(\frac{r}{h}\right)' = 1 \quad \Rightarrow \quad \frac{r}{h} = r + c_2,$$

where c_2 is another integration constant. Set $c_2 = -r_s$, where r_s is a constant with dimensions of length. Thus, we have:

$$\frac{1}{h(r)} = 1 - \frac{r_s}{r} = -f(r) \quad \Rightarrow \quad f(r) = -\left(1 - \frac{r_s}{r}\right), \quad h(r) = \left(1 - \frac{r_s}{r}\right)^{-1}. \tag{19.11}$$

Thus, the metric for a spherically symmetric static vacuum spacetime is given by:

$$\mathrm{d}s^2 = -\left(1 - \frac{r_s}{r}\right)\mathrm{d}t^2 + \left(1 - \frac{r_s}{r}\right)^{-1}\mathrm{d}r^2 + r^2\,\mathrm{d}\Omega^2\,, \tag{19.12}$$

where r_s is called the Schwarzschild radius. This is the Schwarzschild metric, which is the unique spherically symmetric static vacuum solution of Einstein's equations.

§19.1 Christoffel Symbols of Schwarzschild Metric

Using the expressions for $f(r)$ and $h(r)$, let's write down the non-zero Christoffel symbols for the Schwarzschild metric:

$$\Gamma^t_{tr} = \frac{r_s}{2r(r-r_s)} = \Gamma^t_{rt}, \qquad \Gamma^r_{tt} = \frac{r_s}{2r^3}(r - r_s), \qquad \Gamma^r_{rr} = -\frac{r_s}{2r(r-r_s)},$$

$$\Gamma^r_{\theta\theta} = -(r - r_s), \qquad \Gamma^r_{\phi\phi} = -(r - r_s)\sin^2\theta, \qquad \Gamma^\theta_{r\theta} = \frac{1}{r} = \Gamma^\theta_{\theta r},$$

$$\Gamma^\theta_{\phi\phi} = -\sin\theta\cos\theta, \qquad \Gamma^\phi_{r\phi} = \frac{1}{r} = \Gamma^\phi_{\phi r}, \qquad \Gamma^\phi_{\theta\phi} = \cot\theta = \Gamma^\phi_{\phi\theta}.$$

These Christoffel symbols will be useful in our future calculations involving the Schwarzschild metric.

§19.2 Geodesics in Schwarzschild Spacetime

The geodesic equation is given by:

$$\frac{\mathrm{d}^2 x^\mu}{\mathrm{d}\tau^2} + \Gamma^\mu_{\alpha\beta} \frac{\mathrm{d}x^\alpha}{\mathrm{d}\tau} \frac{\mathrm{d}x^\beta}{\mathrm{d}\tau} = 0, \tag{19.13}$$

where τ is the proper time along the geodesic. Let's write down the geodesic equations for each coordinate in the Schwarzschild spacetime.

$$\mu = t: \quad \frac{\mathrm{d}^2 t}{\mathrm{d}\tau^2} + \frac{r_s}{r} \frac{1}{r - r_s} \frac{\mathrm{d}t}{\mathrm{d}\tau} \frac{\mathrm{d}r}{\mathrm{d}\tau} = 0, \tag{19.14}$$

this is the equation for the time coordinate, which at first glance looks weird. As in Newtonian Gravity, time was the absolute parameter, but here time is itself changing along the geodesic. This equation will be useful in calculating the gravitational time dilation effect and redshift.

Lec 20: Schwarzschild Metric

In the last lecture, we ended with the geodesic equation for time t in Schwarzschild spacetime, now let's write down the geodesic equations for the other coordinates r, θ, and ϕ.

$$\mu = r: \quad \frac{\mathrm{d}^2 r}{\mathrm{d}\tau^2} + \frac{r_s}{2r^3}(r - r_s)\left(\frac{\mathrm{d}t}{\mathrm{d}\tau}\right)^2 - \frac{r_s}{2r}\frac{1}{r - r_s}\left(\frac{\mathrm{d}r}{\mathrm{d}\tau}\right)^2 - (r - r_s)\left[\left(\frac{\mathrm{d}\theta}{\mathrm{d}\tau}\right)^2 + \sin^2\theta\left(\frac{\mathrm{d}\phi}{\mathrm{d}\tau}\right)^2\right] = 0, \tag{20.1}$$

$$\mu = \theta: \quad \frac{\mathrm{d}^2\theta}{\mathrm{d}\tau^2} + \frac{2}{r}\frac{\mathrm{d}r}{\mathrm{d}\tau}\frac{\mathrm{d}\theta}{\mathrm{d}\tau} - \sin\theta\cos\theta\left(\frac{\mathrm{d}\phi}{\mathrm{d}\tau}\right)^2 = 0, \tag{20.2}$$

$$\mu = \phi: \quad \frac{\mathrm{d}^2\phi}{\mathrm{d}\tau^2} + \frac{2}{r}\frac{\mathrm{d}r}{\mathrm{d}\tau}\frac{\mathrm{d}\phi}{\mathrm{d}\tau} + 2\cot\theta\frac{\mathrm{d}\theta}{\mathrm{d}\tau}\frac{\mathrm{d}\phi}{\mathrm{d}\tau} = 0. \tag{20.3}$$

In Newtonian gravity, we know that planetary orbits are confined to a plane. We expect a similar result here.

Consider the initial conditions such that the initial position $\vec{\mathbf{r}}_0$ and initial velocity $\vec{\mathbf{v}}_0$ are not parallel, then define the plane spanned by these two vectors (passing through the origin) and choose our coordinate system such that this plane is the equatorial plane $\theta = \frac{\pi}{2}$. Since the Schwarzschild metric is spherically symmetric, we can always choose such a coordinate system without loss of generality. So we set the initial conditions as

$$\theta(\tau = 0) = \frac{\pi}{2}, \quad \left.\frac{\mathrm{d}\theta}{\mathrm{d}\tau}\right|_{\tau=0} = 0.$$

Thus, at $\tau = 0$, the geodesic equation for θ (20.2) reduces to

$$\left.\frac{\mathrm{d}^2\theta}{\mathrm{d}\tau^2}\right|_{\tau=0} = 0.$$

This implies that θ remains constant at $\frac{\pi}{2}$ for all τ. Therefore, the motion of the particle is confined to the equatorial plane.

Recall the geodesic equation for ϕ, we have

$$\frac{\mathrm{d}^2\phi}{\mathrm{d}\tau^2} + \frac{2}{r}\frac{\mathrm{d}r}{\mathrm{d}\tau}\frac{\mathrm{d}\phi}{\mathrm{d}\tau} = 0$$

multiplying by r^2, we get

$$r^2\frac{\mathrm{d}^2\phi}{\mathrm{d}\tau^2} + 2r\frac{\mathrm{d}r}{\mathrm{d}\tau}\frac{\mathrm{d}\phi}{\mathrm{d}\tau} = 0$$

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\left(r^2\frac{\mathrm{d}\phi}{\mathrm{d}\tau}\right) = 0$$

Integrating, we get

$$r^2\frac{\mathrm{d}\phi}{\mathrm{d}\tau} = \frac{l}{m} = \text{constant}. \tag{20.4}$$

This equation is the relativistic generalization of the conservation of angular momentum. Similar to this, we expect to recover the planetary orbit equation in the Newtonian limit from the geodesic equations in Schwarzschild spacetime. So let's first review the Newtonian planetary orbit equation.

§20.1 Review: Newtonian Planetary Orbits

In Classical Mechanics, action for a particle of mass m moving in a gravitational potential $\Phi_{\mathrm{N}}(r) = -\frac{GM}{r}$ is given by:

$$S = \int \mathrm{d}t\, L = \int \mathrm{d}t \left[\frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\dot{\phi}^2 + \frac{GMm}{r}\right]. \tag{20.5}$$

There is no $\dot{\theta}$ term since we are considering motion in the equatorial plane $\theta = \frac{\pi}{2}$. From the Lagrangian, we can see that ϕ is a cyclic coordinate, so the corresponding conjugate momentum is conserved

$$p_\phi = \frac{\partial L}{\partial \dot{\phi}} = mr^2\dot{\phi} = l = \text{constant}. \tag{20.6}$$

The Euler-Lagrange equation for r gives

$$m\ddot{r} - \frac{\partial}{\partial r}\left(\frac{1}{2}mr^2\dot{\phi}^2 + m\Phi_{\rm N}(r)\right) = 0 \quad \Rightarrow \quad \ddot{r} = -\frac{GM}{r^2} + mr\dot{\phi}^2.$$

Using the conservation of angular momentum, we can write $\dot{\phi} = \frac{l}{mr^2}$. Substituting this into the above equation, we get

$$\ddot{r} = -\frac{GM}{r^2} + \frac{l^2}{m^2r^3}.$$

We can rewrite this equation in more compact form as

$$\ddot{r} = -\frac{\partial \Phi_{\rm N}^{\rm eff}(r)}{\partial r}, \quad \text{where } \Phi_{\rm N}^{\rm eff}(r) = \Phi_{\rm N}(r) + \frac{l^2}{2mr^2} = -\frac{GM}{r} + \frac{l^2}{2mr^2}. \tag{20.7}$$

Let's go back to the geodesic equations in Schwarzschild spacetime and see how we can recover this Newtonian planetary orbit equation in the appropriate limit.

§20.2 Recovering Newtonian Planetary Orbits from Schwarzschild Geodesics

Substituting $\theta = \frac{\pi}{2}$ and $\dot{\phi} = \frac{l}{mr^2}$ into the geodesic equation for r (20.1), we get

$$\frac{\mathrm{d}^2r}{\mathrm{d}\tau^2} + \frac{r_s}{2r^2}\left[\left(1 - \frac{r_s}{r}\right)\left(\frac{\mathrm{d}t}{\mathrm{d}\tau}\right)^2 - \left(1 - \frac{r_s}{r}\right)^{-1}\left(\frac{\mathrm{d}r}{\mathrm{d}\tau}\right)^2\right] - (r - r_s)\frac{l^2}{m^2r^4} = 0.$$

Recall the invariant interval in Schwarzschild spacetime, for motion of a massive particle in the equatorial plane, we have

$$-\,\mathrm{d}\tau^2 = \mathrm{d}s^2 = -\left(1 - \frac{r_s}{r}\right)\mathrm{d}t^2 + \left(1 - \frac{r_s}{r}\right)^{-1}\mathrm{d}r^2 + r^2\,\mathrm{d}\phi^2\,.$$

Dividing by $-\,\mathrm{d}\tau^2$, we get

$$1 = \left(1 - \frac{r_s}{r}\right)\left(\frac{\mathrm{d}t}{\mathrm{d}\tau}\right)^2 - \left(1 - \frac{r_s}{r}\right)^{-1}\left(\frac{\mathrm{d}r}{\mathrm{d}\tau}\right)^2 - r^2\left(\frac{\mathrm{d}\phi}{\mathrm{d}\tau}\right)^2.$$

Substituting this into the geodesic equation for r, we get

$$\frac{\mathrm{d}^2r}{\mathrm{d}\tau^2} + \frac{r_s}{2r^2}\left[1 + r^2\left(\frac{l}{mr^2}\right)^2\right] - (r - r_s)\frac{l^2}{m^2r^4} = 0.$$

Simplifying, we get

$$\frac{\mathrm{d}^2r}{\mathrm{d}\tau^2} = -\frac{r_s}{2r^2} + \frac{l^2}{m^2r^3} - \frac{3r_sl^2}{2m^2r^4}. \tag{20.8}$$

Let's now write this equation in terms of some kind of effective potential. We have

$$\frac{\mathrm{d}^2r}{\mathrm{d}\tau^2} = -\frac{\partial \Phi_{\rm GR}^{\rm eff}(r)}{\partial r}, \quad \text{where } \Phi_{\rm GR}^{\rm eff}(r) = -\frac{r_s}{2r} + \frac{l^2}{2m^2r^2} - \frac{r_sl^2}{2m^2r^3}. \tag{20.9}$$

Rewriting the effective potential as

$$\Phi_{\rm GR}^{\rm eff}(r) = -\frac{r_s}{2r} + \frac{l^2}{2m^2r^2}\left(1 - \frac{r_s}{r}\right). \tag{20.10}$$

With this form of the effective potential, we can clearly see that in the limit $r \gg r_s$, the geodesic equation for r takes the same form as the Newtonian planetary orbit equation with the identification

$r_s = 2GM$. This shows that we can recover the Newtonian planetary orbits from the geodesic equations in Schwarzschild spacetime in the appropriate limit.

With this we have the constant of integration r_s identified in terms of the mass M of the central gravitating body. So we have completely determined the static spherically symmetric vacuum solution of Einstein's equations, which is given by the Schwarzschild metric:

$$\mathrm{d}s^2 = -\left(1 - \frac{2GM}{r}\right)\mathrm{d}t^2 + \left(1 - \frac{2GM}{r}\right)^{-1}\mathrm{d}r^2 + r^2\left(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2\right). \tag{20.11}$$

This metric describes the spacetime outside a static, spherically symmetric mass M. The constant of integration $r_s = 2GM/c^2$ is called the Schwarzschild radius.

Lec 21: Planetary Orbits and the Precession of Mercury

In the last lecture, we have simplified the geodesic equations for r in the Schwarzschild spacetime to:

$$\frac{\mathrm{d}^2 r}{\mathrm{d}\tau^2} = -\frac{GM}{r^2} + \frac{l^2}{m^2 r^3} - \frac{3GMl^2}{m^2 r^4}, \tag{21.1}$$

where $mr^2\dot{\phi} = l$ for some constant l. The first term on the right-hand side is the familiar Newtonian gravitational attraction, the second term is the centrifugal repulsion due to angular momentum, and the third term is a relativistic correction term that arises due to the curvature of spacetime. First two terms are the same as in Newtonian gravity, while the third term is "correction" due to general relativity. This correction term is responsible for the famous perihelion precession of Mercury, which was one of the first experimental confirmations of General Relativity.

Since we want to study the orbits of planets not the motion in time, it is convenient to change the independent variable from τ to ϕ. Let's define $u = 1/r$. Using the chain rule, we have:

$$\frac{\mathrm{d}u}{\mathrm{d}\phi} = \frac{\mathrm{d}u}{\mathrm{d}r}\frac{\mathrm{d}r}{\mathrm{d}\phi} = -\frac{1}{r^2}\frac{\mathrm{d}r}{\mathrm{d}\phi} = -u^2\frac{\mathrm{d}r}{\mathrm{d}\phi}.$$

We have

$$\begin{aligned}\frac{\mathrm{d}^2 r}{\mathrm{d}\tau^2} = \frac{\mathrm{d}}{\mathrm{d}\tau}\left[\frac{\mathrm{d}r}{\mathrm{d}\tau}\right] &= \frac{\mathrm{d}\phi}{\mathrm{d}\tau}\frac{\mathrm{d}}{\mathrm{d}\phi}\left[\frac{\mathrm{d}\phi}{\mathrm{d}\tau}\frac{\mathrm{d}r}{\mathrm{d}\phi}\right] \\ &= \left(\frac{l}{mr^2}\right)\frac{\mathrm{d}}{\mathrm{d}\phi}\left[\left(\frac{l}{mr^2}\right)(-r^2)\frac{\mathrm{d}u}{\mathrm{d}\phi}\right] \\ &= -\frac{l^2}{m^2}u^2\frac{\mathrm{d}^2 u}{\mathrm{d}\phi^2}.\end{aligned}$$

Substituting this into the geodesic equation, we get:

$$-\frac{l^2}{m^2}u^2\frac{\mathrm{d}^2 u}{\mathrm{d}\phi^2} = -GMu^2 + \frac{l^2}{m^2}u^3 - 3GM\frac{l^2}{m^2}u^4.$$

Dividing throughout by $-\frac{l^2}{m^2}u^2$, we get:

$$\frac{\mathrm{d}^2 u}{\mathrm{d}\phi^2} = \frac{GMm^2}{l^2} - u + 3GMu^2.$$

Since the first term on the right-hand side is a constant, define $u_\circ = \frac{GMm^2}{l^2} = \frac{1}{r_\circ}$. Thus, we have

$$\frac{\mathrm{d}^2 u}{\mathrm{d}\phi^2} + u = u_\circ + 3GMu^2. \tag{21.2}$$

Introduce a small parameter ϵ, so we write

$$\frac{\mathrm{d}^2 u}{\mathrm{d}\phi^2} + u = u_\circ + \epsilon 3GMu^2, \tag{21.3}$$

and we will set $\epsilon = 1$ at the end of the calculation. Now, we can solve this equation perturbatively by expanding u in powers of ϵ:

$$u(\phi) = u_0(\phi) + \epsilon u_1(\phi) + \epsilon^2 u_2(\phi) + \cdots. \tag{21.4}$$

Substituting this expansion into eqn. (21.2) and equating terms of the same order in ϵ, we get the following equations:

$$\mathcal{O}\left(\epsilon^0\right): \quad \frac{\mathrm{d}^2 u_0}{\mathrm{d}\phi^2} + u_0 = u_\circ, \tag{21.5}$$

$$\mathcal{O}\left(\epsilon^1\right): \quad \frac{\mathrm{d}^2 u_1}{\mathrm{d}\phi^2} + u_1 = 3GMu_0^2, \tag{21.6}$$

and so on. Let's solve these equations one by one. Starting with zeroth-order equation eqn. (21.5), we can rewrite the equation as

$$\frac{\mathrm{d}^2}{\mathrm{d}\phi^2}(u_0 - u_\circ) + (u_0 - u_\circ) = 0.$$

The general solution of this equation is given by:

$$u_0(\phi) = u_\circ + A\cos(\phi - \phi_\circ) = u_\circ[1 + e\cos(\phi - \phi_\circ)], \tag{21.7}$$

where A and $\phi_\circ$ are constants of integration, and in the second equality we have defined $e = \frac{A}{u_\circ}$. Without loss of generality, we can set $\phi_\circ = 0$ by choosing the direction of $\phi = 0$ along the major axis of the orbit. In terms of r, we have:

$$r_0(\phi) = \frac{1}{u_0(\phi)} = \frac{r_\circ}{1 + e\cos\phi}. \tag{21.8}$$

We can this the zeroth-order solution as solution for the case $\epsilon = 0$, which corresponds to the Newtonian limit. This is the equation of a conic section in polar coordinates, with one focus at the origin. The shape of the orbit depends on the value of the eccentricity e, as summarised in table 21.1.

Eccentricity e	Orbit Type
0	Circle
$0 < e < 1$	Ellipse
$e = 1$	Parabola
$e > 1$	Hyperbola

Table 21.1: Types of Orbits based on Eccentricity

For bound orbits (like planets around the Sun), we have $0 \leq e < 1$, which corresponds to elliptical orbits. The point of the closest approach to the sun is called the *perihelion*, and the point of the farthest distance from the sun is called the *aphelion*. To get the perihelion point of an orbit, we can extremise $u_0(\phi)$

$$\frac{\mathrm{d}u_0}{\mathrm{d}\phi} = -A\sin\phi = 0 \quad \Rightarrow \quad \phi = n\pi, \quad n \in \mathbb{Z}.$$

Thus, the perihelion points occur at $\phi = 2n\pi$, as at perihelion u_0 is maximum (so $\mathrm{d}^2u_0/\mathrm{d}\phi^2 < 0$).

Next, let's solve eqn. (21.6). Substituting the expression for $u_0(\phi)$ into the right-hand side, we get:

$$\frac{\mathrm{d}^2 u_1}{\mathrm{d}\phi^2} + u_1 = 3GMu_\circ^2[1 + e\cos\phi]^2$$

For a typical bound orbit, $e \ll 1$, so we can neglect terms of order e^2 and higher. Thus, we have:

$$\frac{\mathrm{d}^2 u_1}{\mathrm{d}\phi^2} + u_1 \approx 3GMu_\circ^2(1 + 2e\cos\phi).$$

We can solve this equation using the method of undetermined coefficients. The homogeneous part is exactly same as zeroth-order equation, so the homogeneous solution is given by:

$$u_{1,\mathrm{hom}}(\phi) = B\cos\phi + C\sin\phi,$$

where B and C are constants of integration. With the right-hand side being a combination of constant and $\cos\phi$, and the homogeneous equation being a simple harmonic oscillator, we try a particular solution of the form:

$$u_{1,\mathrm{p}}(\phi) = D + E\phi\cos\phi + F\phi\sin\phi.$$

Substituting this into the differential equation and equating coefficients, we have

$$u_{1,\mathrm{p}}(\phi)' = E[\cos\phi - \phi\sin\phi] + F[\sin\phi + \phi\cos\phi],$$
$$u_{1,\mathrm{p}}(\phi)'' = E[-2\sin\phi - \phi\cos\phi] + F[2\cos\phi - \phi\sin\phi].$$

Substituting these into the differential equation, we get:

$$E(-2\sin\phi - \phi\cos\phi) + F(2\cos\phi - \phi\sin\phi) + D + E\phi\cos\phi + F\phi\sin\phi = 3GMu_\circ^2(1 + 2e\cos\phi).$$

Equating coefficients of $\sin\phi$, $\cos\phi$, and constant terms, we get:

$$\begin{aligned}
\sin\phi: &\quad -2E - F\phi + F\phi = 0 \quad \Rightarrow \quad E = 0,\\
\cos\phi: &\quad -E\phi + 2F + E\phi = 6GMu_\circ^2 e \quad \Rightarrow \quad 2F = 6GMu_\circ^2 e \quad \Rightarrow \quad F = 3GMu_\circ^2 e,\\
\text{constant}: &\quad D = 3GMu_\circ^2.
\end{aligned}$$

Thus, the particular solution is given by:

$$u_{1,\mathrm{p}}(\phi) = 3GMu_\circ^2(1 + e\phi\sin\phi).$$

Therefore, the general solution for $u_1(\phi)$ is given by:

$$u_1(\phi) = B\cos\phi + C\sin\phi + 3GMu_\circ^2(1 + e\phi\sin\phi).$$

Combining the zeroth-order and first-order solutions, we have:

$$u(\phi) = u_\circ[1 + e\cos\phi] + \epsilon\Big[B\cos\phi + C\sin\phi + 3GMu_\circ^2(1 + e\phi\sin\phi)\Big].$$

Setting $\epsilon = 1$, we get:

$$u(\phi) = u_\circ[1 + e\cos\phi] + B\cos\phi + C\sin\phi + 3GMu_\circ^2(1 + e\phi\sin\phi).$$

The terms $B\cos\phi$ and $C\sin\phi$ can be absorbed into the zeroth-order solution by redefining the constants A and $\phi_\circ$. So we have

$$u(\phi) = u_\circ[1 + e\cos\phi] + 3GMu_\circ^2(1 + e\phi\sin\phi).$$

In the Newtonian limit, the perihelion points occur at $\phi = 2n\pi$. We expect the relativistic correction to slightly shift the perihelion points, so let's write the new perihelion points as $\phi_n = 2n\pi + \delta_n$, where δ_n is a small correction. To get the value of δ_n, we can extremise $u(\phi)$:

$$\left.\frac{\mathrm{d}u}{\mathrm{d}\phi}\right|_{\phi=\phi_n} = -u_\circ e\sin\phi_n + 3GMu_\circ^2 e(\sin\phi_n + \phi_n\cos\phi_n) = 0.$$

Divide through by $u_\circ e$:

$$-\sin\phi_n + 3GMu_\circ(\sin\phi_n + \phi_n\cos\phi_n) = 0.$$

Substituting $\phi_n = 2n\pi + \delta_n$ and using small-angle approximations (*i.e.*, $\sin\delta_n \approx \delta_n$ and $\cos\delta_n \approx 1$), expanding the trigonometric functions, we get:

$$-\sin(2n\pi + \delta_n) + 3GMu_\circ[\sin(2n\pi + \delta_n) + (2n\pi + \delta_n)\cos(2n\pi + \delta_n)] = 0.$$

Using $\sin(2n\pi + \delta_n) = \sin\delta_n \approx \delta_n$ and $\cos(2n\pi + \delta_n) = \cos\delta_n \approx 1$, we have:

$$-\delta_n + 3GMu_\circ(2\delta_n + 2n\pi) = 0 \quad \Rightarrow \quad \delta_n(1 - 6GMu_\circ) = 6\pi mGMu_\circ.$$

For sun $GM \approx 1.5$ km, and for Mercury, $r_\circ \approx 5.8 \times 10^7$ km, so $u_\circ = \frac{1}{r_\circ} \approx 1.72 \times 10^{-8}$ km^{-1}. Thus, $6GMu_\circ \approx 1.55 \times 10^{-7} \ll 1$, so we can approximate $1 - 6GMu_\circ \approx 1$. Therefore, we have:

$$\delta_n \approx 6\pi nGMu_\circ = \frac{6G^2M^2m^2\pi n}{l^2}.$$

Thus, the perihelion points occur at:

$$\phi_n = 2n\pi + \delta_n \approx 2n\pi + \frac{6G^2M^2m^2\pi n}{l^2} = 2n\pi\left(1 + \frac{3G^2M^2m^2}{l^2}\right).$$

The angular shift in the perihelion per revolution is given by:

$$\Delta\phi = \phi_{n+1} - \phi_n - 2\pi = 2\pi\left(1 + \frac{3G^2M^2m^2}{l^2}\right) - 2\pi = \frac{6\pi G^2M^2m^2}{l^2}.$$

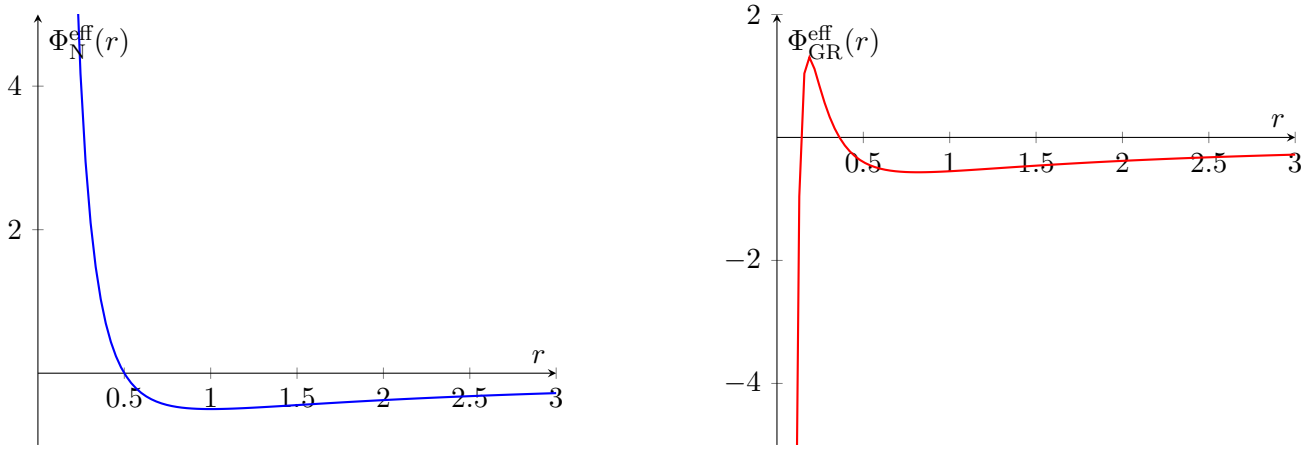
This is the famous result for the perihelion precession of Mercury due to general relativistic effects. When we plug in the values for Mercury's orbit, we find that the angular shift corresponds to approximately 43 arc-seconds per century.

§21.1 Black Holes: Schwarzschild Solution

Recall the effective potential for General Relativity, we have

$$\Phi_{\text{GR}}^{\text{eff}}(r) = -\frac{GM}{r} + \frac{l^2}{2m^2r^2} - \frac{GMl^2}{m^2r^3}.$$

When r is very small, the last term dominates, and since it is negative, the effective potential goes to $-\infty$ as $r \to 0$. This means that there is no stable orbit for very small r, and any particle that comes too close to the central mass will fall into it. This leads to the concept of black holes, which are regions of spacetime where gravity is so strong that nothing, not even light, can escape from it.



(a) Newtonian Effective Potential

(b) General Relativistic Effective Potential

Figure 21.1: Comparison of Effective Potentials in Newtonian Gravity and General Relativity

To formally understand the no escape property of black holes, recall that trajectories of light rays are given by null geodesics, which satisfy $\mathrm{d}s^2 = 0$. Assume a point mass M located at the origin. The spacetime outside this mass is described by the Schwarzschild metric, let a photon is emitted at a distance $r_0 < r_s$ and which is moving radially outward. The Schwarzschild metric for radial motion is given by:

$$\mathrm{d}s^2 = -\left(1 - \frac{r_s}{r}\right)\mathrm{d}t^2 + \left(1 - \frac{r_s}{r}\right)^{-1}\mathrm{d}r^2\,.$$

Setting $\mathrm{d}s^2 = 0$ for null geodesics, we have:

$$0 = -\left(1 - \frac{r_s}{r}\right)\mathrm{d}t^2 + \left(1 - \frac{r_s}{r}\right)^{-1}\mathrm{d}r^2 \quad \Rightarrow \quad \left(1 - \frac{r_s}{r}\right)^{2}\mathrm{d}t^2 = \mathrm{d}r^2\,.$$

Thus, we have:

$$\mathrm{d}t = \pm\left(1 - \frac{r_s}{r}\right)^{-1}\mathrm{d}r\,.$$

Using this equation, let's calculate the time taken by the photon to reach the $r = r_s$ surface from $r = r_0$:

$$\Delta t = \int_{r_0}^{r_s}\mathrm{d}t = \int_{r_0}^{r_s} -\left(1 - \frac{r_s}{r}\right)^{-1}\mathrm{d}r = \int_{r_0}^{r_s}\frac{r}{r_s - r}\,\mathrm{d}r\,.$$

Here, we have taken the negative sign since time interval should be positive, and position of the photon is $r < r_s$. Evaluating the integral, substituting $u = r_s - r$, we have:

$$\Delta t = \int_{r_s - r_0}^{0}\frac{r_s - u}{u}(-\,\mathrm{d}u) = \int_0^{r_s - r_0}\left(\frac{r_s}{u} - 1\right)\mathrm{d}u = r_s[\ln u|_0^{r_s - r_0} - (r_s - r_0).$$

As $u \to 0$, $\ln u \to -\infty$, so the time taken by the photon to reach the $r = r_s$ surface is infinite. Thus, no photon emitted from inside the Schwarzschild radius can ever reach an observer outside the Schwarzschild radius. This is the formal statement of the no escape property of black holes. Any massive object will move with a speed less than the speed of light, so it will take even longer time to reach the Schwarzschild

radius. Thus, nothing can escape from inside the Schwarzschild radius, which is why the surface $r = r_s$ is called the *event horizon*.

Similarly, we can show that a photon will take infinite time to reach the event horizon from outside. Thus, to an external observer, any object falling into a black hole will appear to slow down and never actually cross the event horizon. However, from the perspective of the in-falling object, it will cross the event horizon in a finite amount of proper time.

Any object with its extent entirely within its Schwarzschild radius is called a *black hole i.e.*, if its physical radius R satisfies $R < r_s = 2GM$. For example, the Schwarzschild radius of the Sun is approximately 3 km, while the actual radius of the Sun is about 0.6×10^6 km, so the Sun is not a black hole.

Lec 22: Interior Solutions for Spherically Symmetric Bodies

In the last lecture, we have talked about the precessing of perihelion of Mercury using the exterior Schwarzschild solution, and how the extra term in the effective potential leads to objects like black holes. In the discussion on black holes, we have seen that the photon is taking an infinite time to reach the event horizon. We will first understand this in more detail *i.e.*, time dilation due to gravity

§22.1 Gravitational Time Dilation and Gravitational Redshift

Consider a static metric which is asymptotically flat

$$\mathrm{d}s^2 = g_{\mu\nu}(\vec{\mathbf{x}})\,\mathrm{d}x^\mu\,\mathrm{d}x^\nu = g_{00}(\vec{\mathbf{x}})(\mathrm{d}x^0)^2 + g_{ij}(\vec{\mathbf{x}})\,\mathrm{d}x^i\,\mathrm{d}x^j\,. \tag{22.1}$$

Here, $g_{00}(\vec{\mathbf{x}}) \to -1$ and $g_{ij}(\vec{\mathbf{x}}) \to \delta_{ij}$ as $|\vec{\mathbf{x}}| \to \infty$.

Construct two identical clocks in a flat spacetime, so both the clocks are synchronized *i.e.*, they tick at the same rate. Now, fix one clock at a spatial position $\vec{\mathbf{x}}_{(1)}$ in the gravitational field, and another clock at a different spatial position $\vec{\mathbf{x}}_{(2)}$.

Consider the observer at $\vec{\mathbf{x}}_{(1)}$ claps twice with a time interval of 1 second in his own frame. The proper time interval and the coordinate time interval are related as

$$(\Delta\tau_{(1)})^2 = -g_{00}(\vec{\mathbf{x}}_{(1)})(\Delta {x^0}_{(1)})^2 \quad \Rightarrow \quad \Delta {x^0}_{(1)} = \frac{\Delta\tau_{(1)}}{\sqrt{-g_{00}(\vec{\mathbf{x}}_{(1)})}} = \frac{1\text{ s}}{\sqrt{-g_{00}(\vec{\mathbf{x}}_{(1)})}}. \tag{22.2}$$

The observer at $\vec{\mathbf{x}}_{(2)}$ is watching these claps. Say he observes the first clap at coordinate time ${x^0}_{(2)}$ and the second clap at coordinate time ${x^0}_{(2)} + \Delta {x^0}_{(2)}$. The proper time interval between the two claps as observed by the observer at $\vec{\mathbf{x}}_{(2)}$ is

$$(\Delta\tau_{(2)})^2 = -g_{00}(\vec{\mathbf{x}}_{(2)})(\Delta {x^0}_{(2)})^2. \tag{22.3}$$

Now to compare the time-lapse between the two claps as observed by the observer at $\vec{\mathbf{x}}_{(2)}$, we need to relate the coordinate time intervals at the two spatial positions *i.e.*, $\Delta {x^0}_{(1)}$ and $\Delta {x^0}_{(2)}$. First let's prove a special result for static spacetimes.

> Claim: Let $X^\mu(\lambda)$ be a null geodesic connecting, then $Y^\mu(\lambda) = \begin{pmatrix} X^0(\lambda) + c & X^i(\lambda) \end{pmatrix}$ is also a null geodesic, where c is an arbitrary constant.

Proof:
Since $X^\mu(\lambda)$ is a null geodesic, we have

$$g_{\mu\nu}\frac{\mathrm{d}X^\mu}{\mathrm{d}\lambda}\frac{\mathrm{d}X^\nu}{\mathrm{d}\lambda} = 0, \quad \frac{\mathrm{d}^2X^\mu}{\mathrm{d}\lambda^2} + \Gamma^\mu_{\alpha\beta}\frac{\mathrm{d}X^\alpha}{\mathrm{d}\lambda}\frac{\mathrm{d}X^\beta}{\mathrm{d}\lambda} = 0.$$

As c is a constant, we have $\frac{\mathrm{d}Y^0}{\mathrm{d}\lambda} = \frac{\mathrm{d}X^0}{\mathrm{d}\lambda}$ and $\frac{\mathrm{d}Y^i}{\mathrm{d}\lambda} = \frac{\mathrm{d}X^i}{\mathrm{d}\lambda}$. And since the metric is static, $\Gamma^\mu_{\alpha\beta}$ are independent of x^0. Thus,

$$g_{\mu\nu}\frac{\mathrm{d}Y^\mu}{\mathrm{d}\lambda}\frac{\mathrm{d}Y^\nu}{\mathrm{d}\lambda} = g_{\mu\nu}\frac{\mathrm{d}X^\mu}{\mathrm{d}\lambda}\frac{\mathrm{d}X^\nu}{\mathrm{d}\lambda} = 0,$$

and

$$\frac{\mathrm{d}^2Y^\mu}{\mathrm{d}\lambda^2} + \Gamma^\mu_{\alpha\beta}\frac{\mathrm{d}Y^\alpha}{\mathrm{d}\lambda}\frac{\mathrm{d}Y^\beta}{\mathrm{d}\lambda} = \frac{\mathrm{d}^2X^\mu}{\mathrm{d}\lambda^2} + \Gamma^\mu_{\alpha\beta}\frac{\mathrm{d}X^\alpha}{\mathrm{d}\lambda}\frac{\mathrm{d}X^\beta}{\mathrm{d}\lambda} = 0.$$

Thus, $Y^\mu(\lambda)$ is also a null geodesic. Q.E.D.

Let the null geodesic connecting the two observers be $X^\mu(\lambda)$. For simplicity, choose the affine parameter such that the geodesic starts at $\lambda = 0$ from $\begin{pmatrix} x^0{}_{(1)} & \vec{\mathbf{x}}_{(1)} \end{pmatrix}$ and ends at $\lambda = 1$ at $\begin{pmatrix} x^0{}_{(2)} & \vec{\mathbf{x}}_{(2)} \end{pmatrix}$. Now, using the above claim, we can construct another null geodesic $Y^\mu(\lambda) = \begin{pmatrix} X^0(\lambda) + \Delta x^0{}_{(1)} & X^i(\lambda) \end{pmatrix}$ which starts at $\lambda = 0$ from $\begin{pmatrix} x^0{}_{(1)} + \Delta x^0{}_{(1)} & \vec{\mathbf{x}}_{(1)} \end{pmatrix}$ and ends at $\lambda = 1$ at $\begin{pmatrix} x^0{}_{(2)} + \Delta x^0{}_{(1)} & \vec{\mathbf{x}}_{(2)} \end{pmatrix}$. From setup of the problem, we know that the second clap reaches the observer at $\vec{\mathbf{x}}_{(2)}$ at coordinate time $x^0{}_{(2)} + \Delta x^0{}_{(2)}$. Thus, we have

$$\Delta x^0{}_{(2)} = \Delta x^0{}_{(1)}. \tag{22.4}$$

Using this in the expression for $\Delta\tau_{(2)}$, we get

$$\Delta\tau_{(2)} = \sqrt{-g_{00}(\vec{\mathbf{x}}_{(2)})}\Delta x^0{}_{(2)} = \sqrt{-g_{00}(\vec{\mathbf{x}}_{(2)})}\Delta x^0{}_{(1)} = \sqrt{\frac{-g_{00}(\vec{\mathbf{x}}_{(2)})}{-g_{00}(\vec{\mathbf{x}}_{(1)})}}\Delta\tau_{(1)}. \tag{22.5}$$

This shows that the clock at $\vec{\mathbf{x}}_{(2)}$ runs at a different rate compared to the clock at $\vec{\mathbf{x}}_{(1)}$.

Consider the Schwarzschild metric, and let the clock at $\vec{\mathbf{x}}_{(2)}$ be at infinity *i.e.*, $g_{00}(\vec{\mathbf{x}}_{(2)}) = -1$. And let the clock at $\vec{\mathbf{x}}_{(1)}$ be at a radial distance r from the mass M *i.e.*, $g_{00}(\vec{\mathbf{x}}_{(1)}) = -\left(1 - \frac{2GM}{r}\right)$. Thus, the time interval between the two claps as observed by the observer at infinity is

$$\Delta\tau_{(\infty)} = \frac{1 \text{ s}}{\sqrt{1 - \frac{2GM}{r}}} > 1 \text{ s}. \tag{22.6}$$

This shows that the clock closer to the mass M runs slower compared to the clock at infinity.

§22.1.1 Gravitational Redshift

Consider a light source at $\vec{\mathbf{x}}_{(1)}$ emitting light. The proper time interval between two successive wave crests as measured in the frame of the light source is $\Delta\tau_{(1)} = \frac{1}{\nu_{(1)}}$, where $\nu_{(1)}$ is the frequency of the light source. An observer at $\vec{\mathbf{x}}_{(2)}$ watches these wave crests. Using the previous result, the time interval between the two wave crests as measured by the observer at $\vec{\mathbf{x}}_{(2)}$ is

$$\Delta\tau_{(2)} = \sqrt{\frac{-g_{00}(\vec{\mathbf{x}}_{(2)})}{-g_{00}(\vec{\mathbf{x}}_{(1)})}}\Delta\tau_{(1)} \quad \Rightarrow \quad \nu_{(2)} = \sqrt{\frac{-g_{00}(\vec{\mathbf{x}}_{(1)})}{-g_{00}(\vec{\mathbf{x}}_{(2)})}}\nu_{(1)}.$$

This shows that the frequency of light changes as it moves in a gravitational field. A simpler notation, $\nu_{(\text{e})}$ is the frequency of light when emitted at position $\vec{\mathbf{x}}_{(\text{e})}$ and $\nu_{(\text{o})}$ is the frequency of light when observed at position $\vec{\mathbf{x}}_{(\text{o})}$, then the two frequencies are related as

$$\nu_{(\text{o})} = \sqrt{\frac{-g_{00}(\vec{\mathbf{x}}_{(\text{e})})}{-g_{00}(\vec{\mathbf{x}}_{(\text{o})})}}\nu_{(\text{e})}. \tag{22.7}$$

In terms of wavelength, we have

$$\lambda_{(\text{o})} = \sqrt{\frac{-g_{00}(\vec{\mathbf{x}}_{(\text{o})})}{-g_{00}(\vec{\mathbf{x}}_{(\text{e})})}}\lambda_{(\text{e})}. \tag{22.8}$$

Consider the Schwarzschild metric, and let the light source be at a radial distance r from the mass M and the observer be at infinity. Thus, we have

$$\nu_{(o)} = \nu_{(\infty)} = \sqrt{1 - \frac{2GM}{r}}\nu_{(e)} < \nu_{(e)}, \quad \lambda_{(\infty)} = \frac{\lambda_{(e)}}{\sqrt{1 - \frac{2GM}{r}}} > \lambda_{(e)}. \tag{22.9}$$

This shows that the light is redshifted as it moves away from the mass M.

Assignment 07.1.
Compute the wave-length of light that gets to a distant observer from the following sources:

(a) Light emitted with wave-length 6563 Å (H-α line) by a source at rest where Newtonian potential $\Phi_{\rm N} = -GM/r = -10^6$ (a typical star).

(b) Same as above, but the Newtonian potential $\Phi_{\rm N} = -6 \times 10^{-5}$ (white dwarf).

(c) Same source as above, but is at a radial distance $r = 1.1r_s$ from an object of mass $M = 1M_\odot$.

(d) Same source as above, but is at a radial distance $r = 1.01r_s$ (neutron star).

Till now, we have only discussed the exterior (vacuum) solution of a static spherically symmetric mass distribution. Now, we will discuss the interior solution *i.e.*, the metric inside a spherically symmetric mass distribution.

§22.2 Interior Solution for a Spherically Symmetric Body

Consider a static spherically symmetric mass distribution with radius R. The general form of static spherically symmetric metric is

$$\mathrm{d}s^2 = f(r)\,\mathrm{d}t^2 + h(r)\,\mathrm{d}r^2 + r^2\,\mathrm{d}\Omega^2\,,$$

where $f(r) < 0$ and $h(r) > 0$. Consider these arbitrary functions $f(r)$ and $h(r)$ have the following forms:

$$f(r) = -e^{2\Phi(r)}, \qquad h(r) = e^{2\nu(r)}, \tag{22.10}$$

where $\Phi(r)$ and $\nu(r)$ are functions to be determined. Using the power of symbolic computation (see section B.2), we can compute all the important quantities for this metric. The important non-zero Christoffel symbols are:

$$\Gamma^r_{tt} = e^{2(\Phi-\nu)}\Phi',$$

where the prime denotes derivative with respect to r. The important non-zero components of the Einstein tensor are:

$$G_{tt} = \frac{1}{r^2}e^{2(\Phi-\nu)}\Big[-1 + e^{2\nu} + 2r\nu'\Big], \tag{22.11}$$

$$G_{rr} = \frac{1}{r^2}\Big[1 - e^{2\nu} + 2r\Phi'\Big]. \tag{22.12}$$

Recall that the Einstein field equations are

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}\,.$$

To solve these equations, we need to know the form of the energy-momentum tensor $T_{\mu\nu}$ inside the mass distribution. For simplicity, we will consider the matter inside the mass distribution to be a perfect fluid (no viscosity). The energy-momentum tensor for a perfect fluid is

$$T_{\mu\nu} = (\rho + P)u_\mu u_\nu + P g_{\mu\nu},$$

where $\rho(\bar{\mathbf{x}})$ is the energy density, $P(\bar{\mathbf{x}})$ is the pressure, and $\bar{\mathbf{u}}$ is the 4-velocity of the fluid elements.

Since the mass distribution is static, we have

$$\frac{\mathrm{d}x^i}{\mathrm{d}\tau} = 0 \quad \Rightarrow \quad u^0 = \frac{\mathrm{d}x^0}{\mathrm{d}\tau}, \quad u^i = 0.$$

Using the normalization condition $u^\mu u_\mu = -1$, we get

$$g_{00}\Big(u^0\Big)^2 = -1 \quad \Rightarrow \quad -e^{2\Phi}\Big(u^0\Big)^2 = -1 \quad \Rightarrow \quad u^0 = e^{-\Phi}.$$

So the 4-velocity of the fluid elements is

$$u^\mu = \begin{pmatrix} e^{-\Phi} \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad u_\mu = \begin{pmatrix} -e^{\Phi} \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

Thus, the non-zero components of the energy-momentum tensor are

$$T_{tt} = (\rho + P)u_t u_t + P g_{tt} = (\rho + P)(-e^{\Phi})(-e^{\Phi}) + P(-e^{2\Phi}) = \rho e^{2\Phi}, \tag{22.13}$$

$$T_{rr} = (\rho + P)u_r u_r + P g_{rr} = P e^{2\nu}. \tag{22.14}$$

Let's substitute these in the Einstein field equations. From eqn. (22.11), we have

$$\begin{aligned} \frac{1}{r^2} e^{2(\Phi-\nu)} \left[-1 + e^{2\nu} + 2r\nu' \right] &= 8\pi G \rho e^{2\Phi} \\ \Rightarrow \quad 1 - e^{-2\nu} + 2r\nu' e^{-2\nu} &= 8\pi G \rho r^2. \end{aligned}$$

We can write the left-hand side as

$$\frac{\mathrm{d}}{\mathrm{d}r} \left[r(1 - e^{-2\nu}) \right] = 8\pi G \rho r^2.$$

Define a new function $r(1 - e^{-2\nu}) = 2G\mathcal{M}(r)$, where $\mathcal{M}(r)$ is called the mass function. Thus, we have

$$\frac{\mathrm{d}\mathcal{M}}{\mathrm{d}r} = 4\pi r^2 \rho(r). \tag{22.15}$$

Using the definition of the mass function, we can write

$$e^{-2\nu} = 1 - \frac{2G\mathcal{M}(r)}{r}. \tag{22.16}$$

From the definition of the mass function, it is easy to see that $\mathcal{M}(0) = 0$.

Lec 23: TOV Equations

In the last lecture, we have started discussing the interior solution of a spherically symmetric body. We have re-expressed the Einstein equations for $(0,0)$ component as

$$\frac{\mathrm{d}\mathcal{M}}{\mathrm{d}r} = 4\pi r^2 \rho(r), \tag{23.1}$$

where we have defined the mass function $\mathcal{M}(r)$ as

$$2G\mathcal{M}(r) = r\Big(1 - e^{-2\nu}\Big). \tag{23.2}$$

This allows us to write $e^{-2\nu}$ as

$$e^{-2\nu} = 1 - \frac{2G\mathcal{M}(r)}{r}. \tag{23.3}$$

Now, let's look at the $(1,1)$ component of the Einstein equations, using the Symbolic computation tool we have

$$G_{rr} = \frac{1}{r^2}\Big[1 - e^{2\nu} + 2r\Phi'\Big] = 8\pi G P e^{2\nu}. \tag{23.4}$$

Rearranging this, we get

$$(e^{-2\nu} - 1) + 2r\Phi' e^{-2\nu} = 8\pi G r^2 P.$$

Multiplying both sides by r and substituting for $r(1 - e^{-2\nu}) = 2G\mathcal{M}(r)$, we have

$$-2G\mathcal{M}(r) + 2r^2\Phi' e^{-2\nu} = 8\pi G r^3 P.$$

So we get

$$\Phi' = \frac{\mathrm{d}\Phi}{\mathrm{d}r} = \frac{G\big[\mathcal{M}(r) + 4\pi r^3 P\big]}{r^2\Big(1 - \frac{2G\mathcal{M}(r)}{r}\Big)}. \tag{23.5}$$

Assignment 08.1.
Using the conservation equation $\nabla_\mu T^{\mu\nu} = 0$, show that

$$\frac{\mathrm{d}P}{\mathrm{d}r} = -(\rho + P)\frac{\mathrm{d}\Phi}{\mathrm{d}r}. \tag{23.6}$$

Eqns. (23.1), (23.5) and (23.6) together form the Tolman-Oppenheimer-Volkoff (TOV) equations, which describe the structure of a spherically symmetric body in hydrostatic equilibrium under its own gravity in general relativity.

§23.1 Comparing with Stellar Structure Equations

Let's compare the TOV equations with the Newtonian stellar structure equations. Consider a spherically symmetric star of mass M and radius R. Let's first look at the mass continuity equation in Newtonian gravity, say $M(r)$ is the mass enclosed within radius r. The mass continuity equation is given by

$$\frac{\mathrm{d}M(r)}{\mathrm{d}r} = 4\pi r^2 \rho(r), \tag{23.7}$$

where $\rho(r)$ is the mass density at radius r. Now for hydrostatic equilibrium, the force exerted by the pressure gradient balances the gravitational force. This gives us

$$4\pi r^2 (P(r + \mathrm{d}r) - P(r)) = -\frac{GM(r)\,\mathrm{d}M}{r^2}$$

which simplifies to

$$\frac{\mathrm{d}P}{\mathrm{d}r} = -\frac{GM(r)\rho(r)}{r^2}. \tag{23.8}$$

Recall that the Newtonian gravitational potential Φ_{N} is given by

$$\Phi_{\mathrm{N}}(r) = -\frac{GM}{r}. \tag{23.9}$$

Taking the derivative with respect to r, we have

$$\frac{\mathrm{d}\Phi_{\mathrm{N}}}{\mathrm{d}r} = \frac{GM}{r^2}. \tag{23.10}$$

So we can rewrite the hydrostatic equilibrium equation as

$$\frac{\mathrm{d}P}{\mathrm{d}r} = -\rho(r)\frac{\mathrm{d}\Phi_{\mathrm{N}}}{\mathrm{d}r}. \tag{23.11}$$

So we see that eqns. (23.7), (23.10) and (23.11) are the Newtonian analogues of the TOV equations.

To recover the Newtonian limit from the TOV equations, we consider the following approximations:

- The pressure P is much smaller than the energy density ρ, *i.e.*, $P \ll \rho$.
- The gravitational field is weak, so that $2G\mathcal{M}(r)/r \ll 1$.

The first TOV equation eqn. (23.1) is identical to the Newtonian mass continuity equation eqn. (23.7). In the second TOV equation eqn. (23.5), under weak pressure and weak field approximations, we have

$$\frac{\mathrm{d}\Phi}{\mathrm{d}r} \approx \frac{G\mathcal{M}(r)}{r^2}.$$

Also, the weak pressure approximation in eqn. (23.6) gives

$$\frac{\mathrm{d}P}{\mathrm{d}r} \approx -\rho\frac{\mathrm{d}\Phi}{\mathrm{d}r}.$$

Thus, in the Newtonian limit, the TOV equations reduce to the Newtonian stellar structure equations.

§23.2 Boundary Conditions

To solve the TOV equations, we need appropriate boundary conditions. At the center of the star ($r = 0$), we have

$$2G\mathcal{M}(r) = r(1 - e^{-2\nu}) \quad \Rightarrow \quad \mathcal{M}(0) = 0. \tag{23.12}$$

At the surface of the star ($r = R$), the pressure drops to zero,

$$P(R) = 0. \tag{23.13}$$

Additionally, the interior solution must match the exterior Schwarzschild solution at $r = R$, ensuring continuity of the metric functions for a given spherically symmetric body of mass M, we have

$$e^{2\nu}\Big|_{r=R} = \left(1 - \frac{2GM}{R}\right)^{-1} \Rightarrow \quad \left(1 - \frac{2G\mathcal{M}(R)}{R}\right)^{-1} = \left(1 - \frac{2GM}{R}\right)^{-1} \Rightarrow \quad \mathcal{M}(R) = M. \tag{23.14}$$

Thus, we have two boundary conditions for $\mathcal{M}(r)$:

$$\mathcal{M}(0) = 0, \qquad \mathcal{M}(R) = M. \tag{23.15}$$

Matching for g_{00} component will give us the boundary condition for $\Phi(r)$ at $r = R$,

$$e^{2\Phi}\Big|_{r=R} = 1 - \frac{2GM}{R}. \tag{23.16}$$

Now if the Schwarzschild radius $r_s = 2GM$ is very small compared to the radius of the star R, we can approximate

$$e^{2\Phi}\Big|_{r=R} \approx 1 + 2\Phi(R) \quad \Rightarrow \quad \Phi(R) \approx -\frac{GM}{R}. \tag{23.17}$$

Lec 24: Linearized Gravity

Till now, we have Einstein's field equations in their full non-linear glory. However, in many practical scenarios, especially when dealing with weak gravitational fields, it is beneficial to linearize these equations around a flat spacetime background. This approach simplifies the mathematics significantly and allows us to study phenomena such as gravitational waves.

§24.1 Perturbation of the Metric: Weak Field Approximation

Consider a spacetime metric $g_{\mu\nu}$ which is given by

$$g_{\mu\nu} = \eta_{\mu\nu} + \epsilon h_{\mu\nu}, \tag{24.1}$$

where $\eta_{\mu\nu}$ is the Minkowski metric, ϵ is a small parameter indicating the perturbation's magnitude, and $h_{\mu\nu}$ represents the perturbation tensor. We will work to first order in ϵ, neglecting terms of order ϵ^2 and higher.

Assignment 08.2.
Show that to first order in ϵ, the inverse metric $g^{\mu\nu}$ is given by

$$g^{\mu\nu} = \eta^{\mu\nu} - \epsilon h^{\mu\nu}. \tag{24.2}$$

Solution.
We start with the definition of the inverse metric:

$$g_{\mu\alpha} g^{\alpha\nu} = \delta_\mu{}^\nu. \tag{24.3}$$

Since we want to work to first order in ϵ, we make an ansatz for the inverse metric:

$$g^{\mu\nu} = \eta^{\mu\nu} + \epsilon k^{\mu\nu}, \tag{24.4}$$

where $k^{\mu\nu}$ is an unknown symmetric tensor to be determined. Substituting the expressions for $g_{\mu\nu}$ and $g^{\mu\nu}$ into the definition of the inverse metric, we have

$$(\eta_{\mu\alpha} + \epsilon h_{\mu\alpha})(\eta^{\alpha\nu} + \epsilon k^{\alpha\nu}) = \delta_\mu{}^\nu.$$

Expanding this product and keeping only terms up to first order in ϵ, we get

$$\eta_{\mu\alpha}\eta^{\alpha\nu} + \epsilon\left(\eta_{\mu\alpha}k^{\alpha\nu} + h_{\mu\alpha}\eta^{\alpha\nu}\right) + \mathcal{O}\left(\epsilon^2\right) = \delta_\mu{}^\nu.$$

The zeroth-order term simplifies to $\delta_\mu{}^\nu$, so we have

$$\delta_\mu{}^\nu + \epsilon\left(\eta_{\mu\alpha}k^{\alpha\nu} + h_{\mu\alpha}\eta^{\alpha\nu}\right) = \delta_\mu{}^\nu.$$

This implies that the first-order term must vanish:

$$\eta_{\mu\alpha}k^{\alpha\nu} + h_{\mu\alpha}\eta^{\alpha\nu} = 0 \quad \Rightarrow \quad k_\mu{}^\nu = -h_\mu{}^\nu.$$

Therefore, the inverse metric to first order in ϵ is

$$g^{\mu\nu} = \eta^{\mu\nu} - \epsilon h^{\mu\nu}.$$

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Now that we have the perturbed metric and its inverse, we can proceed to compute other geometric quantities such as the Christoffel symbols, Riemann tensor, Ricci tensor, and Ricci scalar to first order in ϵ. Starting with the Christoffel symbols, we have

$$\Gamma^{\alpha}_{\mu\nu} = \frac{1}{2} g^{\alpha\beta}\Big(\partial_\mu g_{\beta\nu} + \partial_\nu g_{\mu\beta} - \partial_\beta g_{\mu\nu}\Big).$$

We can also write the Christoffel symbols explicitly as

$$\Gamma^{\alpha}_{\mu\nu} = \Gamma^{(0)\alpha}_{\mu\nu} + \epsilon\Gamma^{(1)\alpha}_{\mu\nu}, \tag{24.5}$$

where $\Gamma^{(0)\alpha}_{\mu\nu}$ are the Christoffel symbols for the background metric (which is Minkowski in this case, so they vanish), and $\Gamma^{(1)\alpha}_{\mu\nu}$ are the first-order perturbations given by

$$\Gamma^{(1)\alpha}_{\mu\nu} = \frac{1}{2}\eta^{\alpha\beta}\Big(\partial_\mu h_{\beta\nu} + \partial_\nu h_{\mu\beta} - \partial_\beta h_{\mu\nu}\Big). \tag{24.6}$$

Since $h_{\mu\nu}$ is perturbation, so we have $\Big|h_{\mu\nu}\Big| \ll \Big|\eta_{\mu\nu}\Big| = 1$. So we can use the background metric $\eta_{\mu\nu}$ to raise and lower indices of $h_{\mu\nu}$ in the first-order terms.

The Christoffel symbols is itself of order ϵ. Therefore, when computing the Riemann tensor, we can neglect terms quadratic in the Christoffel symbols. Also, since the background Christoffel symbols vanish, the zeroth-order Riemann tensor also vanishes. Thus, the Riemann tensor to first order in ϵ is given by

$$R^{(1)\rho}{}_{\sigma\mu\nu} = \partial_\mu \Gamma^{(1)\rho}_{\sigma\nu} - \partial_\nu \Gamma^{(1)\rho}_{\sigma\mu}. \tag{24.7}$$

Similarly, the Ricci tensor in zero order vanishes, and the first-order Ricci tensor is given by

$$R^{(1)}_{\mu\nu} = \partial_\rho \Gamma^{(1)\rho}_{\mu\nu} - \partial_\nu \Gamma^{(1)\rho}_{\mu\rho}. \tag{24.8}$$

Finally, the Ricci scalar to first order in ϵ is given by

$$R^{(1)} = \eta^{\mu\nu} R^{(1)}_{\mu\nu}. \tag{24.9}$$

Assume that the spacetime is vacuum, *i.e.*, there is no matter content, so the stress-energy tensor $T_{\mu\nu}(x) = 0$. Then, Einstein's field equations reduce to

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = 0 \quad \Rightarrow \quad R_{\mu\nu} = 0. \tag{24.10}$$

To first order in ϵ, this implies

$$R^{(1)}_{\mu\nu} = 0. \tag{24.11}$$

Now let's substitute the expression for $\Gamma^{(1)\alpha}_{\mu\nu}$ into the expression for $R^{(1)}_{\mu\nu}$:

$$R^{(1)}_{\mu\nu} = \partial_\rho\left[\frac{1}{2}\eta^{\rho\beta}\Big(\partial_\mu h_{\beta\nu} + \partial_\nu h_{\mu\beta} - \partial_\beta h_{\mu\nu}\Big)\right] - \partial_\nu\left[\frac{1}{2}\eta^{\rho\beta}\Big(\partial_\mu h_{\beta\rho} + \partial_\rho h_{\mu\beta} - \partial_\beta h_{\mu\rho}\Big)\right].$$

Notice that the terms marked in red cancel each other out. We are left with

$$R^{(1)}_{\mu\nu} = \frac{1}{2}\Big[\partial_\mu\partial^\beta h_{\beta\nu} - \eta^{\rho\beta}\partial_\rho\partial_\beta h_{\mu\nu} - \partial_\mu\partial_\nu(\eta^{\rho\beta}h_{\beta\rho}) + \partial_\nu\partial^\beta h_{\mu\beta}\Big]. \tag{24.12}$$

Notice that $\eta^{\rho\beta}\partial_\rho\partial_\beta = \Box$ is the d'Alembertian operator (wave operator or generalized Laplacian in spacetime). Also, define the trace of the perturbation tensor as

$$h = h^\rho{}_\rho = \eta^{\rho\beta}h_{\beta\rho}. \tag{24.13}$$

Using these definitions, we can rewrite the expression for $R^{(1)}_{\mu\nu}$ as

$$R^{(1)}_{\mu\nu} = \frac{1}{2}\Big[\partial_\mu\partial^\beta h_{\beta\nu} + \partial_\nu\partial^\beta h_{\mu\beta} - \Box h_{\mu\nu} - \partial_\mu\partial_\nu h\Big]. \tag{24.14}$$

Thus, the linearized Einstein's field equations in vacuum reduce to

$$\partial_\mu\partial^\beta h_{\beta\nu} + \partial_\nu\partial^\beta h_{\mu\beta} - \Box h_{\mu\nu} - \partial_\mu\partial_\nu h = 0. \tag{24.15}$$

§24.2 Analogy with Electromagnetism

Recall the EoM for electromagnetic potentials in presence of sources is given by

$$\partial^\mu F_{\mu\nu} = J_\nu, \tag{24.16}$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is the electromagnetic field strength tensor, and J^ν is the four-current density. Expanding in terms of the potentials, as well as let us choose the space to be source-free ($J_\nu = 0$), we get

$$\partial^\mu\Big(\partial_\mu A_\nu - \partial_\nu A_\mu\Big) = 0 \quad \Rightarrow \quad \Box A_\nu - \partial_\nu\Big(\partial_\mu A^\mu\Big) = 0. \tag{24.17}$$

We know that the electromagnetic potentials have a gauge freedom, *i.e.*, we can perform a gauge transformation

$$A_\mu \to A_\mu + \partial_\mu \Lambda,$$

where Λ is an arbitrary scalar function, without changing the physical electromagnetic fields. So we can choose any gauge we like to simplify the equations. A common choice is the Lorenz gauge, defined by

$$\partial_\mu A^\mu = 0.$$

In this gauge, the equations of motion simplify to

$$\Box A_\nu = 0. \tag{24.18}$$

This is a wave equation for each component of the four-potential in vacuum. We can have plane wave solutions of the form

$$A_\mu(x) = C_\mu \cos(k \cdot x + \phi) + D_\mu \sin(k \cdot x + \phi),$$

where C_μ, D_μ, k_α, and ϕ are constants.

Similarly, in linearized gravity, we have gauge freedom due to diffeomorphism invariance. We can perform an infinitesimal coordinate transformation

$$x^\mu \to x'^\mu = x^\mu + \epsilon\xi^\mu(x),$$

where $\xi^\mu(x)$ is an arbitrary vector field. Under this transformation, the perturbation tensor transforms as

$$h_{\mu\nu} \to h_{\mu\nu} - \partial_\mu \xi_\nu - \partial_\nu \xi_\mu.$$

This gauge freedom allows us to impose additional conditions on $h_{\mu\nu}$ to simplify the linearized Einstein's equations. A common choice is the Transverse-Traceless (TT) gauge, defined by

$$\partial^\beta h_{\beta\nu} = 0, \qquad \text{and} \qquad h = 0.$$

In this gauge, the linearized Einstein's equations reduce to

$$\Box h_{\mu\nu} = 0. \tag{24.19}$$

This is a wave equation for each component of the perturbation tensor in vacuum. Similar to electromagnetism, we can have plane wave solutions of the form

$$h_{\mu\nu}(x) = C_{\mu\nu} \cos(k \cdot x + \phi) + D_{\mu\nu} \sin(k \cdot x + \phi),$$

§24.3 Gravitational Waves

We have seen that in the linearized approximation of General Relativity, perturbations in the metric tensor $h_{\mu\nu}$ satisfy a wave equation in vacuum. These perturbations can propagate as waves through spacetime, analogous to electromagnetic waves in electromagnetism. These are known as gravitational waves.

To see the effects of this oscillating metric, consider two points in spacetime with coordinates x_1^μ and x_2^μ. Define the separation vector between these two points as $\Delta x^\mu = x_2^\mu - x_1^\mu$. The proper distance d between these two points is given by

$$d^2 = g_{\mu\nu}\Delta x^\mu \Delta x^\nu = (\eta_{\mu\nu} + h_{\mu\nu})\Delta x^\mu \Delta x^\nu = \eta_{\mu\nu}\Delta x^\mu \Delta x^\nu + h_{\mu\nu}\Delta x^\mu \Delta x^\nu. \tag{24.20}$$

The first term represents the proper distance in flat spacetime, while the second term represents the perturbation due to the gravitational wave. As the gravitational wave passes through the region, $h_{\mu\nu}$ oscillates, leading to a time-varying proper distance between the two points. This effect can be measured using devices such as laser interferometers, which can detect the tiny changes in distance caused by passing gravitational waves.

Lec 25: Cosmology

Till now, we have talked about how to modify Newtonian gravity to make it compatible with special relativity, and in doing so we have arrived at general relativity (GR). And we have seen one non-trivial solution to Einstein's equations, the Schwarzschild solution, which describes the spacetime around a static, spherically symmetric mass, and for the interior of a static, spherically symmetric star we have the Tolman-Oppenheimer-Volkoff (TOV) equations. Then we have also seen that how perturbations to the Minkowski metric (flat spacetime) can be interpreted as gravitational waves. Now we will move on to another aspect of this course, which is *cosmology*.

§25.1 Introduction to Cosmology

Cosmology is the study of the *cosmos*, *i.e.*, the universe. The '*literal*' cosmology is pre-historic, almost every ancient civilization had studied the cosmos in some way or the other, for example, in our context, the ancient Indian civilization had the concept of *Brahmanda* (the cosmic egg), which goes back to the Vedic period (1700 BC).

In this subject, one can ask multitude of questions, however "*physics*" only deals with those questions which can be answered using the scientific method, *i.e.*, by observation and experiment. So we are going to focus on the *physical cosmology*, which is the study of the universe with respect to the physical quantities (quantities which can be measured directly or indirectly), for example, density, temperature, etc.

§25.1.1 Isotropy and Cosmological Principle

Consider a glass of water, length of the glass is about 10^{-2} m. Now if someone asks what is the density of water in the glass, we can easily measure it by taking a small sample of water from the glass and measuring its mass and volume. We can do this experiment with smaller and smaller scales, but at some point, we will reach the molecular scale, which is about 10^{-10} m. At this scale, water is no longer a continuous fluid, but rather a collection of discrete molecules. So the concept of density becomes meaningless at this scale. So, just by zooming in 10^8 times, we have reached a scale where the concept of density is no longer valid. Thus, we have to be careful about the scale at which we are measuring physical quantities.

Now consider our universe, the average distance between sun and earth is about 1.5×10^{11} m (1 AU). The size of our solar system is about 10^{12} m, the size of our galaxy (Milky Way) is about 30 kpc $\approx 10^{21}$ m, and the size of the observable universe is about 3000 Mpc $\approx 10^{26}$ m. If we consider a solar system as an atom, then observable universe is about 10^{14} times larger than the solar system. So, we can think of the observable universe as '*fluid*' at this scale.

However, we can't directly measure the density of the universe at this scale, because we can't take a sample of the universe. But we can use observations to infer the density of the universe. For example, we can count the number of galaxies in a given volume of space and estimate the average density of matter in the universe.

§25.1.2 Isotropy: Are humans special?

If we employ the method of counting galaxies in different directions from Earth, we find that the density of galaxies is roughly the same in all directions. This observation suggests that the large scale properties of the universe is independent of direction from Earth or around it. This property is known as *isotropy*.

Now one may ask, are we (humans on Earth) special or privileged observers in the universe? Again, we can't answer this question using scientific method. So we have to make an assumption, about the nature of the universe. This assumption is known as the *Cosmological Principle*, which states that *the universe is isotropic on large scales around every point*. In other words, there are no special or privileged observers

in the universe. This principle is a generalization of the Copernican principle, which states that the Earth is not the center of the universe.

Assignment 09.1 (Homogeneity from Isotropy)**.**
Show that if a space appears to be isotropic from just two distinct points in it then the space is homogeneous.

So at large scales, the universe is homogeneous and isotropic.

§25.2 Size and Age of the Universe

Let's first talk about a paradox known as *Olbers' Paradox*, which states that if the universe is infinite, static, and filled with stars, then the night sky should be as bright as the daytime sky.

Consider a star at a distance r from us with absolute luminosity L (total energy emitted per unit time). The apparent luminosity A (energy received per unit time per unit area) of the star is given by

$$A(r) = \frac{L}{4\pi r^2}.$$

Say the number density of stars in the universe is $n(r)$ (number of stars per unit volume). Apparent luminosity as viewed by the observer for stars in a thin spherical shell of radius r and thickness $\mathrm{d}r$ is given by

$$\mathrm{d}E = A(r) \times [(4\pi r^2\,\mathrm{d}r) \times n] = Ln\,\mathrm{d}r\,.$$

Now integrating this from $r = 0$ to $r = \infty$, we get

$$E = \int_0^\infty Ln\,\mathrm{d}r\,,$$

for homogeneous and static universe, L and n are constants, so we get

$$E = Ln\int_0^\infty \mathrm{d}r \to \infty.$$

This means that the night sky should be bright, which is not the case. Now how do we resolve this paradox? There are two possible resolutions:

- Universe is finite in size.
- Light can reach to us only from a finite part of the universe, *i.e.*, the age of the universe is finite.

Observations suggest that the universe is indeed finite in age, about 13.8 billion years old. This means that we can only see light from objects that are within a distance of 13.8 billion light-years from us. Beyond this distance, light has not had enough time to reach us since the beginning of the universe.

Lec 26: FRLW Metric

In the last lecture, we introduced the scale of the universe, and how we can see the whole universe as "*perfect fluid*." Then we discussed the observational evidences which says that the universe is isotropic with respect to Earth, and then we introduced the Cosmological Principle which states that the universe is *isotropic* with respect to every point in the universe. Using this principle, we showed that the universe is also *homogeneous*. Now let's first see if the universe can be static.

§26.1 Can the Universe be Static? – Gravitational Redshift and Expanding Universe

During the early 20th century, it was widely believed that the universe is static and unchanging. However, Edwin Hubble's observations of distant galaxies in the 1920s revealed that the spectrum of light from these galaxies was *redshifted*, *i.e.*, if we observe the Hydrogen spectral lines from these galaxies, we find that the wavelengths are longer (shifted towards the red end of the spectrum) compared to the same lines observed in a laboratory on Earth. Define the redshift parameter z as

$$z = \frac{\lambda_{(\mathrm{o})} - \lambda_{(\mathrm{e})}}{\lambda_{(\mathrm{e})}} \quad \Rightarrow \quad \lambda_{(\mathrm{o})} = (1+z)\lambda_{(\mathrm{e})}, \tag{26.1}$$

where $\lambda_{(\mathrm{e})}$ is the wavelength of light when it was emitted from the source, and $\lambda_{(\mathrm{o})}$ is the wavelength of light when it is observed. Since the spectral lines from distant galaxies are redshifted, we have $z > 0$ for these galaxies. This redshift can be interpreted in two ways:

- **Doppler Redshift:** If the source of light is moving away from the observer, the observed wavelength will be longer due to the Doppler effect.
- **Gravitational Redshift:** If light is emitted from a region of strong gravitational field and observed in a region of weaker gravitational field, the light will be redshifted.

Hubble's observations led to the following empirical relation, between the redshift z and the distance d to the galaxy:

$$z \propto d. \tag{26.2}$$

So the farther away a galaxy is, the more its light is redshifted. Now let's try to interpret this redshift using gravitational redshift. With this interpretation, the gravitational potential gets stronger as we go farther away from Earth (same in all directions as observations show isotropy). This means that Earth is at the center of a spherically symmetric gravitational field, which implies that the Earth is a privileged point in the universe. This contradicts the Cosmological Principle, which states that the universe is isotropic and homogeneous with respect to every point. Therefore, we conclude that the gravitational redshift interpretation is not valid on cosmological scales.

Instead, the redshift observed in distant galaxies is best explained by Doppler redshift, indicating that these galaxies are receding from us. This gives us the linear relationship between recession velocity v and distance d, known as *Hubble's Law*:

$$v = H_0 d, \tag{26.3}$$

where H_0 is the *Hubble constant*. In some observations, we find that $v > c$ for very distant galaxies, which is not possible in Special Relativity. So, we need to reinterpret this recession velocity as a result of the expansion of space itself, rather than the motion of galaxies through space. This leads us to the conclusion that the universe is expanding. Thus, the idea of a static universe is not supported by observational evidence, and we accept that the universe is dynamic and expanding.

Observations point that the Hubble constant H_0 is not actually constant over time, so H_0 is actually just the current value of the *Hubble parameter* $H(t)$.

§26.2 Friedmann-Lemaître-Robertson-Walker (FLRW) Metric

To describe a dynamic universe that is homogeneous and isotropic, the metric can at most depend on time, *i.e.*,

$$g_{\mu\nu} = g_{\mu\nu}(t).$$

Since the universe is isotropic, the direction $+x$ and $-x$ should be equivalent, so a transformation $x \to -x$ should leave the metric invariant. This implies that there can be no cross terms like $\mathrm{d}t\,\mathrm{d}x$ in the metric. Similar arguments hold for y and z. Also, there can be no cross terms like $\mathrm{d}x\,\mathrm{d}y$, $\mathrm{d}y\,\mathrm{d}z$, or $\mathrm{d}z\,\mathrm{d}x$ in the metric due to same isotropy arguments. Therefore, the most general form of the metric that respects homogeneity and isotropy is

$$\mathrm{d}s^2 = g_{tt}(t)\,\mathrm{d}t^2 + g_{xx}(t)\,\mathrm{d}x^2 + g_{yy}(t)\,\mathrm{d}y^2 + g_{zz}(t)\,\mathrm{d}z^2\,,$$

where $i, j = 1, 2, 3$ correspond to the spatial coordinates x, y, z. Due to isotropy, we have $g_{xx}(t) = g_{yy}(t) = g_{zz}(t)$. So we can write the metric as

$$\mathrm{d}s^2 = g_{tt}(t)\,\mathrm{d}t^2 + a^2(t)\Big[\mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2\Big],$$

where we have defined $a^2(t) = g_{xx}(t)$. Now, we can always rescale the time coordinate t such that $g_{tt}(t) = -1$. This is done by defining a new time coordinate τ as

$$\mathrm{d}\tau = \sqrt{-g_{tt}(t)}\,\mathrm{d}t\,.$$

This new time coordinate τ is called the *cosmic time*, and it represents the proper time experienced by comoving observers (observers who are at rest with respect to the expanding space). With this rescaling, the metric becomes

$$\mathrm{d}s^2 = -\,\mathrm{d}\tau^2 + a^2(\tau)\Big[\mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2\Big]. \tag{26.4}$$

This metric is known as the *Friedmann-Lemaître-Robertson-Walker (FLRW) metric* for a spatially flat universe. Here, $a(\tau)$ is known as the *scale factor*, it will come into play when we discuss the expansion of the universe. For convenience, we will simply denote the cosmic time τ as t from now on.

§26.2.1 Christoffel Symbols for FLRW Metric

So the metric can be written as

$$g_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & a^2 & 0 & 0 \\ 0 & 0 & a^2 & 0 \\ 0 & 0 & 0 & a^2 \end{pmatrix} \qquad g^{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & a^{-2} & 0 & 0 \\ 0 & 0 & a^{-2} & 0 \\ 0 & 0 & 0 & a^{-2} \end{pmatrix}. \tag{26.5}$$

Let's now compute the Christoffel symbols for this metric. Since the metric is only time-dependent, thus at least one of the indices of the Christoffel symbols must be 0 (the time index) to get a non-zero value, so we only need to compute Γ^0_{ij}, Γ^i_{0j}, Γ^i_{00}, Γ^0_{i0}, and Γ^0_{00}. Since $g_{00} = -1$ is constant, we have $\Gamma^0_{00} = 0 = \Gamma^i_{00}$. Let's compute Γ^0_{i0}:

$$\Gamma^0_{i0} = \frac{1}{2}g^{00}(\partial_i g_{00} + \partial_0 g_{i0} - \partial_0 g_{0i}) = 0.$$

Now, let's compute Γ^0_{ij}:

$$\Gamma^0_{ij} = \frac{1}{2}g^{00}\Big(\partial_i g_{0j} + \partial_j g_{i0} - \partial_0 g_{ij}\Big) = -\frac{1}{2}(-1)\partial_0(a^2\delta_{ij}) = a\dot{a}\delta_{ij},$$

where the dot denotes derivative with respect to time t. Next, let's compute Γ^i_{0j}:

$$\Gamma^i_{0j} = \frac{1}{2}g^{ii}\Big(\partial_0 g_{ij} + \partial_j g_{i0} - \partial_j g_{00}\Big) = \frac{1}{2}a^{-2}\partial_0(a^2\delta_{ij}) = \frac{\dot{a}}{a}\delta_{ij}.$$

Thus, the non-zero Christoffel symbols for the FLRW metric are

$$\Gamma^0_{ij} = a\dot{a}\delta_{ij}, \qquad \Gamma^i_{0j} = \frac{\dot{a}}{a}\delta_{ij}. \tag{26.6}$$

§26.2.2 Riemann and Ricci Tensors for FLRW Metric

Using the Christoffel symbols, we can compute the Riemann tensor components. The Riemann tensor is given by

$$R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\sigma\nu} - \partial_\nu \Gamma^\rho_{\sigma\mu} + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\sigma\nu} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\sigma\mu}.$$

The Ricci tensor is obtained by contracting the Riemann tensor:

$$R_{\mu\nu} = R^{\lambda}{}_{\mu\lambda\nu}.$$

Let's compute the non-zero components of the Ricci tensor. First, we compute R_{00}:

$$\begin{aligned} R_{00} = R_{tt} = R^\lambda{}_{0\lambda 0} &= \partial_\lambda \Gamma^\lambda_{00}{}^{\nearrow 0} - \partial_0 \Gamma^\lambda_{\lambda 0} + \Gamma^\lambda_{\lambda\sigma}\Gamma^\sigma_{00}{}^{\nearrow 0} - \Gamma^\lambda_{0\sigma}\Gamma^\sigma_{\lambda 0} \\ &= -\partial_0 \Gamma^i_{i0} - \Gamma^i_{0j}\Gamma^j_{i0} \\ &= -\partial_0\left(\frac{\dot a}{a}\delta^i{}_i\right) - \left(\frac{\dot a}{a}\delta^i{}_j\right)\left(\frac{\dot a}{a}\delta^j{}_i\right) \\ &= -3\left(\frac{\ddot a}{a} - \frac{\dot a^2}{a^2}\right) - 3\frac{\dot a^2}{a^2} \\ &= -3\frac{\ddot a}{a}. \end{aligned}$$

Assignment 09.2.
Show that $R_{xx} = R_{yy} = R_{zz} = a\ddot a + 2\dot a^2$.

Solution.
Let's compute for general R_{ij}:

$$\begin{aligned} R_{ij} = R^\lambda{}_{i\lambda j} &= \partial_\lambda \Gamma^\lambda_{ij} - \partial_j \Gamma^\lambda_{i\lambda} + \Gamma^\lambda_{\lambda\sigma}\Gamma^\sigma_{ij} - \Gamma^\lambda_{j\sigma}\Gamma^\sigma_{i\lambda} \\ &= \partial_0 \Gamma^0_{ji} + \Gamma^k_{k0}\Gamma^0_{ij} - \Gamma^0_{jk}\Gamma^k_{i0} - \Gamma^l_{j0}\Gamma^0_{il} \\ &= \partial_0(a\dot a\delta_{ji}) + \left(\frac{\dot a}{a}\delta^k{}_k\right)(a\dot a\delta_{ij}) - (a\dot a\delta_{jk})\left(\frac{\dot a}{a}\delta_i{}^k\right) - (a\dot a\delta^l{}_j)\left(\frac{\dot a}{a}\delta_{il}\right) \\ &= \left[\partial_0(a\dot a) + 3\dot a^2 - \dot a^2 - \dot a^2\right]\delta_{ij} \\ &= \left(a\ddot a + 2\dot a^2\right)\delta_{ij}. \end{aligned}$$

Thus, we have $R_{xx} = R_{yy} = R_{zz} = a\ddot a + 2\dot a^2$. ◂

Thus, the non-zero components of the Ricci tensor for the FLRW metric are

$$R_{tt} = -3\frac{\ddot a}{a}, \qquad R_{ij} = \left(a\ddot a + 2\dot a^2\right)\delta_{ij}. \tag{26.7}$$

§26.2.3 Ricci Scalar for FLRW Metric

The Ricci scalar is obtained by contracting the Ricci tensor with the inverse metric:

$$R = g^{\mu\nu}R_{\mu\nu}.$$

Let's compute the Ricci scalar for the FLRW metric:

$$\begin{aligned} R &= g^{tt}R_{tt} + g^{ij}R_{ij} \\ &= (-1)\left(-3\frac{\ddot a}{a}\right) + a^{-2}\left(a\ddot a + 2\dot a^2\right)\delta^i{}_j\delta_i{}^j \\ &= 3\frac{\ddot a}{a} + 3\left(\frac{\ddot a}{a} + 2\frac{\dot a^2}{a^2}\right) = 6\left(\frac{\ddot a}{a} + \frac{\dot a^2}{a^2}\right). \end{aligned}$$

Thus, the Ricci scalar for the FLRW metric is

$$R = 6\left(\frac{\ddot a}{a} + \frac{\dot a^2}{a^2}\right). \tag{26.8}$$

§26.2.4 Einstein Tensor for FLRW Metric

The Einstein tensor is given by

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}.$$

Let's compute the non-zero components of the Einstein tensor for the FLRW metric. First, we compute G_{tt}:

$$\begin{aligned} G_{tt} &= R_{tt} - \frac{1}{2}Rg_{tt} \\ &= -3\frac{\ddot{a}}{a} - \frac{1}{2}\left[6\left(\frac{\ddot{a}}{a} + \frac{\dot{a}^2}{a^2}\right)\right](-1) \\ &= -3\frac{\ddot{a}}{a} + 3\frac{\ddot{a}}{a} + 3\frac{\dot{a}^2}{a^2} \\ &= 3\frac{\dot{a}^2}{a^2}. \end{aligned}$$

Let's compute G_{ij}:

$$\begin{aligned} G_{ij} &= R_{ij} - \frac{1}{2}Rg_{ij} \\ &= \left(a\ddot{a} + 2\dot{a}^2\right)\delta_{ij} - \frac{1}{2}\left[6\left(\frac{\ddot{a}}{a} + \frac{\dot{a}^2}{a^2}\right)\right](a^2\delta_{ij}) \\ &= \left(a\ddot{a} + 2\dot{a}^2 - 3a\ddot{a} - 3\dot{a}^2\right)\delta_{ij} \\ &= -\left(2a\ddot{a} + \dot{a}^2\right)\delta_{ij}. \end{aligned}$$

Thus, the non-zero components of the Einstein tensor for the FLRW metric are

$$G_{tt} = 3\frac{\dot{a}^2}{a^2}, \qquad G_{ij} = -\left(2a\ddot{a} + \dot{a}^2\right)\delta_{ij}. \tag{26.9}$$

§26.3 Einstein's Field Equations for FLRW Metric

In the last lecture, we have argued that on large scales, the matter content of the universe can be modeled as a perfect fluid. The energy-momentum tensor for a perfect fluid is given by

$$T_{\mu\nu} = (\rho + P)u_\mu u_\nu + Pg_{\mu\nu}, \tag{26.10}$$

where ρ is the energy density, P is the pressure, and u^μ is the four-velocity of the fluid elements. Let's now compute the four-velocity, u^μ, for comoving observers in the FLRW metric. Comoving observers are those who are at rest with respect to the expanding space, so their spatial coordinates x, y, z are constant. Therefore, their four-velocity is given by

$$u^\mu = \frac{\mathrm{d}x^\mu}{\mathrm{d}\tau} = \frac{\mathrm{d}t}{\mathrm{d}\tau}(1,0,0,0) = (1,0,0,0),$$

where we have used the fact that for comoving observers, $\frac{\mathrm{d}t}{\mathrm{d}\tau} = 1$ since t is the proper time for these observers. Thus, the energy-momentum tensor for a perfect fluid in the FLRW metric becomes

$$T_{\mu\nu} = \begin{pmatrix} \rho & 0 & 0 & 0 \\ 0 & Pa^2 & 0 & 0 \\ 0 & 0 & Pa^2 & 0 \\ 0 & 0 & 0 & Pa^2 \end{pmatrix}. \tag{26.11}$$

Now, let's write down $t-t$ component of Einstein's field equations:

$$G_{tt} = 8\pi G T_{tt} \quad \Rightarrow \quad 3\frac{\dot{a}^2}{a^2} = 8\pi G\rho \quad \Rightarrow \quad \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho. \tag{26.12}$$

This equation is known as the *Friedmann equation*.

Lec 27: Minimal Coupling of Scalar Field to Gravity

In the last lecture, we have derived the Friedmann equations using the Einstein field equations. Today, we will first look at the $i-i$ component of the Einstein equations to derive the acceleration equation. Then, we will look at the interpretation of the energy-momentum tensor for a perfect fluid using basic scalar field theory.

§27.1 Einstein's Field Equations for FLRW Metric (continued)

We will now look at the $i-i$ component of the Einstein field equations for the FLRW metric. We have

$$G_{ii} = 8\pi G T_{ii} \quad \Rightarrow \quad -(2a\ddot{a} + \dot{a}^2) = 8\pi G a^2 P \quad \Rightarrow \quad 2\frac{\ddot{a}}{a} + \left(\frac{\dot{a}}{a}\right)^2 = -8\pi G P.$$

Using the first Friedmann equation, we can rewrite this as

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3P). \tag{27.1}$$

This is known as the *Raychaudhuri equation* or the *acceleration equation*.

Assignment 10.1.
Using eqns. (26.12) and (27.1), show that the continuity equation takes the form

$$\dot{\rho} + 3\left(\frac{\dot{a}}{a}\right)(\rho + P) = 0. \tag{27.2}$$

Solution.
Starting with the Friedmann equation eqn. (26.12), we have

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho \quad \xrightarrow{\text{differentiate w.r.t. } t} \quad 2\left(\frac{\dot{a}}{a}\right)\left(\frac{\ddot{a}}{a} - \left(\frac{\dot{a}}{a}\right)^2\right) = \frac{8\pi G}{3}\dot{\rho}.$$

Using the Raychaudhuri equation eqn. (27.1) to substitute for $\frac{\ddot{a}}{a}$, we get

$$2\left(\frac{\dot{a}}{a}\right)\left(-\frac{4\pi G}{3}(\rho + 3P) - \frac{8\pi G}{3}\rho\right) = \frac{8\pi G}{3}\dot{\rho} \quad \Rightarrow \quad -\left(\frac{\dot{a}}{a}\right)((\rho + 3P) + 2\rho) = \dot{\rho}.$$

Rearranging, we get

$$\dot{\rho} + 3\left(\frac{\dot{a}}{a}\right)(\rho + P) = 0,$$

◀

We can also derive the continuity equation directly from the covariant conservation of the energy-momentum tensor, *i.e.*, $\nabla_\mu T^{\mu\nu} = 0$.

Assignment 10.2.
Verify that the continuity equation eqn. (27.2) can also be obtained from the covariant conservation of the stress-energy tensor, $\nabla_\mu T^{\mu\nu} = 0$.

Solution.
The covariant conservation of the stress-energy tensor is set of four equations, one for each value of ν. We will look at the $\nu = t$ equation, as the components of the stress-energy tensor are time-dependent only, we have

$$\nabla_\mu T^{\mu t} = 0 \quad \Rightarrow \quad \partial_t T^{tt} + \Gamma^\mu_{\mu\lambda} T^{\lambda t} + \Gamma^t_{\mu\lambda} T^{\mu\lambda} = 0.$$

Now, we have $T^{tt} = \rho$ and $T^{it} = 0$ for $i = x, y, z$, so only the term with $\lambda = t$ and $\mu = i$ sums will contribute in the second term, for the third term only the term with $\mu = i$ and $\lambda = j$ will contribute. Thus, we have

$$\dot{\rho} + \Gamma^i_{it}(\rho) + \Gamma^t_{ij}(Pa^{-2}\delta^{ij}) = 0.$$

Recall that for the FLRW metric, the non-zero Christoffel symbols are given in eqn. (26.6), so we have

$$\dot{\rho} + \left(\frac{\dot{a}}{a}\delta^i{}_i\right)\rho + (a\dot{a}\delta_{ij})(Pa^{-2}\delta^{ij}) = 0 \quad \Rightarrow \quad \dot{\rho} + 3\frac{\dot{a}}{a}\rho + 3\frac{\dot{a}}{a}P = 0.$$

Rearranging, we get

$$\dot{\rho} + 3\left(\frac{\dot{a}}{a}\right)(\rho + P) = 0,$$

◀

Till now, we have just defined the stress-energy tensor as extra terms obtained by varying the matter action with respect to the metric.

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta S_{\mathrm{M}}}{\delta g^{\mu\nu}}.$$

But what is the physical meaning of this stress-energy tensor? To understand this, we will look at a simple model for matter: a scalar field ϕ.

§27.2 Scalar Field in Curved Spacetime

In flat spacetime (Minkowski spacetime), the action for a scalar field ϕ is given by

$$S_\phi = \int \mathrm{d}^4x \left[-\frac{1}{2}(\partial^\mu\phi)(\partial_\mu\phi) - V(\phi)\right] = \int \mathrm{d}^4x \left[-\frac{1}{2}\eta^{\mu\nu}(\partial_\mu\phi)(\partial_\nu\phi) - V(\phi)\right], \tag{27.3}$$

where $V(\phi)$ is the potential energy density of the scalar field. The minimal coupling principle states that there is no extra coupling between matter fields and gravity other than through the metric, *i.e.*, we have no direct coupling between the matter fields and the curvature tensors.

So to minimally couple the matter action to gravity, we replace the Minkowski metric $\eta^{\mu\nu}$ with the general metric $g^{\mu\nu}$, replace the volume element d^4x with general covariant volume element $\sqrt{-g}\,\mathrm{d}^4x$, and we also replace the partial derivatives with appropriate covariant derivatives. Thus, the action for a scalar field in curved spacetime is given by

$$S_\phi = \int \mathrm{d}^4x\,\sqrt{-g}\left[-\frac{1}{2}g^{\mu\nu}(\nabla_\mu\phi)(\nabla_\nu\phi) - V(\phi)\right]. \tag{27.4}$$

For a scalar field, the covariant derivative is the same as the partial derivative, *i.e.*, $\nabla_\mu\phi = \partial_\mu\phi$. Therefore, we can write the action as

$$S_\phi = \int \mathrm{d}^4x\,\sqrt{-g}\left[-\frac{1}{2}g^{\mu\nu}(\partial_\mu\phi)(\partial_\nu\phi) - V(\phi)\right].$$

For a homogeneous universe, the scalar field ϕ is only a function of time t, *i.e.*, $\phi = \phi(t)$. Therefore, the action reduces to

$$S_\phi = \int \mathrm{d}^4x\,\sqrt{-g}\left[-\frac{1}{2}g^{tt}\dot{\phi}^2 - V(\phi)\right].$$

Define the Lagrangian L_ϕ as

$$L_\phi = \int \mathrm{d}^3\mathbf{x}\,\sqrt{-g}\left[\frac{1}{2}\dot{\phi}^2 - V(\phi)\right] = \left[\frac{1}{2}\dot{\phi}^2 - V(\phi)\right]\left\{a^3(t)\int \mathrm{d}^3\mathbf{x}\right\},$$

where we have used $\sqrt{-g} = a^3(t)$ for the FLRW metric. We have argued that even if the universe is infinite, only finite portion of it is observable (or causally connected to us) due to the finite age of the universe and the finite speed of light. So we can consider a finite comoving volume (coordinate volume) $V_c = \int \mathrm{d}^3\mathbf{x}$. Therefore, we can write the Lagrangian as

$$L_\phi = \left[\frac{1}{2}\dot{\phi}^2 - V(\phi)\right]V_0,$$

where $V_0 = a^3(t)V_c$ is the physical volume. The action can now be written as

$$S_\phi = \int \mathrm{d}t\, L_\phi = \int \mathrm{d}t\, V_0 \left[\frac{1}{2}\dot{\phi}^2 - V(\phi)\right].$$

Since we want to relate this model to matter in the universe, we need a quantity that behaves like energy density. The Hamiltonian H_ϕ is the quantity we are looking for. The conjugate momentum π_ϕ corresponding to the field ϕ is given by

$$\pi_\phi = \frac{\partial L_\phi}{\partial \dot{\phi}} = V_0 \dot{\phi}.$$

The Hamiltonian is given by Legendre transform of the Lagrangian:

$$H_\phi = \pi_\phi \dot{\phi} - L_\phi = V_0 \left[\frac{1}{2}\dot{\phi}^2 + V(\phi)\right] = V_0 \left[\frac{1}{2}\left(\frac{\pi_\phi}{V_0}\right)^2 + V(\phi)\right].$$

Thus, the energy density ρ_ϕ of the scalar field is given by

$$\rho_\phi = \frac{H_\phi}{V_0} = \frac{1}{2}\left(\frac{\pi_\phi}{V_0}\right)^2 + V(\phi) = \frac{1}{2}\dot{\phi}^2 + V(\phi).$$

Similarly, the pressure P_ϕ of the scalar field can be obtained using the standard thermodynamic relation

$$P = -\frac{\partial E}{\partial V} \quad \Rightarrow \quad P_\phi = -\frac{\partial H_\phi}{\partial V_0} = -\frac{\partial}{\partial V_0}\left\{V_0 \left[\frac{1}{2}\left(\frac{\pi_\phi}{V_0}\right)^2 + V(\phi)\right]\right\} = \frac{1}{2}\dot{\phi}^2 - V(\phi).$$

Thus, we have

$$\rho_\phi = \frac{1}{2}\dot{\phi}^2 + V(\phi), \quad P_\phi = \frac{1}{2}\dot{\phi}^2 - V(\phi).$$

We obtained these expressions for energy density and pressure of a scalar field by considering the Hamiltonian and using thermodynamic relations. Now let's derive the same results using the stress-energy tensor.

Assignment 10.3.
Show that the stress-energy tensor for the scalar field is given by

$$T_{\mu\nu} = (\partial_\mu \phi)(\partial_\nu \phi) + g_{\mu\nu}\left[-\frac{1}{2} g^{\alpha\beta}(\partial_\alpha \phi)(\partial_\beta \phi) - V(\phi)\right]. \tag{27.5}$$

Solution.
To compute the stress-energy tensor, we start with the definition

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta S_\phi}{\delta g^{\mu\nu}}.$$

The variation of the action S_ϕ with respect to the metric $g^{\mu\nu}$ is given by

$$\delta S_\phi = \int \mathrm{d}^4x \left\{\delta(\sqrt{-g})\left[-\frac{1}{2} g^{\alpha\beta}(\partial_\alpha \phi)(\partial_\beta \phi) - V(\phi)\right] + \sqrt{-g}\,\delta\left[-\frac{1}{2} g^{\alpha\beta}(\partial_\alpha \phi)(\partial_\beta \phi) - V(\phi)\right]\right\}$$

Using the eqn. (17.6), we have

$$= \int \mathrm{d}^4x \left\{-\frac{1}{2}\sqrt{-g} g_{\mu\nu}\, \delta g^{\mu\nu}\left[-\frac{1}{2} g^{\alpha\beta}(\partial_\alpha \phi)(\partial_\beta \phi) - V(\phi)\right] + \sqrt{-g}\left[-\frac{1}{2}(\partial_\mu \phi)(\partial_\nu \phi)\, \delta g^{\mu\nu}\right]\right\}$$

$$= -\frac{1}{2}\int \mathrm{d}^4x \sqrt{-g}\left\{(\partial_\mu \phi)(\partial_\nu \phi) + g_{\mu\nu}\left[-\frac{1}{2} g^{\alpha\beta}(\partial_\alpha \phi)(\partial_\beta \phi) - V(\phi)\right]\right\}\delta g^{\mu\nu}.$$

Therefore, we have

$$\frac{\delta S_\phi}{\delta g^{\mu\nu}} = -\frac{1}{2}\sqrt{-g}\left\{(\partial_\mu \phi)(\partial_\nu \phi) + g_{\mu\nu}\left[-\frac{1}{2} g^{\alpha\beta}(\partial_\alpha \phi)(\partial_\beta \phi) - V(\phi)\right]\right\}.$$

Plugging this back into the definition of the stress-energy tensor, we get

$$T_{\mu\nu} = (\partial_\mu \phi)(\partial_\nu \phi) + g_{\mu\nu}\left[-\frac{1}{2} g^{\alpha\beta}(\partial_\alpha \phi)(\partial_\beta \phi) - V(\phi)\right].$$

◀

Using this expression for the stress-energy tensor, let's compute the diagonal components for a homogeneous scalar field $\phi = \phi(t)$

$$T_{tt} = \dot{\phi}^2 + g_{tt}\left[-\frac{1}{2}g^{tt}\dot{\phi}^2 - V(\phi)\right] = \frac{1}{2}\dot{\phi}^2 + V(\phi) = \rho_\phi,$$
$$T_{ii} = 0 + g_{ii}\left[-\frac{1}{2}g^{tt}\dot{\phi}^2 - V(\phi)\right] = a^2(t)\left[\frac{1}{2}\dot{\phi}^2 - V(\phi)\right] = a^2 P_\phi.$$

Now recall that for a perfect fluid, the stress-energy tensor is given by

$$T_{\mu\nu} = \text{diag}(\rho, Pa^2, Pa^2, Pa^2).$$

Thus, we see that the energy density and pressure computed using the Hamiltonian and thermodynamic relations match exactly with those obtained from the stress-energy tensor, which also points to the fact the field theory model is fundamental as it reproduces the thermodynamic properties of matter in the universe.

Lec 28: Coupling EM Field to Gravity and Radiation Dominated Universe

In the last lecture, we have seen that how minimal coupling prescription helps us to understand the energy-momentum tensor for a perfect fluid using scalar field Lagrangian. Today, we will solve the Einstein's field equations for the FLRW metric where the matter content is given by electromagnetic field.

§28.1 Electromagnetic Field in Curved Spacetime

The action for free electromagnetic field in flat spacetime is given by

$$S_{\text{EM}} = \int \mathrm{d}^4x \left[-\frac{1}{4}F^{\mu\nu}F_{\mu\nu}\right] = \int \mathrm{d}^4x \left[-\frac{1}{4}\eta^{\mu\alpha}\eta^{\nu\beta}F_{\alpha\beta}F_{\mu\nu}\right], \tag{28.1}$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is the Maxwell field strength tensor. To generalize this to curved spacetime, we use the minimal coupling prescription:

$$S_{\text{EM}} = \int \mathrm{d}^4x \sqrt{-g}\left[-\frac{1}{4}g^{\mu\alpha}g^{\nu\beta}F_{\alpha\beta}F_{\mu\nu}\right], \tag{28.2}$$

where $F_{\mu\nu} = \nabla_\mu A_\nu - \nabla_\nu A_\mu$. However, since the covariant derivative of a vector field is given by

$$\nabla_\mu A_\nu = \partial_\mu A_\nu - \Gamma^\lambda_{\mu\nu}A_\lambda,$$

and the Christoffel symbols are symmetric in the lower indices, we have

$$F_{\mu\nu} = \nabla_\mu A_\nu - \nabla_\nu A_\mu = \left(\partial_\mu A_\nu - \Gamma^\lambda_{\mu\nu}A_\lambda\right) - \left(\partial_\nu A_\mu - \Gamma^\lambda_{\nu\mu}A_\lambda\right) = \partial_\mu A_\nu - \partial_\nu A_\mu. \tag{28.3}$$

Assignment 10.4.
Show that the energy-momentum tensor for electromagnetic field in curved spacetime is given by

$$T_{\mu\nu} = g^{\alpha\beta}F_{\mu\alpha}F_{\nu\beta} - \frac{1}{4}g_{\mu\nu}F^{\alpha\beta}F_{\alpha\beta}. \tag{28.4}$$

Solution.
The energy-momentum tensor is given by

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta S_{\text{EM}}}{\delta g^{\mu\nu}}.$$

To compute the functional derivative, let's first write take the variation of the action:

$$\begin{aligned}
\delta S_{\text{EM}} &= \int \mathrm{d}^4x \left[\delta(\sqrt{-g})\left(-\frac{1}{4}g^{\mu\alpha}g^{\nu\beta}F_{\alpha\beta}F_{\mu\nu}\right) + \sqrt{-g}\delta\left(-\frac{1}{4}g^{\mu\alpha}g^{\nu\beta}F_{\alpha\beta}F_{\mu\nu}\right)\right] \\
&= \int \mathrm{d}^4x \left\{-\frac{1}{2}\sqrt{-g}g_{\mu\nu}\,\delta g^{\mu\nu}\left(-\frac{1}{4}F^{\alpha\beta}F_{\alpha\beta}\right) + \sqrt{-g}\left(-\frac{1}{4}\right)\left[\underbrace{\delta g^{\mu\alpha}\,g^{\nu\beta}F_{\alpha\beta}F_{\mu\nu}}_{\alpha\leftrightarrow\nu} + \underbrace{g^{\mu\alpha}\,\delta g^{\nu\beta}\,F_{\alpha\beta}F_{\mu\nu}}_{\beta\leftrightarrow\mu}\right]\right\}
\end{aligned}$$

Renaming the dummy indices in the second and third terms, as indicated, we have

$$= \int \mathrm{d}^4x \left\{-\frac{1}{2}\sqrt{-g}g_{\mu\nu}\,\delta g^{\mu\nu}\left(-\frac{1}{4}F^{\alpha\beta}F_{\alpha\beta}\right) + \sqrt{-g}\left(-\frac{1}{4}\right)\left[\delta g^{\mu\nu}\,g^{\alpha\beta}F_{\nu\beta}F_{\mu\alpha} + g^{\beta\alpha}\,\delta g^{\nu\mu}\,F_{\alpha\mu}F_{\beta\nu}\right]\right\}$$

Using the symmetry of the metric tensor, $\delta g^{\nu\mu} = \delta g^{\mu\nu}$, we get

$$= \int \mathrm{d}^4x \left(-\frac{1}{2}\sqrt{-g}\right)\left\{g_{\mu\nu}\left(-\frac{1}{4}F^{\alpha\beta}F_{\alpha\beta}\right) + \frac{1}{2}\left[g^{\alpha\beta}F_{\nu\beta}F_{\mu\alpha} + g^{\beta\alpha}F_{\alpha\mu}F_{\beta\nu}\right]\right\}\delta g^{\mu\nu}$$

since the field strength tensor is antisymmetric, $F_{\mu\nu} = -F_{\nu\mu}$, the two terms in the square brackets are equal. Thus, we have

$$= \int \mathrm{d}^4x \left(-\frac{1}{2}\sqrt{-g}\right)\left\{g_{\mu\nu}\left(-\frac{1}{4}F^{\alpha\beta}F_{\alpha\beta}\right) + g^{\alpha\beta}F_{\mu\alpha}F_{\nu\beta}\right\}\delta g^{\mu\nu}\,.$$

Thus, we get

$$\frac{\delta S_{\mathrm{EM}}}{\delta g^{\mu\nu}} = -\frac{1}{2}\sqrt{-g}\left\{g_{\mu\nu}\left(-\frac{1}{4}F^{\alpha\beta}F_{\alpha\beta}\right) + g^{\alpha\beta}F_{\mu\alpha}F_{\nu\beta}\right\}.$$

Therefore, the energy-momentum tensor is given by

$$T_{\mu\nu} = g^{\alpha\beta}F_{\mu\alpha}F_{\nu\beta} + g_{\mu\nu}\left(-\frac{1}{4}F^{\alpha\beta}F_{\alpha\beta}\right).$$

◀

Let's first see a property of this energy-momentum tensor. Taking the trace, we have

$$\begin{aligned} g^{\mu\nu}T_{\mu\nu} &= g^{\mu\nu}\left(g^{\alpha\beta}F_{\mu\alpha}F_{\nu\beta} - \frac{1}{4}g_{\mu\nu}F^{\alpha\beta}F_{\alpha\beta}\right) \\ &= F^{\nu\beta}F_{\nu\beta} - \frac{1}{4}\underbrace{g^{\mu\nu}g_{\mu\nu}}_{=4}F^{\alpha\beta}F_{\alpha\beta} \\ &= F^{\nu\beta}F_{\nu\beta} - F^{\alpha\beta}F_{\alpha\beta} = 0. \end{aligned}$$

Thus, the energy-momentum tensor for electromagnetic field is traceless.

This observation is not just true for electromagnetic field, but for any massless field, *i.e.*, fields whose quanta are massless particles (like photons) have traceless energy-momentum tensor.

§28.2 Radiation Dominated Universe

Till now, we have considered the universe to be dynamic (expanding), isotropic and homogeneous, with matter content modelled as a perfect fluid. Consider the case when the matter content is dominated by radiation (photons). Since the energy-momentum tensor for electromagnetic field is traceless, we impose the traceless condition on the perfect fluid energy-momentum tensor:

$$\begin{aligned} g^{\mu\nu}T_{\mu\nu} &= g^{\mu\nu}\Big[(\rho + P)u_\mu u_\nu + P g_{\mu\nu}\Big] \\ &= (\rho + P)\underbrace{g^{\mu\nu}u_\mu u_\nu}_{=-1} + P\underbrace{g^{\mu\nu}g_{\mu\nu}}_{=4} \\ &= -(\rho + P) + 4P = 0. \end{aligned}$$

This gives us the equation of state for radiation[1]

$$P = \frac{1}{3}\rho. \tag{28.5}$$

This is consistent with the equation of state for a gas of photons in thermodynamic equilibrium. With this, we can define the parameter w known as the equation of state parameter:

$$w = \frac{P}{\rho} = \frac{1}{3}. \tag{28.6}$$

Using this equation of state in the continuity equation eqn. (27.2), we have

$$\dot\rho + 3\frac{\dot a}{a}(\rho + P) = \dot\rho + 3\frac{\dot a}{a}\left(\rho + \frac{1}{3}\rho\right) = \dot\rho + 4\left(\frac{\dot a}{a}\right)\rho = 0.$$

We get

$$\frac{\mathrm{d}\rho}{\rho} = -4\frac{\mathrm{d}a}{a} \quad \Rightarrow \quad \ln\left(\frac{\rho}{\rho_0}\right) = -4\ln\left(\frac{a}{a_0}\right) \quad \Rightarrow \quad \rho = \rho_0\left(\frac{a_0}{a}\right)^4, \tag{28.7}$$

[1] This equation of state (EoS) tells us that radiation is Barotropic fluid.

where ρ_0 and a_0 are the energy density and scale factor at some reference time t_0. Using this in the Friedmann equation eqn. (26.12), we have

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho_0\left(\frac{a_0}{a}\right)^4.$$

Take square root on both sides and consider the positive root (since the universe is expanding, $\dot{a} > 0$):

$$\frac{\dot{a}}{a} = \sqrt{\frac{8\pi G\rho_0}{3}a_0^4}\frac{1}{a^2}.$$

Define the square root term as a constant $c_0^2/2$, we have

$$2a\dot{a} = c_0^2 \quad \Rightarrow \quad \mathrm{d}\left(a^2\right) = c_0^2\,\mathrm{d}t\,.$$

Integrating both sides, we get

$$a^2 = c_0^2(t - t_0) + a_0^2,$$

setting $a_0 = 0$. This gives us the scale factor for a radiation dominated universe:

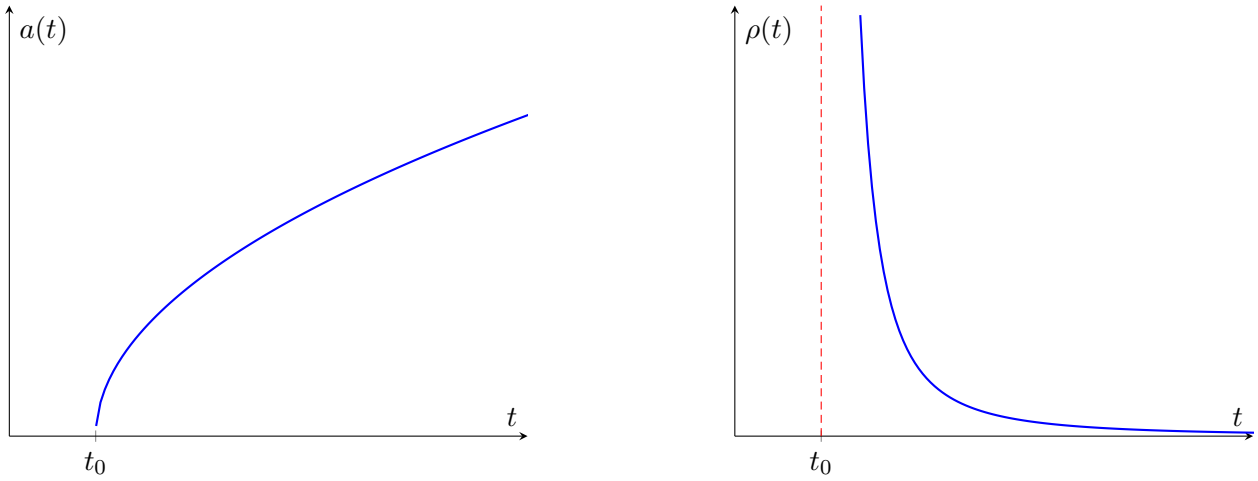
$$a(t) = c_0\sqrt{t - t_0}. \tag{28.8}$$

Recall that the physical distance between two comoving points is given by

$$d(t) = a(t)l,$$

where l is the coordinate (comoving) distance between the two points.

Let's look at the behavior of the scale factor as a function of time.



(a) Scale factor $a(t)$ **(b)** Energy density $\rho(t)$

Figure 28.1: Scale factor and energy density in a radiation dominated universe

From fig. 28.1a, we see that $a(t) \to 0$ as $t \to t_0$ (for some finite time t_0). This implies that the physical distance $d(t) \to 0$ as $t \to t_0$. This means that at time $t = t_0$, all the matter in the universe was concentrated at a single point (since physical distances between all points were zero). Also, the time stops at $t = t_0$ as we cannot extend the scale factor to $t < t_0$ (as it becomes imaginary). So the universe "was born" at $t = t_0$, and has a finite age. This instant $t = t_0$ is called the *Big Bang Singularity*. Note that this singularity is not a point in space, but a point in time when the scale factor vanishes.

Also note that the energy density as a function of time is given by

$$\rho(t) = \rho_0\left(\frac{a_0}{a(t)}\right)^4 = \rho_0\left(\frac{a_0}{c_0\sqrt{t - t_0}}\right)^4 = \frac{\rho_0 a_0^4}{c_0^4}\frac{1}{(t - t_0)^2}.$$

We see from fig. 28.1b that $\rho(t) \to \infty$ as $t \to t_0$. This means that at the Big Bang Singularity, the energy density of the universe was infinite. This indicates that General Relativity (classical theory) breaks down at this point. People believe that a quantum theory of gravity will hopefully rescue it like it did for Hydrogen atom in Quantum Mechanics.

Assignment 10.5.
Solve the Friedmann equations for pressure-less dust matter ($P = 0$). Plot the scale factor $a(t)$ and energy density $\rho(t)$ as functions of time t.

Solution.
For pressure-less dust matter, the equation of state is given by $P = 0$, which gives us the equation of state parameter $w = 0$. Using this in the continuity equation eqn. (27.2), we have

$$\dot{\rho} + 3\frac{\dot{a}}{a}(\rho + P) = \dot{\rho} + 3\frac{\dot{a}}{a}\rho = 0.$$

This gives us

$$\frac{\mathrm{d}\rho}{\rho} = -3\frac{\mathrm{d}a}{a} \quad \Rightarrow \quad \ln\left(\frac{\rho}{\rho_0}\right) = -3\ln\left(\frac{a}{a_0}\right) \quad \Rightarrow \quad \rho = \rho_0\left(\frac{a_0}{a}\right)^3,$$

where ρ_0 and a_0 are the energy density and scale factor at some reference time t_0. Using this in the Friedmann equation eqn. (26.12), we have

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho_0\left(\frac{a_0}{a}\right)^3.$$

Taking square root on both sides and considering the positive root, we have

$$\frac{\dot{a}}{a} = \sqrt{\frac{8\pi G\rho_0}{3}a_0^3}\frac{1}{a^{3/2}} \quad \Rightarrow \quad \frac{\dot{a}}{a} = \frac{2}{3}c_1^{3/2}\frac{1}{a^{3/2}} \quad \Rightarrow \quad \frac{3}{2}a^{1/2}\dot{a} = c_1^{3/2},$$

where we have defined the constant $\sqrt{\frac{8\pi G\rho_0}{3}a_0^3} = \frac{2}{3}c_1^{3/2}$. Integrating both sides, we get

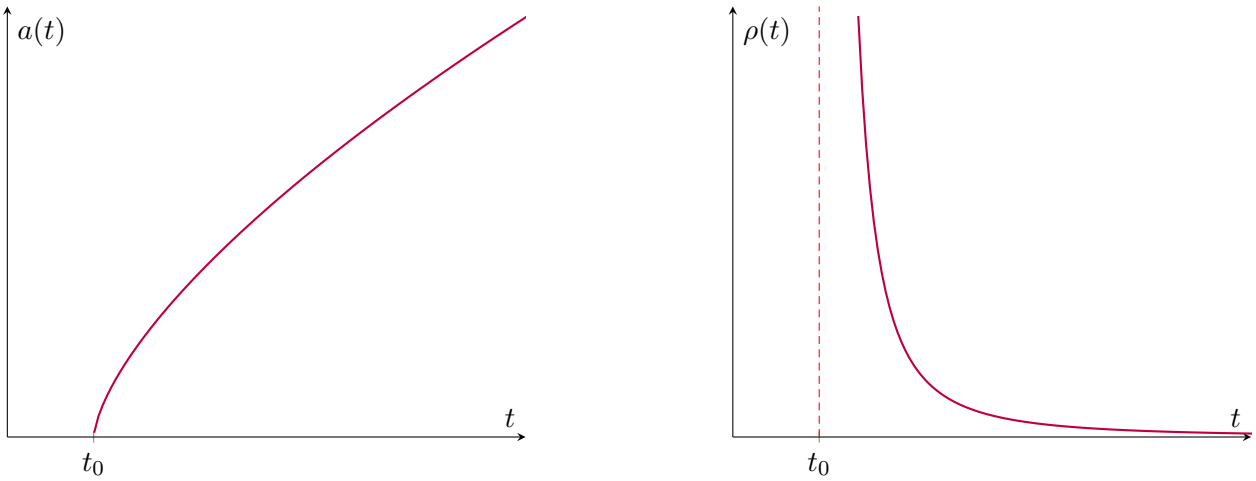
$$\mathrm{d}\left(a^{3/2}\right) = c_1^{3/2}\,\mathrm{d}t \quad \Rightarrow \quad a^{3/2} = c_1^{3/2}(t - t_0) + a_0^{3/2}.$$

Setting $a_0 = 0$, we have the scale factor for a pressure-less dust matter dominated universe:

$$a(t) = c_1(t - t_0)^{2/3}.$$

The energy density as a function of time is given by

$$\rho(t) = \rho_0\left(\frac{a_0}{a(t)}\right)^3 = \rho_0\left(\frac{a_0}{c_1(t - t_0)^{2/3}}\right)^3 = \frac{\rho_0 a_0^3}{c_1^3}\frac{1}{(t - t_0)^2}.$$



(a) Scale factor $a(t)$ **(b)** Energy density $\rho(t)$

Figure 28.2: Scale factor and energy density in a pressure-less dust matter dominated universe

From fig. 28.2a, we see that $a(t) \to 0$ as $t \to t_0$, indicating a Big Bang Singularity at $t = t_0$. Also, from fig. 28.2b, we see that $\rho(t) \to \infty$ as $t \to t_0$, indicating infinite energy density at the Big Bang Singularity. ◂

Lec 29: Accelerating Universe

We have seen that according to Hubble's law, the recession velocity of distant galaxies is proportional to their distance from us, *i.e.*,

$$v = H_0 d,$$

where H_0 is the Hubble constant. Also observations indicate that the Hubble constant is not actually constant but varies with time. To account for this, we introduce the Hubble parameter $H(t)$, defined as

$$H(t) = \frac{v}{d}$$

at any given time t, where v is the recession velocity of a galaxy at distance d at that time. We know that the recession velocity v is the rate of change of the proper distance d with respect to time, *i.e.*,

$$v = \frac{\mathrm{d}d}{\mathrm{d}t} = l\frac{\mathrm{d}a}{\mathrm{d}t} = l\dot{a},$$

where l is the comoving distance of the galaxy and $a(t)$ is the scale factor of the universe at time t. Therefore, we can express the Hubble parameter as

$$H(t) = \frac{\dot{a}}{a}.$$

Recall that the Friedmann equation relates the quantity $(\dot{a}/a)^2$ to the energy density ρ of the universe, *i.e.*,

$$3H^2 = 8\pi G\rho.$$

The current value of the Hubble parameter, known as the Hubble constant H_0, is approximately 70 km s^{-1} Mpc^{-1}.

> Remark 29.1 (Hubble Tension). There is currently a discrepancy in the measured values of the Hubble constant from different observational methods. Measurements based on the cosmic microwave background (CMB) radiation suggest a value of about 67.4 km s^{-1} Mpc^{-1}, while measurements based on local distance indicators, such as Cepheid variables and Type Ia supernovae, suggest a higher value of about 73.5 km s^{-1} Mpc^{-1}. This discrepancy is known as the "Hubble tension" and is an active area of research in cosmology.

Using this value, we can estimate the total energy density of the universe today as

$$1 = \frac{8\pi G}{3H_0^2}\rho_{\text{total}} \quad \Rightarrow \quad \frac{\rho_{\text{total}}}{\rho_H} = 1,$$

where $\rho_H = 3H_0^2/(8\pi G)$. This implies that the total energy density of the universe is equal to the critical density ρ_H. However, the observed (baryonic) matter density of the universe is only $\sim 0.05\rho_H$. This discrepancy suggests the presence of additional components of energy density in the universe.

We know that light bends in a gravitational field, and galaxies (or clusters of galaxies) can act as gravitational lenses, bending the light from more distant objects. Observations of gravitational lensing indicate that there is more mass present in galaxies and clusters than can be accounted for by visible (baryonic) matter alone. This additional mass is referred to as "*dark matter*," to account for the observed gravitational effects it is estimated that dark matter constitutes about 25% of the total energy density of the universe, or $\sim 0.25\rho_H$.

The remaining 70% of the energy density of the universe is attributed to a mysterious form of energy known as "*dark energy*."

The scale factor $a(t)$ quantifies the expansion of the universe over time, and its second derivative $\ddot{a}(t)$ indicates whether the expansion is accelerating or decelerating. Raychaudhuri's equation relates $\ddot{a}/a$ to the energy density ρ and pressure P of the universe:

$$3\left(\frac{\ddot{a}}{a}\right) = -4\pi G(\rho + 3P).$$

Since the gravitational force is attractive, we expect the expansion of the universe to decelerate over time, which implies $\ddot{a} < 0$. Define deceleration parameter q as

$$q = \frac{-(\ddot{a}/a)}{(\dot{a}/a)^2} = \frac{1+3w}{2},$$

where $w = P/\rho$ is the equation of state parameter. For normal matter and energy, we have $w \geq 0$, which leads to $q > 0$, indicating a decelerating expansion.

According to "strong energy condition," we have $w \geq -1/3$, which also implies $q \geq 0$. However, in 1998, observations of distant Type Ia supernovae revealed that the expansion of the universe is actually accelerating, which means $\ddot{a} > 0$ and thus $q < 0$. This unexpected result suggests the presence of a form of energy with negative pressure, which we refer to as dark energy.

§29.1 Cosmological Constant

Recall the Einstein-Hilbert action,

$$S = \frac{1}{16\pi G}\int \mathrm{d}^4x\,\sqrt{-g}(R - 2\Lambda) + S_{\text{matter}},$$

where Λ is the cosmological constant. Till now, we have set $\Lambda = 0$ for simplicity. However, we can consider a non-zero cosmological constant, and treat it as another matter field. Thus, the action becomes

$$S = \frac{1}{16\pi G}\int \mathrm{d}^4x\,\sqrt{-g}R + S_\Lambda + S_{\text{matter}},$$

where

$$S_\Lambda = -\frac{1}{8\pi G}\int \mathrm{d}^4x\,\sqrt{-g}\Lambda.$$

Varying this action with respect to the metric $g_{\mu\nu}$, we obtain the stress-energy tensor associated with new matter field as

$$\delta S_\Lambda = -\frac{1}{8\pi G}\int \mathrm{d}^4x\left(-\frac{1}{2}g_{\mu\nu}\,\delta g^{\mu\nu}\right)\sqrt{-g}\Lambda \quad \Rightarrow \quad T^{(\Lambda)}_{\mu\nu} = -\frac{\Lambda}{8\pi G}g_{\mu\nu}.$$

Consider this matter to be a perfect fluid, we can identify its energy density and pressure as

$$\rho_\Lambda + P_\Lambda = 0, \quad P_\Lambda = -\frac{\Lambda}{8\pi G} \quad \Rightarrow \quad \rho_\Lambda = \frac{\Lambda}{8\pi G},$$

which gives us the equation of state parameter

$$w_\Lambda = \frac{P_\Lambda}{\rho_\Lambda} = -1.$$

Substituting this into the expression for the deceleration parameter, we find

$$q = \frac{1+3w_\Lambda}{2} = -1 < 0,$$

indicating that a non-zero cosmological constant can lead to an accelerating expansion of the universe. For this new matter field, the pressure is negative.

Thus, the cosmological constant Λ provides a simple explanation for the dark energy which can be thought of as the energy density of the vacuum of space itself. Observations suggest that the value of Λ is small but positive, leading to a current acceleration of the universe's expansion.

Lec 30: Isometry

We started this course with reviewing Special Relativity, and in the way we saw that this theory is about connecting different inertial frames via Lorentz transformations. Then we wanted to generalize this idea to arbitrary frames of reference, which forced us to introduce a general coordinate covariant formulation of physics.

Consider a coordinate transformation $x^\mu \to x'^\mu = x'^\mu(x^\nu)$. The differential change in coordinates is given by

$$\mathrm{d}x'^\mu = \Lambda^\mu{}_\nu \, \mathrm{d}x^\nu \,, \quad \text{where} \quad \Lambda^\mu{}_\nu = \frac{\partial x'^\mu}{\partial x^\nu}.$$

Under this transformation, the following quantity transforms as

- Scalar field: $\phi(\bar{\mathbf{x}}) \to \phi'(\bar{\mathbf{x}}') = \phi(\bar{\mathbf{x}})$.
- $(1,0)$ tensor (contravariant vector): $V^\mu(\bar{\mathbf{x}}) \to V'^\mu(\bar{\mathbf{x}}') = \Lambda^\mu{}_\nu V^\nu(\bar{\mathbf{x}})$.
- $(0,2)$ tensor: $T_{\mu\nu}(\bar{\mathbf{x}}) \to T'_{\mu\nu}(\bar{\mathbf{x}}') = (\Lambda^{-1})_\mu{}^\alpha (\Lambda^{-1})_\nu{}^\beta T_{\alpha\beta}(\bar{\mathbf{x}})$.

A diffeomorphism (smooth, invertible coordinate transformation) that preserves the form of the metric tensor is called an *isometry*. Mathematically, a coordinate transformation $x^\mu \to x'^\mu = x'^\mu(x^\nu)$ is an isometry if the metric tensor remains invariant under the transformation, *i.e.*,

$$g_{\mu\nu}(\bar{\mathbf{x}}) = g_{\alpha\beta}(\bar{\mathbf{x}}') \frac{\partial x'^\alpha}{\partial x^\mu} \frac{\partial x'^\beta}{\partial x^\nu}. \tag{30.1}$$

Now, the question is: what are the possible isometries for a given metric tensor $g_{\mu\nu}$? To answer this, we need to think about change in the metric tensor under a general coordinate transformation.

§30.1 Lie Derivative

Consider an infinitesimal coordinate transformation of the form

$$x^\mu \to x'^\mu = x^\mu + \epsilon \xi^\mu(\bar{\mathbf{x}}), \quad \epsilon > 0 \;\&\; \epsilon \ll 1.$$

Under this transformation, the transformation matrix is given by

$$\Lambda^\mu{}_\nu = \frac{\partial x'^\mu}{\partial x^\nu} = \frac{\partial}{\partial x^\nu}(x^\mu + \epsilon \xi^\mu(\bar{\mathbf{x}})) = \delta^\mu{}_\nu + \epsilon \partial_\nu \xi^\mu.$$

So the inverse transformation matrix is

$$(\Lambda^{-1})_\mu{}^\nu = \delta_\mu{}^\nu - \epsilon \partial_\mu \xi^\nu + \mathcal{O}(\epsilon^2).$$

Till now, we have only compared a tensor in two different coordinate systems at a given point (as each tensor component is defined for a given tangent space). However, if we compare a tensor at two different points (on manifold), two things happen:

1. Actual change in the tensor as we move from one point to another.
2. Change in the tensor components due to change in the coordinate system (because the tangent space itself changes as we move from one point to another).

If we want to study the change in a tensor due to only the first effect, we need to use the concept of *Lie derivative*. The Lie derivative of a tensor field along a vector field ξ^μ is denoted by $\pounds_{\boldsymbol{\xi}} T$ and captures the change in the tensor field due to the flow generated by the vector field ξ^μ.

Let T be a general (p, q) tensor field. The Lie derivative of T along the vector field ξ^μ is given by

$$\pounds_{\boldsymbol{\xi}} T = \lim_{\epsilon \to 0} \frac{1}{\epsilon}(\phi_\epsilon^* T - T), \tag{30.2}$$

where $\phi_\epsilon^* T$ is the pullback of the tensor field T under the flow generated by ξ^μ for a parameter ϵ. In coordinate form, we can express the Lie derivative of a (p, q) tensor field $T^{\mu_1\mu_2\cdots\mu_p}{}_{\nu_1\nu_2\cdots\nu_q}$ along the vector field ξ^μ as

$$\pounds_{\boldsymbol{\xi}} T^{\mu_1\mu_2\cdots\mu_p}{}_{\nu_1\nu_2\cdots\nu_q} = \lim_{\epsilon \to 0} \frac{1}{\epsilon}\Big(T^{\mu_1\mu_2\cdots\mu_p}{}_{\nu_1\nu_2\cdots\nu_q}(\bar{\mathbf{x}}') - T'^{\mu_1\mu_2\cdots\mu_p}{}_{\nu_1\nu_2\cdots\nu_q}(\bar{\mathbf{x}}')\Big). \tag{30.3}$$

Note that here, the arguments of both tensors are the same ($\bar{\mathbf{x}}' = \bar{\mathbf{x}} + \epsilon\bar{\boldsymbol{\xi}}$), so we are comparing the tensor at the same point in the manifold, thus isolating the effect of the flow generated by ξ^μ. Let's now compute the Lie derivative for few specific cases.

§30.1.1 Lie Derivative of a Scalar Field

Let $\phi(\bar{\mathbf{x}})$ be a scalar field. Under the infinitesimal transformation, we have

$$\phi'(\bar{\mathbf{x}}') = \phi(\bar{\mathbf{x}}).$$

So the Lie derivative of ϕ along ξ^μ is given by

$$\begin{aligned}
\pounds_{\boldsymbol{\xi}} \phi &= \lim_{\epsilon \to 0} \frac{1}{\epsilon}(\phi(\bar{\mathbf{x}}') - \phi'(\bar{\mathbf{x}}')) \\
&= \lim_{\epsilon \to 0} \frac{1}{\epsilon}\Big(\phi(\bar{\mathbf{x}} + \epsilon\bar{\boldsymbol{\xi}}) - \phi(\bar{\mathbf{x}})\Big) \\
&= \lim_{\epsilon \to 0} \frac{1}{\epsilon}\Big(\phi(\bar{\mathbf{x}}) + \epsilon\xi^\mu \partial_\mu \phi(\bar{\mathbf{x}}) - \phi(\bar{\mathbf{x}})\Big) \\
&= \xi^\mu \partial_\mu \phi \equiv \xi^\mu \nabla_\mu \phi.
\end{aligned} \tag{30.4}$$

Thus, the Lie derivative of a scalar field along a vector field ξ^μ is given by the directional derivative of the scalar field in the direction of ξ^μ.

§30.1.2 Lie Derivative of a Contravariant Vector Field

Let $V^\mu(\bar{\mathbf{x}})$ be a contravariant vector field. Under the infinitesimal transformation, we have

$$V'^\mu(\bar{\mathbf{x}}') = \Lambda^\mu{}_\nu V^\nu(\bar{\mathbf{x}}) = (\delta^\mu{}_\nu + \epsilon\partial_\nu \xi^\mu) V^\nu(\bar{\mathbf{x}}).$$

The Lie derivative of V^μ along ξ^μ is given by

$$\begin{aligned}
\pounds_{\boldsymbol{\xi}} V^\mu &= \lim_{\epsilon \to 0} \frac{1}{\epsilon}(V^\mu(\bar{\mathbf{x}}') - V'^\mu(\bar{\mathbf{x}}')) \\
&= \lim_{\epsilon \to 0} \frac{1}{\epsilon}\Big(V^\mu(\bar{\mathbf{x}} + \epsilon\bar{\boldsymbol{\xi}}) - (\delta^\mu{}_\nu + \epsilon\partial_\nu \xi^\mu) V^\nu(\bar{\mathbf{x}})\Big) \\
&= \lim_{\epsilon \to 0} \frac{1}{\epsilon}(V^\mu(\bar{\mathbf{x}}) + \epsilon\xi^\nu \partial_\nu V^\mu(\bar{\mathbf{x}}) - V^\mu(\bar{\mathbf{x}}) - \epsilon\partial_\nu \xi^\mu V^\nu(\bar{\mathbf{x}})) \\
&= \xi^\nu \partial_\nu V^\mu - V^\nu \partial_\nu \xi^\mu \\
&= \xi^\nu \Big(\partial_\nu V^\mu + \Gamma^\mu_{\nu\lambda} V^\lambda\Big) - V^\nu \Big(\partial_\nu \xi^\mu + \Gamma^\mu_{\nu\lambda} \xi^\lambda\Big) \\
&= \xi^\nu \nabla_\nu V^\mu - V^\nu \nabla_\nu \xi^\mu.
\end{aligned} \tag{30.5}$$

§30.1.3 Lie Derivative of a $(0,2)$ Tensor Field

Let $T_{\mu\nu}(\bar{\mathbf{x}})$ be a $(0,2)$ tensor field. Under the infinitesimal transformation, we have

$$T'_{\mu\nu}(\bar{\mathbf{x}}') = (\Lambda^{-1})_{\mu}{}^{\alpha}(\Lambda^{-1})_{\nu}{}^{\beta}T_{\alpha\beta}(\bar{\mathbf{x}}) = \Big(\delta_{\mu}{}^{\alpha} - \epsilon\partial_{\mu}\xi^{\alpha}\Big)\Big(\delta_{\nu}{}^{\beta} - \epsilon\partial_{\nu}\xi^{\beta}\Big)T_{\alpha\beta}(\bar{\mathbf{x}})$$
$$T'_{\mu\nu}(\bar{\mathbf{x}}') = T_{\mu\nu}(\bar{\mathbf{x}}) - \epsilon\Big(\partial_{\mu}\xi^{\alpha}T_{\alpha\nu} + \partial_{\nu}\xi^{\beta}T_{\mu\beta}\Big) + \mathcal{O}(\epsilon^2).$$

The Lie derivative of $T_{\mu\nu}$ along ξ^{μ} is given by

$$\begin{aligned}
\pounds_{\boldsymbol{\xi}}T_{\mu\nu} &= \lim_{\epsilon\to 0}\frac{1}{\epsilon}\Big(T_{\mu\nu}(\bar{\mathbf{x}}') - T'_{\mu\nu}(\bar{\mathbf{x}}')\Big)\\
&= \lim_{\epsilon\to 0}\frac{1}{\epsilon}\Big(T_{\mu\nu}(\bar{\mathbf{x}} + \epsilon\bar{\boldsymbol{\xi}}) - T_{\mu\nu}(\bar{\mathbf{x}}) + \epsilon\Big(\partial_{\mu}\xi^{\alpha}T_{\alpha\nu} + \partial_{\nu}\xi^{\beta}T_{\mu\beta}\Big)\Big)\\
&= \lim_{\epsilon\to 0}\frac{1}{\epsilon}\Big(T_{\mu\nu}(\bar{\mathbf{x}}) + \epsilon\xi^{\lambda}\partial_{\lambda}T_{\mu\nu}(\bar{\mathbf{x}}) - T_{\mu\nu}(\bar{\mathbf{x}}) + \epsilon\Big(\partial_{\mu}\xi^{\alpha}T_{\alpha\nu} + \partial_{\nu}\xi^{\beta}T_{\mu\beta}\Big)\Big)\\
&= \xi^{\lambda}\partial_{\lambda}T_{\mu\nu} + T_{\alpha\nu}\partial_{\mu}\xi^{\alpha} + T_{\mu\beta}\partial_{\nu}\xi^{\beta}\\
&= \xi^{\lambda}\Big(\partial_{\lambda}T_{\mu\nu} - \Gamma^{\alpha}_{\lambda\mu}T_{\alpha\nu} - \Gamma^{\beta}_{\lambda\nu}T_{\mu\beta}\Big) + T_{\alpha\nu}\Big(\partial_{\mu}\xi^{\alpha} + \Gamma^{\alpha}_{\mu\lambda}\xi^{\lambda}\Big) + T_{\mu\beta}\Big(\partial_{\nu}\xi^{\beta} + \Gamma^{\beta}_{\nu\lambda}\xi^{\lambda}\Big)\\
&= \xi^{\lambda}\nabla_{\lambda}T_{\mu\nu} + T_{\alpha\nu}\nabla_{\mu}\xi^{\alpha} + T_{\mu\beta}\nabla_{\nu}\xi^{\beta}. \qquad (30.6)
\end{aligned}$$

§30.2 Killing Vector Fields

A vector field ξ^{μ} is called a *Killing vector field* if the Lie derivative of the metric tensor along ξ^{μ} vanishes, *i.e.*,

$$\pounds_{\boldsymbol{\xi}}g_{\mu\nu} = 0. \qquad (30.7)$$

Using the expression for the Lie derivative of a $(0,2)$ tensor field, we have

$$\pounds_{\boldsymbol{\xi}}g_{\mu\nu} = \xi^{\lambda}\nabla_{\lambda}g_{\mu\nu} + g_{\lambda\nu}\nabla_{\mu}\xi^{\lambda} + g_{\mu\lambda}\nabla_{\nu}\xi^{\lambda}.$$

Now, since the covariant derivative of the metric tensor vanishes ($\nabla_{\lambda}g_{\mu\nu} = 0$), we get

$$\pounds_{\boldsymbol{\xi}}g_{\mu\nu} = g_{\lambda\nu}\nabla_{\mu}\xi^{\lambda} + g_{\mu\lambda}\nabla_{\nu}\xi^{\lambda} = \nabla_{\mu}\xi_{\nu} + \nabla_{\nu}\xi_{\mu} = 0. \qquad (30.8)$$

This equation is known as the *Killing equation*. Solutions to the Killing equation give the Killing vector fields of the spacetime.

Let's now compare this with the condition for isometry derived earlier. Under an infinitesimal coordinate transformation, we have

$$\begin{aligned}
g_{\mu\nu}(\bar{\mathbf{x}}) &= g_{\alpha\beta}(\bar{\mathbf{x}}')\frac{\partial x'^{\alpha}}{\partial x^{\mu}}\frac{\partial x'^{\beta}}{\partial x^{\nu}}\\
&= \Big(g_{\alpha\beta}(\bar{\mathbf{x}}) + \epsilon\xi^{\lambda}\partial_{\lambda}g_{\alpha\beta}(\bar{\mathbf{x}})\Big)\Big(\delta^{\alpha}{}_{\mu} + \epsilon\partial_{\mu}\xi^{\alpha}\Big)\Big(\delta^{\beta}{}_{\nu} + \epsilon\partial_{\nu}\xi^{\beta}\Big)\\
&= g_{\mu\nu}(\bar{\mathbf{x}}) + \epsilon\Big(\xi^{\lambda}\partial_{\lambda}g_{\mu\nu} + g_{\alpha\nu}\partial_{\mu}\xi^{\alpha} + g_{\mu\beta}\partial_{\nu}\xi^{\beta}\Big) + \mathcal{O}(\epsilon^2).
\end{aligned}$$

For the metric tensor to remain invariant under the transformation, the term in the parentheses must vanish, we get

$$\begin{aligned}
0 &= \xi^{\lambda}\partial_{\lambda}g_{\mu\nu} + g_{\alpha\nu}\partial_{\mu}\xi^{\alpha} + g_{\mu\beta}\partial_{\nu}\xi^{\beta}\\
&= \xi^{\lambda}\Big(\partial_{\lambda}g_{\mu\nu} - \Gamma^{\alpha}_{\lambda\mu}g_{\alpha\nu} - \Gamma^{\beta}_{\lambda\nu}g_{\mu\beta}\Big) + g_{\alpha\nu}\Big(\partial_{\mu}\xi^{\alpha} + \Gamma^{\alpha}_{\mu\lambda}\xi^{\lambda}\Big) + g_{\mu\beta}\Big(\partial_{\nu}\xi^{\beta} + \Gamma^{\beta}_{\nu\lambda}\xi^{\lambda}\Big)\\
&= \xi^{\lambda}\nabla_{\lambda}g_{\mu\nu} + g_{\alpha\nu}\nabla_{\mu}\xi^{\alpha} + g_{\mu\beta}\nabla_{\nu}\xi^{\beta} = \nabla_{\mu}\xi_{\nu} + \nabla_{\nu}\xi_{\mu},
\end{aligned}$$

for last step, we used $\nabla_{\lambda}g_{\mu\nu} = 0$. This is exactly the Killing equation, showing that the vector fields generating isometries are precisely the Killing vector fields of the spacetime.

Thus, we conclude that the isometries of a spacetime are generated by its Killing vector fields. Each Killing vector field corresponds to a continuous symmetry of the spacetime, and the number of independent Killing vector fields gives the dimension of the isometry group of the spacetime.

Lec 31: Killing Vector Field for 2D Flat Space

In last lecture, we derived the Killing equations for a general spacetime metric. Now, we will solve these equations for a simple case of 2D flat space, whose line element is given by

$$\mathrm{d}s^2 = \mathrm{d}x^2 + \mathrm{d}y^2 \,.$$

The corresponding metric tensor and its inverse are

$$g_{ij} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad\qquad g^{ij} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Christoffel symbols for this metric are all zero, *i.e.*, $\Gamma^i_{jk} = 0$. The Killing equations in this case reduce to

$$\nabla_i \xi_j + \nabla_j \xi_i = \partial_i \xi_j + \partial_j \xi_i = 0.$$

Writing out these equations explicitly, we have

$$\partial_x \xi_x + \partial_x \xi_x = 2\frac{\partial \xi_1}{\partial x} = 0, \tag{31.1}$$

$$\partial_y \xi_y + \partial_y \xi_y = 2\frac{\partial \xi_2}{\partial y} = 0, \tag{31.2}$$

$$\partial_x \xi_y + \partial_y \xi_x = \frac{\partial \xi_2}{\partial x} + \frac{\partial \xi_1}{\partial y} = 0. \tag{31.3}$$

The first two equations (eqns. (31.1) and (31.2)) imply that ξ_1 is independent of x and ξ_2 is independent of y. Therefore, we can write

$$\xi_1 = \xi_1(y), \qquad\qquad \xi_2 = \xi_2(x).$$

Take partial derivatives with respect to y and x respectively of the eqn. (31.3) to get

$$\partial_x^2 \xi_2 + \partial_x \partial_y \xi_1 = 0, \qquad\qquad \partial_y \partial_x \xi_2 + \partial_y^2 \xi_1 = 0$$
$$\Rightarrow \quad \partial_x^2 \xi_2 = 0, \qquad\qquad \Rightarrow \quad \partial_y^2 \xi_1 = 0.$$

The general solutions to these equations are

$$\xi_1(y) = A_1 y + B_1, \qquad\qquad \xi_2(x) = A_2 x + B_2,$$

where A_1, A_2, B_1, B_2 are constants. Substituting these back into eqn. (31.3), we get

$$A_1 + A_2 = 0 \quad \Rightarrow \quad A_2 = -A_1.$$

Thus, the components of the Killing vector field are

$$\xi_1 = -A_2 y + B_1, \qquad\qquad \xi_2 = A_2 x + B_2.$$

We have found the components of the Killing vector field. To describe the vector field itself, we need to take linear combinations of the basis vector fields. So we have $\boldsymbol{\xi} = \xi^i \hat{\mathbf{e}}_i = g^{ij}\xi_j \hat{\mathbf{e}}_i$, where $\hat{\mathbf{e}}_i$ are the basis vector fields. In Differential Geometry, we have a natural choice for the basis vector fields given by $\hat{\mathbf{e}}_i = \partial_i$. Therefore, the Killing vector field can be written as

$$\boldsymbol{\xi} = g^{1j}\xi_j \partial_x + g^{2j}\xi_j \partial_y = \xi_1 \partial_x + \xi_2 \partial_y = (-A_2 y + B_1)\partial_x + (A_2 x + B_2)\partial_y.$$

We can identify three linearly independent Killing vector fields by choosing appropriate values of the constants:

$$\text{(i)}\ A_2 = 0,\ B_1 = 1,\ B_2 = 0 \quad \Rightarrow \quad \boldsymbol{\xi}^{(1)} = \partial_x,$$
$$\text{(ii)}\ A_2 = 0,\ B_1 = 0,\ B_2 = 1 \quad \Rightarrow \quad \boldsymbol{\xi}^{(2)} = \partial_y,$$
$$\text{(iii)}\ A_2 = 1,\ B_1 = 0,\ B_2 = 0 \quad \Rightarrow \quad \boldsymbol{\xi}^{(3)} = -y\partial_x + x\partial_y.$$

These Killing vector fields are generators of the isometries of 2D flat space. To see this, we can compute the finite transformations generated by these vector fields by exponentiating them.

§31.1 Isometries Generated by Killing Vector Fields

Let $\mathcal{G}$ be a generator of a continuous transformation. The finite transformation generated by $\mathcal{G}$ can be obtained by exponentiating it:

$$\exp(\epsilon\mathcal{G}) = \sum_{n=0}^{\infty} \frac{\epsilon^n}{n!}\mathcal{G}^n,$$

where ϵ is amount of transformation parameter.

§31.1.1 Translation along x-axis

For the first Killing vector field $\boldsymbol{\xi}^{(1)} = \partial_x$, the finite transformation is

$$\exp(\epsilon\partial_x) = \sum_{n=0}^{\infty} \frac{\epsilon^n}{n!}\partial_x^n = \mathsf{T}_x[\epsilon].$$

Applying this operator to the coordinates (x, y), we get

$$\mathsf{T}_x[\epsilon]x = \sum_{n=0}^{\infty} \frac{\epsilon^n}{n!}\frac{\partial^n x}{\partial x^n} = x + \epsilon,$$
$$\mathsf{T}_x[\epsilon]y = \sum_{n=0}^{\infty} \frac{\epsilon^n}{n!}\frac{\partial^n y}{\partial x^n} = y.$$

This corresponds to a translation along the x-axis by an amount ϵ.

§31.1.2 Translation along y-axis

Similarly, for the second Killing vector field $\boldsymbol{\xi}^{(2)} = \partial_y$, the finite transformation is

$$\exp\Big(\epsilon\partial_y\Big) = \sum_{n=0}^{\infty} \frac{\epsilon^n}{n!}\partial_y^n = \mathsf{T}_y[\epsilon].$$

Applying this operator to the coordinates (x, y), we get

$$\mathsf{T}_y[\epsilon]x = \sum_{n=0}^{\infty} \frac{\epsilon^n}{n!}\frac{\partial^n x}{\partial y^n} = x,$$
$$\mathsf{T}_y[\epsilon]y = \sum_{n=0}^{\infty} \frac{\epsilon^n}{n!}\frac{\partial^n y}{\partial y^n} = y + \epsilon.$$

This corresponds to a translation along the y-axis by an amount ϵ.

§31.1.3 Rotation about the Origin

For the third Killing vector field $\boldsymbol{\xi}^{(3)} = -y\partial_x + x\partial_y$, the finite transformation is

$$\exp\Big(\theta\Big(-y\partial_x + x\partial_y\Big)\Big) = \sum_{n=0}^{\infty} \frac{\theta^n}{n!}\Big(y\partial_x - x\partial_y\Big)^n = \mathsf{R}[\theta].$$

Applying this operator to the coordinates (x, y), we get

$$\mathsf{R}[\theta]x = \sum_{n=0}^{\infty} \frac{\theta^n}{n!}\Big(-y\partial_x + x\partial_y\Big)^n x$$

observe that $\Big(-y\partial_x + x\partial_y\Big)x = -y$, and applying the operator again gives $\Big(-y\partial_x + x\partial_y\Big)(-y) = -x$. Continuing this process, we can separate the series into even and odd terms:

$$\begin{aligned} &= \sum_{m=0}^{\infty} (-1)^m \frac{\theta^{2m}}{(2m)!} x - \sum_{m=0}^{\infty} (-1)^m \frac{\theta^{2m+1}}{(2m+1)!} y \\ &= x\cos\theta - y\sin\theta, \end{aligned}$$

and

$$\mathsf{R}[\theta]y = \sum_{n=0}^{\infty} \frac{\theta^n}{n!} \Big(-y\partial_x + x\partial_y\Big)^n y$$

similarly, we have $\Big(-y\partial_x + x\partial_y\Big)y = x$, and applying the operator again gives $\Big(-y\partial_x + x\partial_y\Big)x = -y$. Continuing this process, we can separate the series into even and odd terms:

$$\begin{aligned} &= \sum_{m=0}^{\infty} (-1)^m \frac{\theta^{2m}}{(2m)!} y + \sum_{m=0}^{\infty} (-1)^m \frac{\theta^{2m+1}}{(2m+1)!} x \\ &= x\sin\theta + y\cos\theta. \end{aligned}$$

This corresponds to a rotation about the origin by an angle θ.

Thus, we have shown that the Killing vector fields in 2D flat space generate the symmetry transformations of translations along the x and y axes, and rotations about the origin.

Lec 32: Killing Vectors for Minkowski metric

In the last lecture, we derived all the Killing vectors for the 2D Euclidean metric. Now, let's study the isometries of the (3+1)-dimensional Minkowski metric given by

$$\mathrm{d}s^2 = -\,\mathrm{d}t^2 + \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2\,, \qquad g_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Since the Minkowski metric is constant, all Christoffel symbols vanish, *i.e.*, $\Gamma^{\alpha}_{\mu\nu} = 0$. Therefore, the Killing equations reduce to

$$\nabla_\mu \xi_\nu + \nabla_\nu \xi_\mu = \partial_\mu \xi_\nu + \partial_\nu \xi_\mu = 0.$$

Differentiating this equation with respect to x^σ, we get

$$\partial_\sigma \partial_\mu \xi_\nu + \cancel{\partial_\sigma \partial_\nu \xi_\mu} = 0. \tag{32.1}$$

Now, by cyclically permuting the indices (σ, μ, ν), we obtain two more equations:

$$\cancel{\partial_\mu \partial_\nu \xi_\sigma} + \partial_\mu \partial_\sigma \xi_\nu = 0, \tag{32.2}$$

$$\cancel{\partial_\nu \partial_\sigma \xi_\mu} + \cancel{\partial_\nu \partial_\mu \xi_\sigma} = 0. \tag{32.3}$$

Adding eqns. (32.1) and (32.2) and subtracting eqn. (32.3), we get

$$\partial_\sigma \partial_\mu \xi_\nu + \partial_\mu \partial_\sigma \xi_\nu = 0 \quad \Rightarrow \quad \partial_\mu \partial_\sigma \xi_\nu = 0.$$

In this equation, the indices σ, μ, ν are all arbitrary. Therefore, we can conclude that $\partial_\sigma \xi_\nu$ is at most a constant, *i.e.*,

$$\partial_\sigma \xi_\nu = B_{\sigma\nu},$$

where $B_{\sigma\nu}$ is a constant tensor. Integrating this equation with respect to x^σ, we get

$$\xi_\nu = A_\nu + B_{\sigma\nu} x^\sigma,$$

where A_ν is another constant tensor. Thus, the components of the Killing vector field are linear functions of the coordinates $\xi_\nu = A_\nu + B_{\mu\nu} x^\mu$. Substituting this back into the Killing equations lecture 32, we get

$$B_{\mu\nu} + B_{\nu\mu} = 0.$$

This implies that the tensor $B_{\mu\nu}$ is antisymmetric. In (3+1)-dimensions, an antisymmetric tensor has 6 independent components. We don't have any restriction on the components of A_ν, which has 4 independent components. Therefore, the total number of independent Killing vector fields for the Minkowski metric is $4 + 6 = 10$.

Similar to the 2D Euclidean case, we can express the Killing vector fields in terms of the basis vector fields ∂_μ as

$$\boldsymbol{\xi} = g^{\mu\nu} \xi_\nu \partial_\mu = g^{\mu\nu} (A_\nu + B_{\sigma\nu} x^\sigma) \partial_\mu.$$

We can identify the 10 linearly independent Killing vector fields by choosing appropriate values of the constants:

- $A_0 = 1,\ A_i = 0,\ B_{\mu\nu} = 0 \quad \Rightarrow \quad \boldsymbol{\xi}^{(1)} = -\partial_t,$

  For this Killing vector field, the finite transformation generated is

  $$\exp\!\left(-\varepsilon \boldsymbol{\xi}^{(1)}\right) = \exp(\varepsilon \partial_t) = \sum_{n=0}^{\infty} \frac{\varepsilon^n}{n!} \partial_t^n = \mathsf{T}_t[\varepsilon].$$

  Applying this operator to the coordinates, we get

  $$\mathsf{T}_t[\varepsilon] \begin{pmatrix} t & x & y & z \end{pmatrix} = \begin{pmatrix} t+\varepsilon & x & y & z \end{pmatrix},$$

  which represents a translation in the time coordinate by an amount ε.

- $A_0 = 0,\ A_1 = 1,\ A_2 = 0,\ A_3 = 0,\ B_{\mu\nu} = 0 \quad \Rightarrow \quad \boldsymbol{\xi}^{(2)} = \partial_x,$

  For this Killing vector field, the finite transformation generated is

$$\exp\Big(\epsilon\boldsymbol{\xi}^{(2)}\Big) = \exp(\epsilon\partial_x) = \sum_{n=0}^{\infty} \frac{\varepsilon^n}{n!}\partial_x^n = \mathsf{T}_x[\varepsilon].$$

  Applying this operator to the coordinates (t, x, y, z), we get

$$\mathsf{T}_x[\varepsilon]\begin{pmatrix} t & x & y & z \end{pmatrix} = \begin{pmatrix} t & x+\varepsilon & y & z \end{pmatrix},$$

  which represents a translation in the x coordinate by an amount ε.

- $A_0 = 0,\ A_1 = 0,\ A_2 = 1,\ A_3 = 0,\ B_{\mu\nu} = 0 \quad \Rightarrow \quad \boldsymbol{\xi}^{(3)} = \partial_y,$
- $A_0 = 0,\ A_1 = 0,\ A_2 = 0,\ A_3 = 1,\ B_{\mu\nu} = 0 \quad \Rightarrow \quad \boldsymbol{\xi}^{(4)} = \partial_z,$

  Similarly, the finite transformations generated by $\boldsymbol{\xi}^{(3)}$ and $\boldsymbol{\xi}^{(4)}$ correspond to translations in the y and z coordinates respectively.

- $A_\mu = 0,\ B_{01} = 1 = -B_{10},\ \text{others } 0 \quad \Rightarrow \quad \boldsymbol{\xi}^{(5)} = t\partial_x + x\partial_t,$

  Considering the Killing vector field $\boldsymbol{\xi}^{(5)} = t\partial_x + x\partial_t$, the finite transformation generated is

$$\exp\Big(\phi\boldsymbol{\xi}^{(5)}\Big) = \sum_{n=0}^{\infty} \frac{\phi^n}{n!}(t\partial_x + x\partial_t)^n = \mathsf{L}_x[\phi],$$

  where ϕ is the rapidity parameter. Applying this operator to the coordinates (t, x), we get

$$\mathsf{L}_x[\phi]t = \sum_{n=0}^{\infty} \frac{\phi^n}{n!}(t\partial_x + x\partial_t)^n t$$

  observe that $(t\partial_x + x\partial_t)t = x$, and $(t\partial_x + x\partial_t)x = t$, so we can separate even and odd terms in the series

$$\begin{aligned} &= t\sum_{m=0}^{\infty} \frac{\phi^{2m}}{(2m)!} + x\sum_{m=0}^{\infty} \frac{\phi^{2m+1}}{(2m+1)!} \\ &= t\cosh\phi + x\sinh\phi, \end{aligned}$$

  and similarly,

$$\mathsf{L}_x[\phi]x = \sum_{n=0}^{\infty} \frac{\phi^n}{n!}(t\partial_x + x\partial_t)^n x$$

  using the same reasoning as above, we separate even and odd terms in the series

$$\begin{aligned} &= x\sum_{m=0}^{\infty} \frac{\phi^{2m}}{(2m)!} + t\sum_{m=0}^{\infty} \frac{\phi^{2m+1}}{(2m+1)!} \\ &= x\cosh\phi + t\sinh\phi. \end{aligned}$$

  Combining these results, we have

$$\mathsf{L}_x[\phi]\begin{pmatrix} t & x & y & z \end{pmatrix} = \begin{pmatrix} t\cosh\phi + x\sinh\phi & x\cosh\phi + t\sinh\phi & y & z \end{pmatrix},$$

  which represents a boost along the x-axis with rapidity ϕ.

- $A_\mu = 0,\ B_{02} = 1 = -B_{20},\ \text{others } 0 \quad \Rightarrow \quad \boldsymbol{\xi}^{(6)} = t\partial_y + y\partial_t,$
- $A_\mu = 0,\ B_{03} = 1 = -B_{30},\ \text{others } 0 \quad \Rightarrow \quad \boldsymbol{\xi}^{(7)} = t\partial_z + z\partial_t,$

  Similar to $\boldsymbol{\xi}^{(5)}$, the finite transformations generated by $\boldsymbol{\xi}^{(6)}$ and $\boldsymbol{\xi}^{(7)}$ correspond to boosts along the y and z axes respectively.

- $A_\mu = 0,\ B_{12} = 1 = -B_{21},$ others $0 \quad \Rightarrow \quad \boldsymbol{\xi}^{(8)} = -y\partial_x + x\partial_y,$

  In the 2D Euclidean case, we have already seen this Killing vector field generates rotations in the xy-plane. Extending this to (3+1)-dimensions, the finite transformation generated by $\boldsymbol{\xi}^{(8)}$ is

$$\exp\Big(\theta\boldsymbol{\xi}^{(8)}\Big) = \sum_{n=0}^{\infty} \frac{\theta^n}{n!}(-y\partial_x + x\partial_y)^n = \mathsf{R}_z[\theta],$$

  where θ is the rotation angle about the z-axis. Applying this operator to the coordinates, we get

$$\mathsf{R}_z[\theta]\Big(t \quad x \quad y \quad z\Big) = \Big(t \quad x\cos\theta - y\sin\theta \quad x\sin\theta + y\cos\theta \quad z\Big),$$

  which represents a rotation about the z-axis by an angle θ.

- $A_\mu = 0,\ B_{23} = 1 = -B_{32},$ others $0 \quad \Rightarrow \quad \boldsymbol{\xi}^{(9)} = -z\partial_y + y\partial_z,$
- $A_\mu = 0,\ B_{31} = 1 = -B_{13},$ others $0 \quad \Rightarrow \quad \boldsymbol{\xi}^{(10)} = -x\partial_z + z\partial_x.$

  Similar to $\boldsymbol{\xi}^{(8)}$, the finite transformations generated by $\boldsymbol{\xi}^{(9)}$ and $\boldsymbol{\xi}^{(10)}$ correspond to rotations about the x and y axes respectively.

Thus, we have identified all 10 independent Killing vector fields for the (3+1)-dimensional Minkowski metric, corresponding to 4 translations (in time and space), 3 boosts (along each spatial axis), and 3 rotations (about each spatial axis). These Killing vector fields generate the Poincaré group of isometries of Minkowski spacetime.

Lec 33: Parallel Transport

We have seen that in curved space, the partial derivative ∂_μ does not behave well under general coordinate transformations (*i.e.*it does not transform as a tensor). To fix this, we had introduced the covariant derivative ∇_μ which does transform as a tensor. The main reason for this was that in a general manifold, the coordinate basis vectors $\hat{\mathbf{e}}_\mu$ vary from point to point. Similar issues arise when we try to compare vectors (or tensors) at different points in the manifold.

Consider a vector $\bar{\mathbf{V}}$ at a point p in the manifold, so the vector lives in the tangent space $T_p\mathcal{M}$. Now, say we want to compare this vector with another vector $\bar{\mathbf{V}}'$ at a different point q. Since $\bar{\mathbf{V}}'$ lives in a different tangent space $T_q\mathcal{M}$, there is no natural way to compare $\bar{\mathbf{V}}$ and $\bar{\mathbf{V}}'$. To compare vectors at different points, we need a way to "*transport*" vectors from one point to another in a manner that respects the manifold's geometry. This process is known as *parallel transport*.

§33.1 Definition of Parallel Transport

Before defining parallel transport in curved space, let's first understand it in flat (Euclidean) space. In flat space, parallel transport is straightforward: we can move a tensor along a straight line without changing its direction or magnitude.

Let a curve $\gamma(\lambda)$ parameterized by λ connect points p and q in the manifold, and the coordinates of the points along the curve are given by $x^\mu(\lambda)$. Then the condition for parallel transport for a tensor $T^{\mu_1\mu_2\cdots\mu_k}{}_{\nu_1\nu_2\cdots\nu_l}$ along the curve γ is that it is constant with respect to the parameter λ:

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}T^{\mu_1\mu_2\cdots mu_k}{}_{\nu_1\nu_2\cdots\nu_l} = 0 \quad \Rightarrow \quad \frac{\mathrm{d}x^\rho}{\mathrm{d}\lambda}\partial_\rho T^{\mu_1\mu_2\cdots\mu_k}{}_{\nu_1\nu_2\cdots\nu_l} = 0. \quad \text{(Flat Space)} \tag{33.1}$$

To generalize this to curved space, we replace the partial derivative with the covariant derivative:

$$\frac{\mathrm{d}x^\rho}{\mathrm{d}\lambda}\nabla_\rho T^{\mu_1\mu_2\cdots\mu_k}{}_{\nu_1\nu_2\cdots\nu_l} = 0. \quad \text{(Curved Space)} \tag{33.2}$$

This equation states that a tensor is parallel transported along the curve γ if its covariant derivative along the tangent vector to the curve is zero. With this, we can define directional covariant derivatives

$$\mathrm{D}_{\bar{\mathbf{u}}} = \bar{\mathbf{u}}\cdot\nabla = u^\rho\nabla_\rho, \tag{33.3}$$

where $\bar{\mathbf{u}}$ is the tangent vector to the curve γ, and $u^\rho = \frac{\mathrm{d}x^\rho}{\mathrm{d}\lambda}$. Thus, the condition for parallel transport can be succinctly written as

$$(\mathrm{D}_{\bar{\mathbf{u}}}T)^{\mu_1\mu_2\cdots\mu_k}{}_{\nu_1\nu_2\cdots\nu_l} = 0. \tag{33.4}$$

This equation is covariant and holds true in any coordinate system, as the tangent vector $\bar{\mathbf{u}}$ and the covariant derivative ∇_ρ both transform appropriately under coordinate changes. For a vector $\bar{\mathbf{V}}$, the parallel transport condition becomes

$$\mathrm{D}_{\bar{\mathbf{u}}}\bar{\mathbf{V}} = \left(\mathrm{D}_{\bar{\mathbf{u}}}\bar{\mathbf{V}}\right)^\mu\hat{\mathbf{e}}_\mu = 0 \quad \Rightarrow \quad u^\rho\nabla_\rho V^\mu = \frac{\mathrm{d}x^\rho}{\mathrm{d}\lambda}\left(\partial_\rho V^\mu + \Gamma^\mu_{\rho\sigma}V^\sigma\right) = \frac{\mathrm{d}V^\mu}{\mathrm{d}\lambda} + \Gamma^\mu_{\rho\sigma}\frac{\mathrm{d}x^\rho}{\mathrm{d}\lambda}V^\sigma = 0. \tag{33.5}$$

Notice that the above is a first-order ordinary differential equation (ODE) for the components of the tensor being parallel transported. Given initial conditions at point p, there will be a unique continuation of the tensor along the curve γ that satisfies the parallel transport condition. This allows us to define a mapping from the tensor space at point p to the tensor space at point q along the curve γ, effectively enabling us to compare tensors at different points in the manifold.

However, it is important to note that the result of parallel transport generally depends on the path taken between the two points. This path-dependence is a manifestation of the manifold's curvature. In flat space, parallel transport is path-independent, but in curved space, the curvature can lead to different results when transporting a tensor along different paths connecting the same two points (see fig. 33.1).

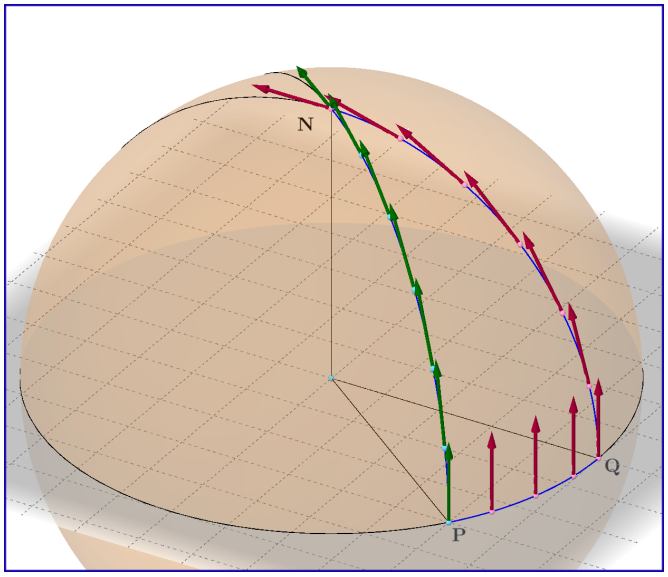


Figure 33.1: Path dependence of parallel transport in curved space. Transporting a vector $\bar{\mathbf{V}}$ from point P to point N along different paths results in different final vectors.

§33.2 Geodesics: Reinterpreting the Straightest Possible Path

Using the concept of parallel transport, we can now reinterpret geodesics as the "straightest possible paths" in curved space. A geodesic is defined as a curve $\gamma(\lambda)$ whose tangent vector $\bar{\mathbf{u}} = \frac{\mathrm{d}\bar{\mathbf{x}}}{\mathrm{d}\lambda}$ is parallel transported along itself. Mathematically, this condition can be expressed as

$$\mathrm{D}_{\bar{\mathbf{u}}}\bar{\mathbf{u}} = 0 \quad \Rightarrow \quad u^\mu \nabla_\mu u^\alpha = \frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}\left(\partial_\mu u^\alpha + \Gamma^\alpha_{\mu\nu} u^\nu\right) = \frac{\mathrm{d}^2 x^\alpha}{\mathrm{d}\lambda^2} + \Gamma^\alpha_{\mu\nu}\frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda} = 0. \tag{33.6}$$

This is precisely the geodesic equation we derived earlier using the variational principle. Thus, we see that geodesics can be understood as curves along which the tangent vector remains parallel to itself, making them the "*straightest*" possible paths in a curved manifold.

This geodesic equation is invariant under affine transformations of the parameter λ, *i.e.*, if λ is replaced by $\lambda' = a\lambda + b$ (with a, b constants and $a \neq 0$), the form of the geodesic equation remains unchanged. That's why λ is often referred to as an *affine parameter* along the geodesic.

A Examinations

§A.1 Class Test 01

40 minutes Weightage: 10 Marks

Question A.1.1.
Show that the quantity $g_{\alpha\beta}V^\alpha V^\beta$ is a scalar under general coordinate transformations, where $g_{\alpha\beta}$ is the metric tensor and $\bar{\mathbf{V}}$ is a contravariant vector. [3 Marks]

Solution.
We know that under a general coordinate transformation, the contravariant vector V^α transforms as:

$$V'^\mu = \frac{\partial x'^\mu}{\partial x^\alpha} V^\alpha.$$

The metric tensor $g_{\alpha\beta}$ transforms as:

$$g'_{\mu\nu} = \frac{\partial x^\rho}{\partial x'^\mu}\frac{\partial x^\sigma}{\partial x'^\nu} g_{\rho\sigma}.$$

Thus, the quantity $g_{\alpha\beta}V^\alpha V^\beta$ transforms as:

$$\begin{aligned} g'_{\mu\nu}V'^\mu V'^\nu &= \left[\frac{\partial x^\rho}{\partial x'^\mu}\frac{\partial x^\sigma}{\partial x'^\nu} g_{\rho\sigma}\right]\left[\frac{\partial x'^\mu}{\partial x^\alpha}V^\alpha\right]\left[\frac{\partial x'^\nu}{\partial x^\beta}V^\beta\right] \\ &= \left[\frac{\partial x^\rho}{\partial x'^\mu}\frac{\partial x'^\mu}{\partial x^\alpha}\right]\left[\frac{\partial x^\sigma}{\partial x'^\nu}\frac{\partial x'^\nu}{\partial x^\beta}\right] g_{\rho\sigma}V^\alpha V^\beta \\ &= \delta^\rho{}_\alpha \delta^\sigma{}_\beta g_{\rho\sigma} V^\alpha V^\beta \\ &= g_{\alpha\beta}V^\alpha V^\beta. \end{aligned}$$

Thus, the quantity $g_{\alpha\beta}V^\alpha V^\beta$ is invariant under general coordinate transformations, and hence it is a scalar. ◂

Question A.1.2.
Consider the coordinate transformation

$$\bar{\mathbf{x}} = \begin{pmatrix} r & \theta \end{pmatrix} \quad \to \quad \bar{\mathbf{x}}' = \begin{pmatrix} x & y \end{pmatrix} = \begin{pmatrix} \frac{1}{2}r\cos(2\theta) & \frac{1}{2}r\sin(2\theta) \end{pmatrix}.$$

1. Compute the transformation matrix $\Lambda^\alpha{}_\beta$ such that $\mathrm{d}x'^\alpha = \Lambda^\alpha{}_\beta\, \mathrm{d}x^\beta$. [3 Marks]
2. Let the invariant distance in the $\bar{\mathbf{x}}'$ coordinates system be given by $\mathrm{d}s'^2 = \mathrm{d}x^2 + \mathrm{d}y^2$. Compute the metric tensor $g_{\alpha\beta}$ in the $\bar{\mathbf{x}}$ coordinates system. [4 Marks]

§A.2 Mid-Semester Exam

`90 minutes` `Weightage: 20 Marks`

Question A.2.1 (4-Velocity)**.**
Show that the 4-velocity $u^\mu = \frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}$ of a particle is a contra-variant vector under general coordinate transformations, where τ is the proper time of the particle. Further, show that the squared magnitude of the vector u^μ in an arbitrary spacetime with metric $g_{\mu\nu}$ is $-c^2$. `[3 + 3 = 6 Marks]`

Solution.
We know that under a general coordinate transformation, the coordinates x^μ transform as:

$$x'^\mu = \frac{\partial x'^\mu}{\partial x^\alpha} x^\alpha.$$

Taking the derivative with respect to the proper time τ, we get:

$$\frac{\mathrm{d}x'^\mu}{\mathrm{d}\tau} = \frac{\partial x'^\mu}{\partial x^\alpha} \frac{\mathrm{d}x^\alpha}{\mathrm{d}\tau}.$$

Thus, the 4-velocity $u^\mu = \frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}$ transforms as:

$$u'^\mu = \frac{\partial x'^\mu}{\partial x^\alpha} u^\alpha.$$

This shows that the 4-velocity is a contravariant vector under general coordinate transformations.

Next, we need to show that the squared magnitude of the 4-velocity in an arbitrary spacetime with metric $g_{\mu\nu}$ is $-c^2$. The squared magnitude is given by:

$$g_{\mu\nu} u^\mu u^\nu.$$

In the local inertial frame of the particle, the metric is the Minkowski metric $\eta_{\mu\nu} = \mathrm{diag}(-1, 1, 1, 1)$. In this frame, the 4-velocity is given by:

$$u^\mu = \gamma(c, \mathbf{v}),$$

where $\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$ is the Lorentz factor and $\mathbf{v}$ is the 3-velocity of the particle. The squared magnitude in this frame is:

$$\begin{aligned}
\eta_{\mu\nu} u^\mu u^\nu &= -\gamma^2 c^2 + \gamma^2 v^2 \\
&= \gamma^2(-c^2 + v^2) \\
&= \gamma^2 \left(-c^2 \left(1 - \frac{v^2}{c^2} \right) \right) \\
&= -c^2.
\end{aligned}$$

Since the squared magnitude is a scalar, it remains invariant under general coordinate transformations. Therefore, we have:

$$g_{\mu\nu} u^\mu u^\nu = -c^2.$$

◀

Question A.2.2 (Geodesic Equation)**.**
The motion of a particle of rest mass m under the influence of gravity is described by the action $S = -\int \mathrm{d}\tau\, mc^2$, where τ is the proper time. Find the limiting form of the corresponding Lagrangian for a non-relativistic particle in a spacetime with the metric $\eta_{\mu\nu} = \mathrm{diag}(-1, 1, 1, 1)$. Find the Euler-Lagrange equation of motion for the particle in a general spacetime with metric $g_{\mu\nu}$. `[3 + 4 = 7 Marks]`

Solution.
We have done this in the class, see section 10.3. ◀

Question A.2.3 (Geodesic in Plane Polar Coordinates)**.**
By computing the Christoffel symbols, find the geodesics on a space with the metric $\mathrm{d}s^2 = \mathrm{d}r^2 + r^2\,\mathrm{d}\theta^2$. Show that these geodesics represent the straight lines. [2 + 3 + 2 = 7 Marks]

Solution.
The metric is given by:

$$g_{\mu\nu} = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix} \equiv \begin{pmatrix} g_{rr} & g_{r\theta} \\ g_{\theta r} & g_{\theta\theta} \end{pmatrix}.$$

The non-zero Christoffel symbols are given by:

$$\Gamma^r_{\theta\theta} = -r, \quad \Gamma^\theta_{r\theta} = \Gamma^\theta_{\theta r} = \frac{1}{r}.$$

The geodesic equations are given by:

$$\frac{\mathrm{d}^2 r}{\mathrm{d}\lambda^2} - r\left(\frac{\mathrm{d}\theta}{\mathrm{d}\lambda}\right)^2 = 0,$$
$$\frac{\mathrm{d}^2\theta}{\mathrm{d}\lambda^2} + \frac{2}{r}\frac{\mathrm{d}r}{\mathrm{d}\lambda}\frac{\mathrm{d}\theta}{\mathrm{d}\lambda} = 0.$$

The second equation can be rewritten as:

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\left(r^2\frac{\mathrm{d}\theta}{\mathrm{d}\lambda}\right) = 0 \Rightarrow r^2\frac{\mathrm{d}\theta}{\mathrm{d}\lambda} = L,$$

where L is a constant. Substituting this into the first equation, we get:

$$\frac{\mathrm{d}^2 r}{\mathrm{d}\lambda^2} - \frac{L^2}{r^3} = 0.$$

Since we want the shape of the geodesic, we need to eliminate the affine parameter λ. We have:

$$\frac{\mathrm{d}}{\mathrm{d}\lambda} = \frac{\mathrm{d}\theta}{\mathrm{d}\lambda}\frac{\mathrm{d}}{\mathrm{d}\theta} = \frac{L}{r^2}\frac{\mathrm{d}}{\mathrm{d}\theta}.$$

Thus, we have:

$$\frac{L}{r^2}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\frac{L}{r^2}\frac{\mathrm{d}r}{\mathrm{d}\theta}\right) = \frac{L^2}{r^3},$$

which simplifies to:

$$\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\frac{1}{r^2}\frac{\mathrm{d}r}{\mathrm{d}\theta}\right) = \frac{1}{r}.$$

Let $u = \frac{1}{r}$. Then, we have:

$$\frac{\mathrm{d}^2 u}{\mathrm{d}\theta^2} + u = 0.$$

The general solution to this equation is:

$$u(\theta) = A\cos(\theta + \theta_0),$$

where A and θ_0 are constants. This represents a straight line in polar coordinates, confirming that the geodesics in this space are indeed straight lines. ◀

§A.3 Class Test 02

35 minutes Weightage: 10 Marks

Question A.3.1.
The geodesic equation for a massive test particle in a curved spacetime is given by

$$\frac{\mathrm{d}^2 r}{\mathrm{d}\tau^2} = -\frac{A^2}{r^4} + \frac{B^2}{r^3} - \frac{C^2}{r^2},$$

where A, B, and C are real constants, and τ is the proper time along the particle's worldline.

(a) Making connections with Newton's 2nd law of motion, identify the effective potential $V_1(r)$ governing the motion of the particle. [2 Marks]

(b) Plot the effective potential $V_1(r)$. [2 Marks]

(c) Now, consider another effective potential given by

$$V_2(r) = -\frac{D^2}{r} + \frac{E^2}{r^2}$$

where D and E are positive real constants. Plot the effective potential $V_2(r)$. [2 Marks]

(d) Comment whether the effective force due to $V_1(r)$ and $V_2(r)$ are attractive or repulsive in nature in [2 + 2 Marks]

(i) in small r limit,

(ii) in large r limit.

Solution. (a) Rewriting the geodesic equation as

$$\frac{\mathrm{d}^2 r}{\mathrm{d}\tau^2} = -\frac{\mathrm{d}V_1}{\mathrm{d}r},$$

we can identify the effective potential $V_1(r)$ as

$$V_1(r) = -\frac{A^2}{3r^3} + \frac{B^2}{2r^2} - \frac{C^2}{r}.$$

(b) The plot of the effective potential $V_1(r)$ is shown below:

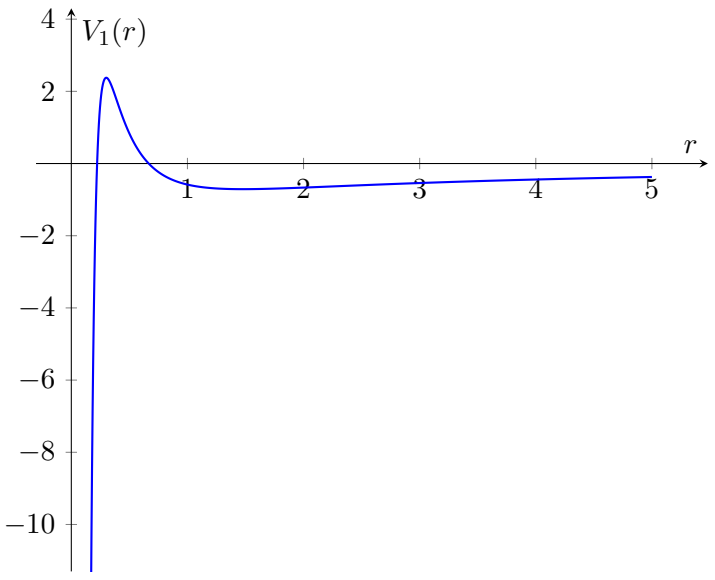


Figure A.1: Plot of the effective potential $V_1(r) = -\frac{A^2}{3r^3} + \frac{B^2}{2r^2} - \frac{C^2}{r}$.

(c) The plot of the effective potential $V_2(r)$ is shown below:

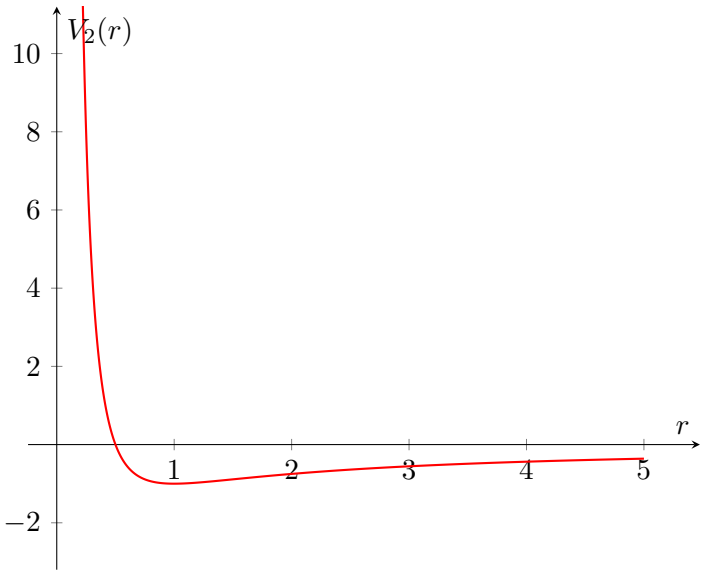


Figure A.2: Plot of the effective potential $V_2(r) = -\frac{D^2}{r} + \frac{E^2}{r^2}$.

(d) Let's compute the effective forces due to both potentials:

$$F_1(r) = -\frac{\mathrm{d}V_1}{\mathrm{d}r} = -\left(\frac{A^2}{r^4} - \frac{B^2}{r^3} + \frac{C^2}{r^2}\right), \qquad F_2(r) = -\frac{\mathrm{d}V_2}{\mathrm{d}r} = -\left(\frac{D^2}{r^2} - \frac{2E^2}{r^3}\right).$$

(i) In the small r limit, both $F_1(r)$ and $F_2(r)$ are dominated by the terms with the highest negative powers of r:

$$F_1(r) \approx -\frac{A^2}{r^4} \quad \text{(attractive)}, \qquad F_2(r) \approx \frac{2E^2}{r^3} \quad \text{(repulsive)}.$$

(ii) In the large r limit, both $F_1(r)$ and $F_2(r)$ are dominated by the terms with the lowest negative powers of r:

$$F_1(r) \approx -\frac{C^2}{r^2} \quad \text{(attractive)}, \qquad F_2(r) \approx -\frac{D^2}{r^2} \quad \text{(attractive)}.$$

◀

§A.4 Class Test 03

35 minutes Weightage: 10 Marks

Question A.4.1.
Solve the Friedmann equations for pressure-less dust matter ($P = 0$). Plot the scale factor $a(t)$ and energy density $\rho(t)$ as functions of time t. [6 Marks]

Solution.
See the solution to assignment 10.5. ◂

Question A.4.2.
Compute the equation of state (EoS) and energy density for the given stress-energy tensors

$$T_{\mu\nu} = -\frac{1}{l^2} g_{\mu\nu},$$

where l is a constant with dimensions of length.

(Hint: Consider matter to be a perfect fluid.) [4 Marks]

Solution.
See the discussion on Cosmological Constant in lecture 29. ◂

§A.5 End-Semester Exam

150 minutes Weightage: 50 Marks

Question A.5.1 (Coordinate Transformation)**.**
Consider the coordinate transformation $\bar{\mathbf{x}} \equiv (r, \theta) \to \bar{\mathbf{x}}' \equiv (x, y)$ such that $x = r\cos\theta$ and $y = r\sin\theta$. Compute the elements of the transformation matrix Λ if $\mathrm{d}\bar{\mathbf{x}}' = \Lambda\,\mathrm{d}\bar{\mathbf{x}}$. If the invariant distance in coordinate system is $\mathrm{d}s^2 = 2\,\mathrm{d}x^2 + 2\,\mathrm{d}y^2$ then compute the metric tensor in $\bar{\mathbf{x}}'$ coordinate system. [4 + 4 = 8 Marks]

Solution.
See section 07.1. ◂

Question A.5.2 (Riemann Tensor Properties)**.**
For a metric tensor $g_{\mu\nu}$, if $\nabla_\alpha g^{\mu\nu} = 0$ and $[\nabla_\alpha, \nabla_\beta]\,u^\mu = R^\mu{}_{\nu\alpha\beta}u^\nu$ then show that $R_{\mu\nu\alpha\beta} = -R_{\nu\mu\alpha\beta}$ where $g^{\mu\nu}$ is the inverse metric and u^μ is a contravariant vector. Further, if $\partial_\alpha g = g g^{\mu\nu}\partial_\alpha g_{\mu\nu}$ where g is the determinant of the metric tensor then prove the following identity

$$\nabla_\nu u^\nu = \frac{1}{\sqrt{-g}}\partial_\nu(\sqrt{-g}u^\nu). \tag{A.1}$$

[4 + 4 = 8 Marks]

Solution.
For the first part, see the derivation in Riemann Curvature Tensor. For the second part, see assignment 04.3. ◂

Question A.5.3 (Geodesics on a Sphere)**.**
For the metric $\mathrm{d}s^2 = \mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2$, compute all Christoffel symbols. By expressing the equation for great circles in (θ, ϕ) coordinates show that the geodesics on a unit two-sphere with the given metric are great circles. [4 + 3 + 4 = 11 Marks]

Solution.
See the discussion on Geodesics on a 2-Sphere S^2. ◂

Question A.5.4 (Constant of Motion)**.**
A particle of mass m is moving along a geodesic in a spacetime spanned by the coordinates (t, r, θ, ϕ). The non-vanishing Christoffel symbols, of the form $\Gamma^\theta_{\mu\nu}$ and $\Gamma^\phi_{\mu\nu}$, are $\Gamma^\theta_{r\theta} = \Gamma^\phi_{r\phi} = 1/r$, $\Gamma^\theta_{\phi\phi} = -\sin\theta\cos\theta$ and $\Gamma^\phi_{\theta\phi} = \cot\theta$. Find the associated constant of motion for the particle. [5 Marks]

Question A.5.5 (Radiation-dominated Universe)**.**
The stress-energy tensor for Maxwell fields is trace-free. Show that the corresponding equation of state is $w = 1/3$ when treated as a perfect fluid. Solve the Friedmann equations for a flat FRW universe containing such a fluid. Plot the corresponding scale factor $a(t)$ and the energy density $\rho(t)$ and $\rho(a)$. [3 + 4 + 3 = 10 Marks]

Solution.
See Radiation Dominated Universe. ◂

Question A.5.6 (Killing Vectors)**.**
Write down the Killing equation for the metric $\mathrm{d}s^2 = -c^2\,\mathrm{d}t^2 + \mathrm{d}x^2$. Find the general solution of the corresponding Killing equation. Show that there are three independent Killing vectors and write them explicitly. [1 + 3 + 4 = 8 Marks]

B Appendix

This chapter contains some additional material or derivations that supplement the main content of the notes.

§B.1 Jacobi's Formula: Derivative of the Determinant

In this section, we will derive the Jacobi's formula, which gives the relation for the derivative of the determinant of a matrix with its trace. We will **not use Einstein summation convention** in this section for clarity.

Let M be a square invertible matrix, and let each of its elements be a function of a set of variables $(x^1, x^2, \ldots, x^n)$. We want to find the derivative of the determinant of M with respect to one of these variables, say x^μ.

Define the adjoint of M, denoted by $\operatorname{adj} M$, as the transpose of the cofactor matrix of M. More explicitly, if C_{ij} is the cofactor of the element M_{ij}, then the adjoint matrix is given by:

$$(\operatorname{adj} M)_{ij} = C_{ji}. \tag{B.1}$$

And the cofactor C_{ij} is defined as:

$$C_{ij} = (-1)^{i+j} \det\Big(M_{(ij)}\Big), \tag{B.2}$$

where $M_{(ij)}$ is the $(n-1) \times (n-1)$ matrix obtained by removing the i-th row and j-th column from M. From definition, we can see that the cofactor C_{ij} does not depend on the element M_{ij}.

The inverse of M can be expressed in terms of its adjoint as:

$$M^{-1} = \frac{1}{\det(M)} \operatorname{adj} M \quad \Rightarrow \quad (M^{-1})_{ij} = \frac{(\operatorname{adj} M)_{ij}}{\det(M)} = \frac{C_{ji}}{\det(M)}. \tag{B.3}$$

The determinant of M can be expressed as:

$$\det(M) = \sum_{i,j} M_{ij} C_{ij} \tag{B.4}$$

Now, taking the derivative of $\det(M)$ with respect to x^μ, so using chain rule, we have:

$$\begin{aligned}
\partial_\mu \det(M) &= \sum_{a,b} \left\{ \frac{\partial}{\partial M_{ab}} \left(\sum_{i,j} M_{ij} C_{ij} \right) \partial_\mu M_{ab} \right\} \\
&= \sum_{a,b} \left\{ \frac{\partial}{\partial M_{ab}} \left(M_{ab} C_{ab} + \sum_{(i,j) \neq (a,b)} M_{ij} C_{ij} \right) \partial_\mu M_{ab} \right\}
\end{aligned}$$

using the fact that C_{ij} for $(i,j) \neq (a,b)$ does not depend on M_{ab}, we have:

$$\begin{aligned}
&= \sum_{a,b} \Big\{ C_{ab} \partial_\mu M_{ab} \Big\} \\
&= \sum_{a,b} \Big\{ \det(M) (M^{-1})_{ba} \partial_\mu M_{ab} \Big\} \\
&= \det(M) \sum_{a,b} \Big\{ (M^{-1})_{ba} \partial_\mu M_{ab} \Big\} \\
&= \det(M) \operatorname{tr}\Big(M^{-1} \partial_\mu M \Big).
\end{aligned}$$

Thus, we have derived the relation:

$$\frac{1}{\det(M)} \partial_\mu \det(M) = \operatorname{tr}\Big(M^{-1} \partial_\mu M \Big). \tag{B.5}$$

§B.2 Symbolic Computation of Curvature Tensors using Mathematica

```
(* Define the coordinates and dimension *)
coordinates = {t, r, \[Theta], \[Phi]};
timeDim = 1;
d = Length[coordinates] - timeDim;

(* Define the metric tensor for a 4D spacetime *)
metric = {
   {-(1 - Rs/r), 0, 0, 0},
   {0, (1 - Rs/r)^-1, 0, 0},
   {0, 0, r^2, 0},
   {0, 0, 0, r^2 Sin[\[Theta]]^2}
   };
metricInv = Simplify@Inverse[metric];

(* Function to compute Christoffel symbols *)
\[CapitalGamma][\[Alpha]_, \[Mu]_, \[Nu]_] := Simplify[
   1/2*Sum[metricInv[[\[Alpha] + 1, \[Sigma] + 1]]*(
      D[metric[[\[Sigma] + 1, \[Nu] + 1]], coordinates[[\[Mu] + 1]]] +
      D[metric[[\[Mu] + 1, \[Sigma] + 1]], coordinates[[\[Nu] + 1]]] -
      D[metric[[\[Mu] + 1, \[Nu] + 1]], coordinates[[\[Sigma] + 1]]]),
     {\[Sigma], 0, d}]
     ];

Christoffel = Table[\[CapitalGamma][\[Alpha], \[Mu], \[Nu]], {\[Alpha], 0, d}, {\[Mu],
↪  0, d}, {\[Nu], 0, d}];

(* Function to compute Riemann curvature tensor *)
R[\[Rho]_, \[Sigma]_, \[Mu]_, \[Nu]_] := Simplify[
   D[Christoffel[[\[Rho] + 1, \[Sigma] + 1, \[Nu] + 1]], coordinates[[\[Mu] + 1]]] -
    D[Christoffel[[\[Rho] + 1, \[Mu] + 1, \[Sigma] + 1]], coordinates[[\[Nu] + 1]]] +
    Sum[Christoffel[[\[Rho] + 1, \[Mu] + 1, \[Lambda] + 1]] * Christoffel[[\[Lambda] +
     ↪  1, \[Sigma] + 1, \[Nu] + 1]], {\[Lambda], 0, d}] -
    Sum[Christoffel[[\[Rho] + 1, \[Nu] + 1, \[Lambda] + 1]] * Christoffel[[\[Lambda] +
     ↪  1, \[Mu] + 1, \[Sigma] + 1]], {\[Lambda], 0, d}]
   ];

Riemann = Table[R[\[Rho], \[Sigma], \[Mu], \[Nu]], {\[Rho], 0, d}, {\[Sigma], 0, d},
↪  {\[Mu], 0, d}, {\[Nu], 0, d}];

(* Function to compute Ricci tensor *)
R[\[Mu]_, \[Nu]_] := Simplify[
   Sum[Riemann[[\[Lambda] + 1, \[Mu] + 1, \[Lambda] + 1, \[Nu] + 1]], {\[Lambda], 0,
    ↪  d}]
   ];

RicciTensor = Table[R[\[Mu], \[Nu]], {\[Mu], 0, d}, {\[Nu], 0, d}];

(* Function to compute Ricci scalar *)
R[] := Simplify[
   Sum[metricInv[[\[Mu] + 1, \[Nu] + 1]] * RicciTensor[[\[Mu] + 1, \[Nu] + 1]],
    ↪  {\[Mu], 0, d}, {\[Nu], 0, d}]
   ];

RicciScalar = R[];
```

```
(* Function to compute Einstein tensor *)
G[\[Mu]_, \[Nu]_] := Simplify[
   RicciTensor[[\[Mu] + 1, \[Nu] + 1]] - 1/2 * metric[[\[Mu] + 1, \[Nu] + 1]] *
    ↪  RicciScalar
   ];

Einstein = Table[G[\[Mu], \[Nu]], {\[Mu], 0, d}, {\[Nu], 0, d}];


(* Printting the results *)

(* Printing Christoffel Symbol *)
Style["Non-zero Christoffel \
Symbol:", 22, FontFamily -> "CMU Sans Serif", Orange]
Do[
 If[(\[Mu] <= \[Nu]) && (Christoffel[[\[Alpha] + 1, \[Mu] +
        1, \[Nu] + 1]] =!= 0),
  Print[Row[{
     Style[
      Subsuperscript["\[CapitalGamma]",
       Row[{coordinates[[\[Mu] + 1]], coordinates[[\[Nu] + 1]]}],
       coordinates[[\[Alpha] + 1]]], 20, FontFamily -> "CMU Concrete"],
     Style[" = ", 20, FontFamily -> "CMU Concrete"],
     Style[Christoffel[[\[Alpha] + 1, \[Mu] + 1, \[Nu] + 1]], 20,
      FontFamily -> "CMU Concrete"]
     }]]
  ], {\[Alpha], 0, d}, {\[Mu], 0, d}, {\[Nu], 0, d}]

(* Printing Riemann Tensor *)
Style["Non-zero Riemann Tensor \
Components:", 22, FontFamily -> "CMU Sans Serif", Orange]
Do[
 If[(\[Rho] < \[Sigma]) && (\[Mu] < \[Nu]) && (Riemann[[\[Rho] +
        1, \[Sigma] + 1, \[Mu] + 1, \[Nu] + 1]] =!= 0),
  Print[Row[{
     Style[
      Subscript[Superscript["R", coordinates[[\[Rho] + 1]]],
       Row[{coordinates[[\[Sigma] + 1]], coordinates[[\[Mu] + 1]],
         coordinates[[\[Nu] + 1]]}]], 20,
      FontFamily -> "CMU Concrete"],
     Style[" = ", 20, FontFamily -> "CMU Concrete"],
     Style[Riemann[[\[Rho] + 1, \[Sigma] + 1, \[Mu] + 1, \[Nu] + 1]],
      20, FontFamily -> "CMU Concrete"]
     }]]
  ], {\[Rho], 0, d}, {\[Sigma], 0, d}, {\[Mu], 0, d}, {\[Nu], 0, d}]

(* Printing Ricci Tensor *)
Style["Non-zero Ricci Tensor \
Components:", 22, FontFamily -> "CMU Sans Serif", Orange]
Do[
 If[(\[Mu] <= \[Nu]) && (RicciTensor[[\[Mu] + 1, \[Nu] + 1]] =!= 0),
  Print[Row[{
     Style[
      Subscript["R",
       Row[{coordinates[[\[Mu] + 1]], coordinates[[\[Nu] + 1]]}]],
      20, FontFamily -> "CMU Concrete"],
     Style[" = ", 20, FontFamily -> "CMU Concrete"],
```

```
     Style[RicciTensor[[\[Mu] + 1, \[Nu] + 1]], 20,
      FontFamily -> "CMU Concrete"]
     }]]
  ], {\[Mu], 0, d}, {\[Nu], 0, d}]

(* Printing Ricci Scalar *)
Row[{Style["Ricci Scalar: ", 22, Orange,
   FontFamily -> "CMU Sans Serif"],
  Style["R = ", 20, FontFamily -> "CMU Concrete"],
  Style[RicciScalar, 20, FontFamily -> "CMU Concrete"]}]

(* Printing Einstein Tensor *)
Style["Non-zero Einstein Tensor \
Components:", 22, FontFamily -> "CMU Sans Serif", Orange]
Do[
 If[(\[Mu] <= \[Nu]) && (Einstein[[\[Mu] + 1, \[Nu] + 1]] =!= 0),
  Print[Row[{
     Style[
      Subscript["G",
       Row[{coordinates[[\[Mu] + 1]], coordinates[[\[Nu] + 1]]}]],
      20, FontFamily -> "CMU Concrete"],
     Style[" = ", 20, FontFamily -> "CMU Concrete"],
     Style[Einstein[[\[Mu] + 1, \[Nu] + 1]], 20,
      FontFamily -> "CMU Concrete"]
     }]]
  ], {\[Mu], 0, d}, {\[Nu], 0, d}]
```

Bibliography

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